Drilling Holes Like a Building and Shop Technician

A physical robot environment: concrete anchoring, the drill press, the cordless drill and the centre punch

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Abstract

Two Franka Research 3 arms on a mobile cart work in a basement maintenance workshop: they drill anchor holes into a reinforced concrete wall with an SDS-plus rotary hammer, clean them and set wedge anchors; at a steel bench they centre-punch, clamp and drill steel and aluminium at a drill press and drill sheet and timber with a cordless drill. Every tool is an articulated mechanism operated only by contact: a trigger is pulled by a finger, a feed handle by a hand. A hole is a physical record that becomes contact geometry for whatever enters it; how fast a bit advances follows from the force on it through published cutting laws for steel, aluminium, wood and concrete. Newton 1.5 with MuJoCo owns the dynamics at a 1 ms tick; a separate Isaac Sim process renders the recorded states and never steps physics. This report states the mechanisms, where every parameter comes from, the component experiments that test them against published data, the robot-performed demonstrations (the arms themselves holding and using the tools, with declared assistance), the measured failures behind the design, stability, runtime and the interfaces. No autonomous task solution and no physical calibration is claimed.

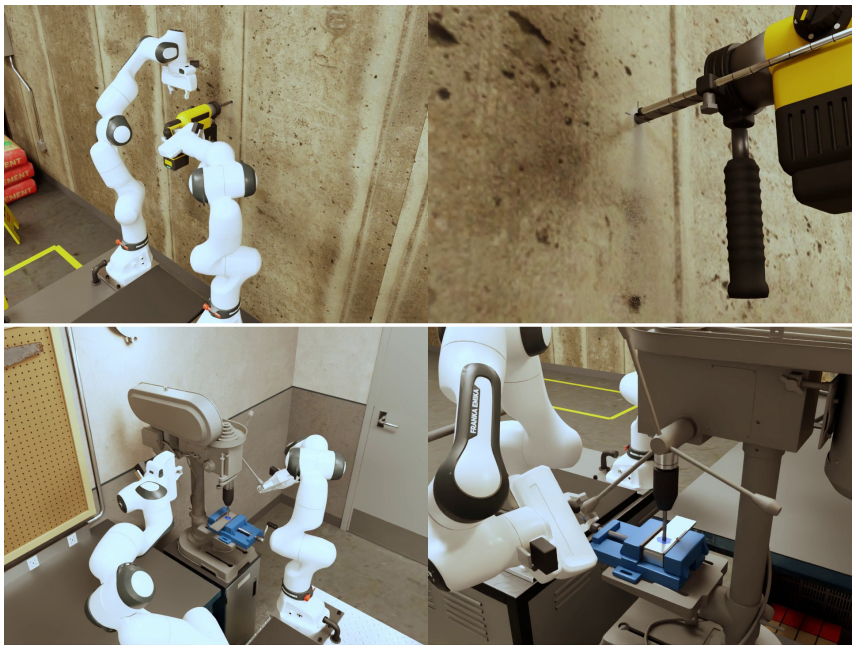


Figure 1: Two of the robot demonstrations, each a frame of its film (ASSISTED: scripted inverse-kinematics targets and planned paths, inside the FR3's own limits). Above: the right hand holds the rotary hammer by its handle, a finger on the trigger, and drills an anchor hole into the concrete wall; the left hand, which helped to carry it, has let go. Below: at the drill press the bar is clamped in the vise and one hand pulls the feed handle while the 9 mm drill cuts. Section 7 says what each run did and where it failed.

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1 The job, the tasks and the acceptance boundary

Drilling a hole is one physical act performed by electricians, plumbers, HVAC and maintenance technicians, carpenters and machinists: a rotating cutter is advanced into a material by a force until a hole of a given diameter, depth and quality exists, and then the hole is used for an anchor, a bolt or a pin. The action group behind this environment (84 occupational task statements) is covered by three scenes in which the physics that decides success in that job differs:

- **Concrete anchoring.** A rotary hammer drills ½ in holes 108–125 mm deep into a cast-in-place wall. Blows land only while the operator pushes; a hidden rebar can stop the bit and must not be cut; the drilling dust is removed where it would stop the anchor short (the maker's instructions for this wedge anchor in a hammer-drilled hole require no cleaning sequence, only that the anchor reaches its embedment), a wedge anchor is hammered until its washer seats, and its nut is tightened to the installation torque, which draws the cone into the expansion clip.
- **Shop metal and timber.** Steel and aluminium are centre-punched, clamped in a drill-press vise and drilled through, blind to depth or matched through a stack; sheet and timber are drilled with a cordless drill. An unseated drill walks, a thin sheet grabs, an exit burrs or tears out, a hand-held part spins. (Of these the grab of thin sheet is not modelled here: the sheet task is drilled the way the trade avoids it, piloted and on a backer, and is a placing and enlarging task. Section 10 lists it with the other missing mechanisms.)
- **Operating the tools.** Triggers, selectors, the press's buttons and feed handle, the vise crank and the automatic punch are mechanisms that the robot operates with its fingers. Nothing is switched on by an action.

Table 1 lists the eleven tasks. Each start state is settled before hand-over; the evaluator reads only physical state (hole records, part poses, joint states, contact forces), never the commands that produced it. This report establishes that the environment is built and physically meaningful. It does not claim an autonomous solution: every demonstration below is labelled ASSISTED, and nothing assisted is reachable through the policy interface.

Table 1: The eleven tasks: the instruction, and the checks the evaluator returned on an correct end state (E17), read from physical state only. A task succeeds when every goal and every constraint holds.

preset / task	instruction given to the policy	goals (score = fraction met)	constraints (must all hold)
bench / take-drill	Take the cordless drill in both hands, one hand on its grip and the other on its motor housing, and hold it ready to drill straight down into the bench, the bit tip 30 to 70 mm above the bench, for 3 seconds. Do not run it.	held only by the robot's two hands for 3 s; one hand on the grip and the other on the motor housing for 3 s; bit points straight down (within 5 deg); tip 30-70 mm above the bench	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); trigger never pressed past 1 mm; the bit touched nothing
bench / punch-layout	Centre-punch the four layout marks on the steel bar.	each mark: dimple within 0.3 mm	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); each mark: no stray dimple within 10 mm; bar moved less than 2 mm
bench / press-drill	Clamp the bar in the drill-press vise and drill a 9 mm hole through it at the punched mark. Leave the spindle stopped and the quill up.	hole A: through hole at the mark; hole A: entry within 0.5 mm of the mark; hole A: axis within 1 deg of the face normal; hole A: diameter at most 9.25 mm	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); bar turned less than 2 deg while drilling; spindle switched off and stopped; quill within 2 mm of the top; bit clear of the work
bench / press-blind	Drill the M8 tap-drill hole at the punched mark in the aluminium block to 15 mm full-diameter depth. Do not drill through. Leave the spindle stopped and the quill up.	full-diameter depth 15.0-15.5 mm; not through; entry within 0.5 mm of the mark; axis within 1 deg	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); spindle switched off and stopped; quill within 2 mm of the top; bit clear of the work

preset / task	instruction given to the policy	goals (score = fraction met)	constraints (must all hold)
bench / match-bolt	Drill the two 9 mm bolt holes through the angle bracket and the flat bar at the marks so that they line up, then push the two M8 bolts through both parts.	angle each hole: through hole at the mark; angle each hole: entry within 0.5 mm of the mark; angle each hole: axis within 1 deg of the face normal; angle each hole: diameter at most 9.25 mm; two through holes in the flat bar; hole spacing within 0.5 mm of 60 mm; each bolt passes through both parts	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); each bolt pushed in with less than 50 N; angle_bracket not spun (< 2 deg); flat_bar not spun (< 2 deg)
bench / panel-holes	Drill four 6 mm holes through the enclosure cover at the four marks. The cover is strapped on a plywood backer; a 3 mm and a 6 mm drill are on the bench.	each hole: through hole at the mark; each hole: entry within 1.0 mm of the mark; each hole: axis within 1 deg of the face normal; each hole: roundness at most 1.10	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand)
bench / timber-bolt	Bore a 9/16-inch hole through both boards at the mark, square to their faces (within 1.2 degrees), and pass the 1/2-inch carriage bolt through by hand.	through hole in the top board; entry within 2 mm of the mark; square to the faces (axis within 1.19 deg of the face normal); no charred wall (< 200 C); through hole in the bottom board; carriage bolt through both boards	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); bolt pushed in with less than 50 N
wall / anchor-holes	Drill four 1/2-inch anchor holes at the marked crosses, square to the wall, 108 to 125 mm deep.	each hole: within 3 mm of the mark; each hole: axis within 3 deg of the wall normal; each hole: depth 108-125 mm; each hole: reinforcement damage at most 0.5 mm	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); bit not chipped on steel
wall / rebar-relocate	Drill the four 1/2-inch anchor holes at the crosses on the two channel lines, square to the wall, 108 to 125 mm deep. If a hole hits reinforcement, stop without cutting the bar and leave that hole unfilled; drill its replacement on the same channel line at least twice the stopped hole's depth from it and no more than 150 mm from it.	each hole: within 3 mm of the mark; each hole: axis within 3 deg of the wall normal; each hole: depth 108-125 mm; each hole: reinforcement damage at most 0.5 mm; each hole (over the bar): started within 3 mm of its mark; each hole (over the bar): reached the bar; each hole (over the bar): the bar was not cut (damage at most 0.5 mm); a replacement on the channel line 90-150 mm from the stopped hole meeting the anchor-hole criteria	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand)
wall / clean-and-set	Set the four 1/2-inch wedge anchors in the four predrilled holes: drive each to its nominal embedment (95 mm, its washer on the wall), then tighten its nut with the impact wrench on the kit trolley (its adaptive-torque module is set for these anchors: it stops with a green light at their installation torque, 68 N m). The holes still hold their drilling dust: remove it with the shop vacuum's nozzle (switch the vacuum on; the brush is optional) so that the anchors reach their embedment.	each hole: anchor driven to h_nom (95 +- 2 mm); each hole: anchor straight (within 3 deg); each hole: nut torqued to T_inst (61-116 N m); each hole: expansion engaged (cone drawn into a gripping clip)	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); each hole: anchor did not spin (<= 45 deg)
wall / ledger-anchor	At the four marked points on the ledger, drill through the ledger and 108 to 125 mm into the concrete, remove the drilling dust as needed with the shop vacuum's nozzle (switch the vacuum on), and set the four 1/2-inch wedge anchors: drive each to its nominal embedment (95 mm) with its washer on the ledger, then tighten its nut with the impact wrench (its adaptive-torque module is set for these anchors: it stops with a green light at their installation torque, 68 N m).	each hole: within 3 mm of the mark; each hole: axis within 3 deg of the wall normal; each hole: depth 108-125 mm; each hole: reinforcement damage at most 0.5 mm; each hole: anchor driven to h_nom (95 +- 2 mm); each hole: anchor straight (within 3 deg); each hole: nut torqued to T_inst (61-116 N m); each hole: expansion engaged (cone drawn into a gripping clip); four holes through the ledger	no fault; no bit broken; no arm carried more than its rated payload (2.27 kg beyond its hand); each hole: anchor did not spin (<= 45 deg); ledger still level within 3 mm over its length; bit not chipped on steel

1.1 Whether each task can be done at all

A task that no operator could complete is a defect of the world, not a hard task. Two probes check this without any operator. The first reads every seed's layout (seeds 0–999 of each task) and tests the geometric preconditions of completion: every mark clear of the hidden reinforcement by more than the evaluator's own placement and axis tolerances allow a hole to stray, the bit long enough for the deepest hole asked, the deepest admissible hole short of the far mat, the anchors long enough for the fixture, a replacement site for the hole stopped by the bar, and the washer seating with the tip at the nominal embedment. The second puts the rotary hammer in the robot's hands at every wall mark and asks for a collision-free two-arm configuration at drilling start and at full depth, on the world's own collision geometry, and for a fingertip reach to the vacuum's switch. It is a static check of reach: only the robot demonstration of Section 7 drills a hole with the arms, and it drills one. That demonstration's own planner (one grasp per hand, the hammer upright) finds no arm configuration for the lower marks; that is a limit

of the planner, not of the station.

Table 2: Feasibility of every task over seeds 0–999: the geometric preconditions that make the task completable in principle, read from the task's own layout (no physics, no robot).

task	conditions	seeds failing	geometric preconditions checked
anchor-holes	4	0	marks clear of rebar; marks inside the drillable region; bit reaches the minimum depth; deepest admissible hole stays in front of the far mat
rebar-relocate	6	0	marks clear of rebar; marks inside the drillable region; bit reaches the minimum depth; deepest admissible hole stays in front of the far mat; the mark over the bar cannot miss it; a replacement site clear of every bar (relocation window)
clean-and-set	8	0	marks clear of rebar; marks inside the drillable region; bit reaches the minimum depth; deepest admissible hole stays in front of the far mat; predrilled holes 108-125 mm at full diameter; seated washer puts the tip at h_nom +- 2 mm; anchor length takes the fixture (Hilti Table 47) and the nut; the shop vacuum stands clear of the cart, the trolley and the wall
ledger-anchor	7	0	marks clear of rebar; marks inside the drillable region; bit reaches the minimum depth; deepest admissible hole stays in front of the far mat; seated washer puts the tip at h_nom +- 2 mm; anchor length takes the fixture (Hilti Table 47) and the nut; the shop vacuum stands clear of the cart, the trolley and the wall
take-drill	1	0	drill bit long enough
punch-layout	1	0	no geometric precondition beyond the station
press-drill	2	0	press quill travel covers the gap and the cut; press bit long enough
press-blind	2	0	press quill travel covers the gap and the cut; press bit long enough
match-bolt	4	0	press quill travel covers the gap and the cut; press bit long enough; m8_bolt0 longer than bracket + bar; m8_bolt1 longer than bracket + bar
panel-holes	10	0	pilot_drill bit long enough; drill bit long enough; mark 0 at least 25 mm from the clamped part's edge; mark 0 at least 30 mm clear of a clamp's toe; mark 1 at least 25 mm from the clamped part's edge; mark 1 at least 30 mm clear of a clamp's toe; mark 2 at least 25 mm from the clamped part's edge; mark 2 at least 30 mm clear of a clamp's toe; mark 3 at least 25 mm from the clamped part's edge; mark 3 at least 30 mm clear of a clamp's toe
timber-bolt	4	0	drill bit long enough; mark 0 at least 25 mm from the clamped part's edge; mark 0 at least 30 mm clear of a clamp's toe; carriage bolt longer than both boards

Table 3: Wall reach with the rotary hammer in hand: for every mark, a collision-free two-arm configuration holding the hammer by its trigger grip and a second grasp, with the bit on the mark, at drilling start and at full depth (static inverse kinematics on the world's collision geometry; no hole is drilled by this probe).

task	robot	seed	marks reachable	hammer roll used (°)	one-handed at the end	vacuum switch
anchor-holes	dual-fr3	0	4/4	0	1	–
anchor-holes	dual-panda	0	4/4	0	1	–
rebar-relocate	dual-fr3	0	6/6	0	1	–
rebar-relocate	dual-panda	0	6/6	0	1	–
clean-and-set	dual-fr3	0	4/4	0	1	yes
clean-and-set	dual-panda	0	4/4	0	1	yes
ledger-anchor	dual-fr3	0	4/4	0, 30	0	yes
ledger-anchor	dual-panda	0	4/4	0, 30	0	yes

2 The scene

2.1 Room, stations and robot

The room is a basement maintenance workshop, 8 m by 6 m and 3 m high (Figure 2). Two stations share it. At the north wall a cast-in-place concrete wall 200 mm thick (C25/30 [20], board-formed finish) carries the layout marks; a mobile robot cart docks in a taped box 0.6 m from it. At the east wall a steel workbench (top 0.90 m) carries the parts and hand tools, and a benchtop drill press stands on its steel cabinet at the bench's south end. The default robot is two Franka Research 3 arms with Franka Hands on that cart; a Panda pair and a single right FR3 or Panda (the press and punch tasks only) are the supported alternatives (Section 5).

Everything a robot can touch is a physical body with colliders, mass and friction, built in `scene.py`, `stations.py`, `tools.py`, `parts.py`, `vacuum.py` and `clamps.py`. That includes the shop vacuum beside the kit trolley and the strap clamps that hold the panel and the boards on the bench. Everything else is visual dressing with no collision: the ceiling with its fluorescent troffers, sprinkler main, copper line, conduit and supply duct; shelving with stock, cement bags, a crate of anchor boxes, a ladder, a roller cabinet and tool chest; the pegboard of hand tools behind

the bench, a bench vise at its far end, an outlet strip, a steel door. Dressing is kept out of every volume an arm, tool or part can sweep, so a robot never passes visibly through an object that is not there. Models and textures are pinned CC0 Poly Haven [48] assets (SHA-256 in `visual_assets.json`); the robot cart, the workbench, the press cabinet, the kit trolley and the rotary hammer's housing are authored meshes fitted to their colliders.



Figure 2: The two stations as rendered from the recorded start states: the wall bay (left) and the drill-press bench (right). Isaac Sim RTX path tracing of the simulator's own poses; the renderer takes no physics step.

2.2 Cameras

Four RGB cameras are available to a policy: a fixed station camera (at the wall, on a mast at the cart's rear, 1.7 m above its deck; at the bench, beside the press), a fixed overview of the cart and work, and one camera on each wrist (the D405-class case on link 7). The station camera frames the work point; the kit trolley, the shop vacuum and the parts lying on the bench are outside it. A session therefore observes the station camera and the overview unless it selects otherwise with the `cameras` option, and the wrist cameras are added the same way; each camera raises the shortest policy command (Section 9). Diagnostic cameras (hero, macro views, the room) are for review only. Figure 3 shows the four images at the start of every task. Camera placement lives in `cameras.py`, apart from the physics, so a reframed view changes no physical result.

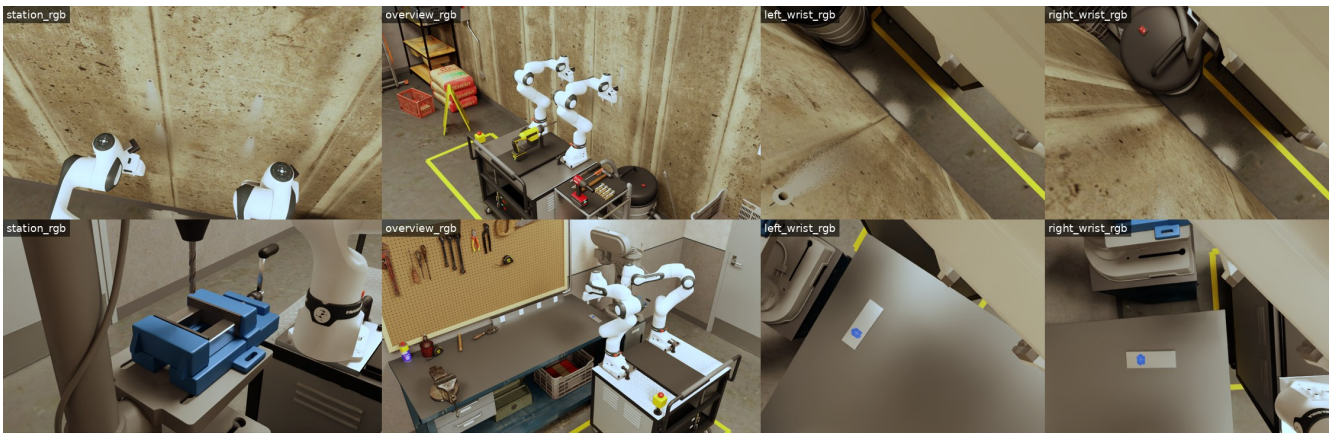


Figure 3: The four policy cameras at the start of each task: station and overview (960 x 540, the default observation) and both wrist cameras (640 x 480, launch options). Isaac RTX.

2.3 Tools and parts at real scale

Table 4 gives every interactable object with the product it represents, its dimensions and mass and where each value comes from. Tools are articulated assemblies whose moving parts are real joints (Section 3).

Table 4: Interactable objects at real scale (`params.py`, `layout.py`).

object	represents	size, mass, rating	basis
Robot	two Franka Research 3 arms with Franka Hands on a mobile cart (deck 0.80 m, bases 0.56 m apart)	FR3 limits 87/12 N m, 3 kg payload; hand 70 N continuous grasp	read (FR3 datasheet); cart authored
Cordless drill	DeWalt DCD800 class 20 V drill/driver, 2.0 Ah pack	1.68 kg; 650/2000 rpm; stall 27/8.8 N m (the manual's soft-joint torque, per gear); clutch collar 1.7-8.5 N m	read (manual); colliders fitted to Drill_01
Rotary hammer	DeWalt DCH273 class SDS-plus, 5 Ah pack, 13.3 x 160 mm carbide bit (0.113 kg)	3.12 kg (2.5 + 0.62); 2.1 J at 4600 bpm; 0-1100 rpm; 400 W, stall 13.9 N m; electronic overload clutch	read (manual); stall and striker derived; clutch level estimate
Drill press	Grizzly G7943 class benchtop press on a steel cabinet	table 0.97 m; three-spoke handle R = 0.21 m; rack pinion 20.2 mm; 1043 rpm at the task's belt step	read; profile measured on drill_press_01
Press vise	4 in drill-press vise (WEN DPA424 class), split bed	opening 75 mm (50 mm for press-blind); screw pitch 2.5 mm; thread mu 0.15, collar mu 0.12: 396 N of clamp per N m at the crank	read / literature (Shigley) / estimate
Centre punch	Rennsteig 430 130 automatic punch	125 mm long, 14 mm body; point 90 deg; cap setting 60 to 130 N (a seeded start state; 95 N and 0.14 J where no seed sets it)	read (maker's listing, search level); energy derived from the spring
Flat bar	S235 mild steel	150 x 50 x 6 mm, 0.35 kg	authored size; material read
Block	6061-T6 aluminium	50 x 50 x 25 mm	authored
Angle	50 x 50 x 5 steel equal angle (EN 10056-1), the bar's width	120 mm long	authored length; section read
Cover panel	1.5 mm steel enclosure cover on an 18 mm plywood backer, under two strap clamps	200 x 250 mm	authored
Boards	two Douglas-fir 2 x 6 (12 % MC) on an 18 mm plywood backer, under two strap clamps	300 x 140 x 38 mm each	read (nominal lumber); backer authored
Hold-down clamps	plain strap clamps (TE-CO 30402) on 3/8-16 studs with spherical flange nuts and heel blocks	strap 102 x 25 x 16 mm; nut torque 10.8 N m, stud tension 5.8 kN	read (maker's catalogue, Carr Lane table); torque authored
Fasteners	M8 x 30 hex bolts (ISO 4017); 1/2 x 4 in carriage bolt (ASME B18.5)	head 13 AF x 5.3 mm; 0.6 mm chamfer; the carriage bolt's flat end on three points 0.3 mm proud	read; the three points authored (engine)
Wedge anchors	Hilti KB-TZ2 1/2 in: 4-1/2 in on the wall, 7 in through the ledger; stud, 1/2-13 nut, washer, expansion clip on a cone	washer at h_nom = 95 mm from the tip; installation torque 68 N m; axial stiffness 27.8 kN/mm	read (ESR-4266, maker's supplement); clip drag and free diameter open
Impact wrench	Hilti SIW 6AT-A22 class with the SI-AT-A22 adaptive-torque module, 3/4 in deep socket	2.33 kg with its 0.83 kg pack, 0.26 kg module and socket; stops at the anchor's installation torque	read (data sheet, patent); stop rule authored
Shop vacuum and kit	14 gal drum vacuum with rocker switch, hose and nozzle; hole brush; 32 oz ball-peen hammer	tank 0.46 x 0.45 m; nozzle 10 x 200 mm; brush 13 mm head on 250 mm rod; hammer head 0.907 kg on a 340 mm handle (1.04 kg)	estimate (class sizes); rocker read
Wall	cast-in-place C25/30 wall, 200 mm, two mats of #4 rebar at 300 mm each way: 25 mm cover at the room face, 38 mm at the far face	drillable region 2.4 x 1.6 m; 19 mm aggregate packed by seed	strength class read (EN 206); covers from ACI 318-19 Table 20.5.1.3.1 (near authored above its 3/4 in, far read 1-1/2 in); layout authored
Workbench	steel workbench	1.83 x 0.76 m, top 0.90 m	authored (commercial 36 in)

3 Tools as mechanisms

Every tool is an articulated assembly whose moving parts are joints; it is operated only by contact. Each tick, before the solver step, a tool reads its own joint states and sets its drives (`tools.py`). No action or key switches anything on.

Cordless drill (DeWalt DCD800 [9] class, 1.68 kg). The trigger slides 8 mm against a 4 N preload rising to 10 N (estimate) and sets the speed command $s = \mathrm{clamp}((x - 1\,\mathrm{mm})/6\,\mathrm{mm}, 0, 1)$; the brushless motor is a velocity actuator $M = s\,M_{\mathrm{stall}}(1 - \omega/(s\,\omega_0))$ integrated implicitly. The stall torque is the figure the drill's own manual gives for a yielding joint, "Max. torque (hard/soft) 90/27 Nm": 27 N m in gear 1 (650 rpm) and, through gear 2's ratio, 8.8 N m at 2000 rpm; the motor line then peaks at 459 W. The hard-joint 90 N m is a peak that includes the rotor's kinetic energy. (The figure matters to a robot: a stall torque of 12.3 N m in gear 2 would let a jammed bit trip an FR3 wrist's 11.4 N m reflex, and 8.8 N m cannot.) Releasing the trigger brakes the chuck. A gear slider, a forward/lock/reverse button and a 15-setting clutch collar (1.7–8.5 N m, then drill mode) are joints too.

Rotary hammer (DeWalt DCH273 [10] class: 2.5 kg bare and a 0.62 kg 5 Ah pack, 3.12 kg, both from its manual; the 13.3 × 160 mm bit weighs 0.113 kg, scaled from bits KIT weighed). D-handle trigger as above, side handle 0.22 m ahead of the grip, depth rod, and the SDS bit sliding 12 mm in the chuck against a return spring (5 N with the bit out, 9.8 N with it pushed back). Rotation is a motor line whose stall torque is what the manual's 400 W maximum output allows, 13.9 N m at 1100 rpm. When the bit jams, the electronic overload clutch interrupts the drive, as the manual describes, and releasing the trigger re-arms it; its trip level is not published and is set at the motor slowed to a tenth of its speed for 50 ms. The three-position mode knob turns only while its release button is held in, as the manual states ("there are no operable positions in between"); without that lock a robot link resting on the knob turned it from hammer-drill to rotary. Blows happen only while the bit is pushed back into its idle-stroke window: no push, no blows. Blow rate $f_b = s \cdot 76.7$ Hz, energy $E_b = 2.1\,s^{2.4}$ J (the exponent derived from KIT's measured 0.59/1.65/2.95 J at 24.6/36.9/50 Hz). Each blow removes material (Section 4) and returns the striker's momentum $p_s = \sqrt{2 m_s E_b}$ to the housing. The striker mass has no published value and is calibrated to the manual, which asks for "a gentle pressure (approx. 5 kg)" and adds that more does not speed up drilling. In steady drilling the push must balance the mean recoil $f_b p_s$, the bit's return spring (9.8 N with the bit pushed back) and the force the bottomed housing passes to the bit (5 N, measured), so 49 N leaves 34 N for the recoil: $m_s = 0.047$ kg at 9.4 m/s, the largest striker with which the simulated hammer keeps its blows at that push (Section 6.3). The recoil reaches the housing as a half-sine 0.30 of the blow period wide (3.9 ms at full rate), the half-sine with the impulse and peak of the measured air-spring pressure pulse of an electro-pneumatic hammer (Wang 2014 [42]).

Drill press (Grizzly G7943 [6] class). The quill is a prismatic joint tied to the feed shaft by a rack and pinion ($x = -r_p\varphi$, $r_p = 20.2$ mm, a joint equality; E1a measures 519.9 N on the quill for 10 N m on the shaft against 519.6 N). Three spokes with knobs at $R = 0.21$ m, a return spring, a depth-stop nut, and start/stop push buttons that latch the motor (1043 rpm at the task's belt step, 1 s coast-down).

Vise. The moving jaw is tied to the screw by a joint equality (2.5 mm pitch). The screw's friction is load dependent on the thread ($\mu = 0.15$, oiled steel, Shigley [27]) and on the thrust collar where the screw's shoulder bears on the body ($\mu = 0.12$ at a 9.6 mm mean radius), so the thread self-locks and the vise gives 396 N of clamp per N m at the crank; a Kurt D688, which has a roller thrust bearing, is published at 327–370 N per N m. Without the collar's friction the vise would give 724 N per N m, twice too strong. The bed is split around a 30 mm slot so a through drill clears it.

Automatic centre punch (Rennsteig [8] 430 130: 125 mm long, 14 mm body). The point slides into the body against the main spring; at the release force the internal hammer strikes with the spring's stored energy. The release force is the tool's own adjustable setting, 60 to 130 N on the cap, and is part of the start state: each seed draws it from that range (95 N and 0.14 J where a scene does not set it), not from one value chosen inside one FR3's axial reach. A punch found set to 130 N needs more push than one FR3 arm delivers along its axis, and that is the robot's limit, not the tool's. The dimple takes the share of the strike's energy the struck part's support does not absorb, so a bar backed by the bench takes all of it and a part free to recoil takes less; a full strike leaves a conical dimple $h = \left(3E_p/(\pi \tan^2\alpha\, H)\right)^{1/3}$, 0.40 to 0.52 mm in S235 over the cap's range. Lifting re-cocks it. The point's contact tip is rounded, and the point's spring is critically damped against the punch body: a flat-tipped, lightly damped punch balances on its point and bounces without end when dropped.

Fasteners and kit. M8 hex bolts (ISO 4017) with hexagonal heads, a ½ in carriage bolt, a hole brush, a 32 oz

ball-peen hammer, and a drum shop vacuum with a hose and nozzle: a nozzle pass removes dust only while the vacuum runs, and it runs after a finger turns its rocker switch over. The wedge anchors and the impact wrench that sets them are described in Section 3.1.

3.1 The wedge anchor and the impact wrench

A wedge anchor holds because a nut is tightened. The stud's tip is a tapered mandrel (the cone) carrying a split expansion clip; hammering puts the anchor in its hole with the clip dragging on the wall, and tightening the nut against the washer draws the stud back out by a millimetre or two, the cone into the clip, and the clip into the concrete. The anchor is an assembly (Hilti KB-TZ2 ½ in, ICC-ES ESR-4266 [18]):

- **Nut and thread.** The nut slides along the stud by the thread's pitch per turn (½–13 UNC, 1.954 mm), a joint equality whose solved force is the stud's tension F. Turning it on costs the torque $T = F\,(P/2\pi + 0.577\,\mu_G d_2 + \mu_K r_b)$ (ISO 16047 [26]), applied as friction on the nut's turn and on the washer under it, with $\mu_G = \mu_K = 0.151$, the value at which this split gives the handbook torque factor $K = T/(Fd) = 0.20$ of zinc-plated threads (an estimate for this anchor, whose nuts carry a proprietary coating). The thread is self-locking.

- **Clip and cone.** The clip's ring presses on the wall of any hole smaller than its free diameter (13.7 mm, estimate) and drags at 500 N in the nominal 13.3 mm hole (open: no value in newtons is published). At 150 N an anchor is seated with a steady 160 N push in 0.4 s, which the approval document does not allow for: it requires the anchor to be hammered in [18], a setting-tool patent says the clip "requires that a significant force be applied" [38], and an industrial arm of 13 kg payload is reported to have pushed a 12 mm wedge anchor 7 mm in before it stuck [39]. 500 N is above what either arm of the kit can push (their reflex stops a push near 90 N); it is still an authored value, bounded only from below by those reports and from above by the blows a hammer needs (E20). The drag is the model's only resistance to driving the anchor, does not depend on speed, and falls in proportion in a hole the bit wore wider (113 N in E9's 13.61 mm hole, where an anchor can again be pushed in: such a hole is also the one in which an anchor spins or pulls up). The kit's hammer is a 32 oz ball-peen (0.907 kg head, 1.04 kg with its handle); a 16 oz hammer would need about three times the blows [41]. It is taken to sit far firmer on concrete than on the cone ($\mu = 0.6$ against 0.16: the assumed inputs of a finite-element study of another anchor, Noh et al. 2014, not measurements), so when the nut pulls, the clip stays and the stud moves through it: after 0.5 mm of play (estimate) the cone meets the clip, and from there the stud's tension rises with its travel by the anchor's published axial stiffness, 27.8 kN/mm in uncracked concrete (ESR-4266). A clip that does not touch the wall holds nothing: in a hole wider than the clip the stud turns and pulls up with the nut and no torque builds, the installer's sign of a bad hole.

- **The impact wrench** (Hilti SIW 6AT-A22 [24] class with the SI-AT-A22 adaptive-torque module, the setting tool ESR-4266 permits). Its trigger runs a motor; the anvil turns with it while the socket meets little resistance (the nut running down), and above the cam's release torque (5 N m, estimate) a hammer on the drive shaft strikes the anvil's jaws at the blow rate, as the maker's patent describes [25]. A blow is a hammer of inertia I_h moving at the shaft's speed ω that meets the anvil through a one-way torsion spring k (jaws, anvil and socket). It carries the hammer's kinetic energy $I_h\omega^2/2$ and no more, it cannot drive the anvil faster than about the hammer moves, and against a nut that does not turn its torque peaks at $\omega\sqrt{kI_h}$. The product kI_h is fixed by the maker's largest torque at its largest speed, 213 N m at 2300 rpm; the three gears' rated torques (90, 135 and 213 N m at 1000, 1500 and 2300 rpm) are proportional to speed, as this law is. I_h itself is an estimate (1.45×10^{-4} kg m^2: 4.2 J a blow at full speed). The torque a blow leaves on a nut is a result: the nut turns on while the spring's torque exceeds its thread and bearing friction.

- **The module** (a start state: its box scanned for this anchor) does what the patent's first phase describes: it runs the motor at a stored speed (authored: the wrench's full speed; the maker's table is not published), senses the torque on the anvil's shaft, and stops the motor with a green light when the torque a blow passed to the socket reaches the stored threshold, the anchor's installation torque of 68 N m. A run that does not get there in 5 s of impacting ends with a red light. The patent's later phases (a waiting time, further blows, a timed phase at a lower speed) answer a relaxation of the preload that this anchor model does not have, and are not modelled. The patent describes sensed torque, not a rule on the anvil's turn per blow.

- **The stud can fail.** Its tension is carried up to the steel strength ESR-4266 gives for this anchor, 50.0 kN; beyond it the stud is broken, the nut draws it out, and the evaluator fails the anchor. Without that limit no upper torque bound could be reached by anything.

What is an input here and what is a result. If the wrench's pulses were made to peak at the installation torque, the torque would be an input, the preload would follow from it through the nut factor ($F = T/(Kd)$, 26.8 kN for $K = 0.20$), and the rise from that preload through the published axial stiffness and the clip's play: three inputs would fix all three numbers. Here the torque the nut is left with is where the module's reading stops the blows, 67 to 70 N m in E20 for a threshold of 68; the preload and the rise still follow from the nut factor (a generic handbook value: the anchor's own is not published) and the published stiffness, and are reported as such (Table 18). What the model does decide by itself: how many blows a setting takes, that a nut in a hole wider than the clip never builds torque, that every further pull of the trigger on a set anchor adds a blow (the nut passes 1.7 times the installation torque at the 66th pull; in a longer run on an earlier version of the model the stud broke at the 95th), and that the same wrench without its module passes that bound after 2.5 s and breaks the stud after 3.0 s, at 50.2 kN.

Declared devices of the blow. A real blow lasts under a millisecond, which the 1 ms tick cannot resolve. Each blow is stretched sixteen times: the hammer's inertia $\times 16^2$ at a sixteenth of its speed on the same spring, so its energy and its peak torque are unchanged while it lasts 5 to 50 ms; the anvil and socket carry 16^2 times their own inertia for the duration of a blow. The hammer is not a body: its state is an energy and the spring's compression, and each tick its energy falls by the spring's torque times the rotation the solver gave the anvil, so the work a blow does on a nut can never exceed the hammer's energy. A blow written as a torque on the anvil's 8×10^{-5} kg m^2 takes the anvil to 574 rad/s on the first blow while the nut still runs freely, and the socket's hex passes the nut's corners three times in nine blows. With the hammer's inertia in the blow the anvil stays below 21 rad/s while impacting (traces of single blows) and the socket never passes a corner: on every tick of every setting in E20 the two hexes are within 6.9° of each other, their play.

Three numerical settings of this assembly are labelled *engine* in the ledger. The nut's slide along the stud carries an added inertia of 10 kg, which acts only on the nut's motion relative to the stud: with it the nut stays within 0.002–0.007 mm of its thread position through a hammer blow. The nut's turn carries 10^{-4} kg m^2, about that of the anvil and socket that drive it: without it the bare nut's friction stops it within a tick whenever the socket's flats lose contact, and in a fast run-down the socket then stands up to 11° into the nut for one tick; its cost is that the stud turns a little with the nut during a setting, 3 to 7° in the cases that set the anchor (Table 18). The washer carries no added inertia and the wrench no drag between blows. The nut's hex and the socket's walls meet at MuJoCo's stiffest contact.

3.2 The hold-down clamps

The enclosure cover on its plywood backer and the two fir boards are free bodies; nothing is welded. Each stack is held on the bench by two plain strap clamps as a machinist sets them: a TE-CO [29] 30402 strap on a 3/8-16 stud, a spherical flange nut on a spherical washer, and a steel heel block of the stack's height. The strap is a free body that lies with its toe on the work and its heel on the block, the stud standing in its slot. The nut slides on the stud against the elasticity of stud and strap, and the washer rocks under it, so the stud's pull meets the strap on the stud's axis however the strap lies. Before hand-over each nut is run down until the stud tension reads the registered preload, 5,783 N: Carr Lane [28]'s table gives 2,600 lb at the recommended 16 ft lb, and the nuts are tightened to 8 ft lb (10.8 N m), because at the full torque the 1 in toe would bear on Douglas fir at 10 MPa, above its 5.5 MPa strength across the grain. From then on the clamp is passive. The strap puts 48 % of the stud's pull on the work (its lever arms are 49 mm and 45.6 mm), and the work is held by the friction of its faces under that load; pushed or twisted harder it slips. E21 measures the rest, the clamp force and the load at which each stack slips (Section 6.6).

Not modelled: the stud's bending (the rod is rigid on the bench, so across the straps the tall timber stack is held better than a real one); the nut's thread (it has no turn, and no task asks an operator to release a clamp); the spherical seat's friction. Under the two clamps the timber stack shortens by 0.04 mm in all (a strap's toe stands 0.03 mm in the board), where a hand estimate for the real stacks gives 0.08–0.14 mm. It is 0.3–0.4 mm with the stiff clamping impedance of hardened steel faces (Section 8.4), and 1.0–2.3 mm with the engine's default contact (ledger entries `HOLD_DOWN_STACK_STIFF`, `HOLD_DOWN_SOLIMP`).

4 Holes

A hole is a record in its workpiece's body frame: entry point, axis, entry-face normal, one stage per bit diameter, burr, exit damage, entry spall, scorch, removed volume, dust (loose, packed, in transit), the lateral offset profile every 0.5 mm and an energy ledger. The evaluator and the renderer read these records; the policy does not.

4.1 Feed from the solved force

While a bit cuts, a floor plug sits under its tip at the hole's depth s; MuJoCo solves the tip/plug contact implicitly and gives the thrust T over the tick. The cutting law inverts to the feed per revolution the thrust supports, $f = F^{-1}(T)$, zero below a threshold, and the floor advances

$$\Delta s = \min(f,\, 0.2\,d)\,\frac{\Delta\theta}{2\pi}, \qquad \Delta\theta = (\omega_{\text{bit}} - \omega_{\text{work}}) \cdot a\,\Delta t, \tag{1}$$

never ahead of the tip. The bit advances only as fast as the material yields to the applied force. The cutting torque follows the chip actually cut and is applied equal and opposite to bit and workpiece, bounded by the torque that would stop their relative spin within the tick.

A stack (a sheet on a backer, a bracket on a bar) is cut part by part. While the point is in the part below and the lips are still cutting the part above, the upper part's remaining ring is a second cut on the same tip: its thrust and torque are added by its own material's law at the same feed, the feed is the one at which the summed thrust equals the solved thrust, and each part's hole advances by what was cut in it. An enlarging drill's stage starts where its point geometrically meets the bore the smaller drill left (the pilot's rim, 0.90 mm below the face for a 6 mm point on a 3 mm bore), not at the tip's position. Setting the upper part's stage to the tip's depth once the point is in the backer, and charging nothing for it, would cut the four 6 mm holes of the sheet task in 0.2 s each by the plywood's law, let a drill that is not turning sink through a 3 mm pilot, and open a second hole record in the backer beside the pilot's. The holes take 1.9 to 2.0 s of cutting at 80 N, 42.4 mm^3 of steel each, and a stopped 6 mm drill rests on the pilot's rim.

4.2 Cutting laws

Metals follow the Machinery's Handbook [1] (Shaw–Oxford) form, in SI (d in mm, f in mm/rev, K_d in MPa):

$$T = 7.294\,K_d B W\,(fd)^{0.8}\lambda_T + K_d d^2 J W \chi \quad [\text{N}], \tag{2}$$

$$M = 0.003647\,K_d A W\,f^{0.8} d^{1.8}\lambda_M \quad [\text{N m}], \tag{3}$$

with $A = 1.085$, $B = 1.355$, $J = 0.030$; the second thrust term is the chisel edge's indentation, a threshold below which the drill rubs (231 N for a 9 mm drill in steel; a pilot at least as wide as the chisel edge removes it, $\chi = 0$). $K_d = 95$ MPa for S235-class steel (fitted to Rubenstein [2]'s AISI 1020 data), 35 MPa for 6061-T6; wear W grows with removed volume and above 450 °C. **Wood** uses Szwajka's [11] measured MDF form scaled by diameter, $T = \kappa(13.0 + 66.7\,c_T f)(d/10)$, $M = \kappa(0.093 + 0.747\,c_M f)(d/10)^2$. For solid softwood the two slopes are fixed by the one solid-wood measurement found (McMillin and Woodson 1972 [35], Table 1: a 25.4 mm spur bit across the grain of southern pine at 1.02 mm/rev and 10.7 % moisture, 667 N and 3.84 N m): $c_T = 3.67$, $c_M = 0.66$, with the MDF intercepts kept ($\kappa = 1$; no measurement of the thrust a brad point needs before it cuts at all was found). The paper's averages over 0 to 80 % moisture, 618 N and 3.73 N m, are not its air-dry row and are not used. With these slopes a 14.3 mm bit feeds 0.061 mm/rev under 40 N; at the MDF slopes it would feed 0.22 mm/rev. The plywood backer keeps the MDF slopes ($\kappa = 1.2$). Exit tear-out follows Hocheng–Dharan [12], $F^* = \pi\sqrt{8 G_c E h^3/(3(1-\nu^2))}$, and a backer board removes it. **Concrete** loses volume per blow, $\Delta V = E_b \sum_i \varphi_i / e_{v,i}$, with the mortar, aggregate and steel fractions under the bit face from the wall's fields. The mortar's e_v depends on the bit's diameter, as a maker's rated drilling performance does: 28, 45 and 49 cm^3/min at 8, 12 and 16 mm with 1.8 J blows at 5,100 a minute [36] are 0.328, 0.204 and 0.187 J/mm^3, interpolated between and held constant outside (0.198 J/mm^3 for the anchor's 13.3 mm bit). The table is a brochure figure of another hammer of nearly the modelled one's blow power (153 W against 161 W), with the concrete's grade and the push not stated. One value at every diameter, 0.34 J/mm^3 derived from KIT's measured rates with a 10 mm bit [17], would make a 13.3 mm hole cost 1.7 times the blows the maker's table gives. The two sources do not agree where they overlap: at 10 mm the table gives 0.266 J/mm^3, KIT's rates 28 % more. Aggregate takes 1.3× the mortar's energy; steel 30 J/mm^3: carbide does not cut rebar.

4.3 Seating, breakthrough, debris and heat

An unseated spinning tip slides under the tangential part of the push by the chisel-edge law

$$F_{\text{lat}} = -\mu_t T \frac{v_{\text{lat}}}{\sqrt{v_{\text{lat}}^2 + (\omega r_c)^2}}; \tag{4}$$

a dimple or pilot seats it. Near the exit the remaining metal cap collapses when $h < \sqrt{2T/(\pi\sigma_y)}$ and its thickness becomes the burr. The grab of thin sheet (the lips catching the collapsing ring) is not modelled (Section 10). Concrete spalls at the entry under a lateral push. Concrete dust is produced at 1.6× the removed volume; 0.29 of it rides the flutes and settles when the bit stops; it forms a column at the bottom that raises the floor every inserted part meets. A vacuum pass removes 85% of the loose dust, a brush stroke loosens 60% of the packed (both open values). What has left a hole is drawn, not simulated as bodies: curled chips and shavings around the mouth, and for concrete a pale ring and streak of dust clinging below the mouth with a heap and a scatter at the wall's foot. The drawn area follows the volume (a 113 mm anchor hole leaves about 25 cm^3 of loose dust). Dust in the air is not drawn. One lumped heat node per bit tip, $C\dot{\theta} = k_e(M\omega + Tv) - (\theta - \theta_a)/R_{\text{th}}$, with the 550 °C burn threshold.

4.4 Holes as contact geometry

MuJoCo cannot change a body's mesh at run time, and a box with a hole is not convex. Each drillable workpiece therefore carries a welded pool of proxy slots; a slot is a floor plug, a 12-sided wall cage, a rim ring and a 45° chamfer cone, placed at a hole when an item enters it and hidden otherwise. The workpiece's solid box stops colliding with the item that is inside the hole, so a bolt, an anchor, a brush or a bit meets only the hole's walls and floor. E10 measures what this gives: an M8 bolt passes two 9 mm holes up to 1.0 mm out of line and jams beyond, against the geometric 1.0 mm.

5 Contact, solver and the robot

Newton 1.5 with its MuJoCo 3.11 CPU solver at a 1 ms tick, elliptic friction cones, implicit integration; the renderer is a separate Isaac Sim 5.1 process that takes no physics steps. The constraint solve is MuJoCo's Newton method (50 iterations) with a line search of 200 iterations at tolerance 10^{-6}, followed by the no-slip pass (100 iterations); the engine's default line search (20 iterations, tolerance 0.01) left the main solve unfinished at hard contacts under a blow (Section 8.4). Friction in the main solve is 100 times harder than the normal contact (MuJoCo's `impratio`). MuJoCo's friction holds a load only against a slip acceleration, and the no-slip pass then removes that slip with the normal forces held fixed: with a softer setting such as 3 the main solve has a clamped stack accelerating before the second pass stops it, and a part lying on the stack gets its forces for a support that then does not move (Section 8.4). Rigid contact uses a 4 ms time constant; hardened steel faces (vise jaws, parallels, washers) use a 2 ms time constant with impedance 0.99/0.999; the hold-down clamps and the wooden parts of a clamped stack use 2 ms with 0.999/0.9999, and their load-bearing faces carry no rolling resistance; metal workpiece faces use 2 ms with 0.97/0.99 (Section 8.4). Where a small loose part lies on one of these faces (a bolt's head, the centre punch), its own 4 ms contact decides how the contact gives, as the softer of two springs in series does. Robot finger pads have a rubber time constant scaled by finger mass and torsional friction 4.2 mm, and rule every contact they make. These are numerical settings with no material meaning; each is in the ledger with the measurement that set it. The world raises a fault, rather than continuing, on any non-finite state, an object faster than 20 m/s or spinning faster than 1000 rad/s, or MuJoCo's bad-acceleration warning.

The robot is built from the shared catalogue (`robots/catalog.py`, entry `drilling`): two FR3 arms on the cart with Franka Hands. Each joint is a position servo (600 N m/rad) whose target follows a time-optimal jerk-limited trajectory within the FR3's own limits: joint velocity 2.62–5.26 rad/s, acceleration 10 rad/s^2 and jerk 5000 rad/s^3 on every joint (Franka's limit table and libfranka [21]'s `rate_limiting.h`; the trajectory is Ruckig's, the generator of the FR3 library franky). The desired joint torque, the servo's output plus any `cartesian_wrench` feed-forward, is capped as one at the joint's rating of 87 or 12 N m, and the feed-forward changes at most at 1000 N m/s. Franka states the rating for the desired torque without gravity: the robot compensates the weight of its own links and of its configured load beside it. Here the configured load is the carried tool that both fingers of a hand pinch, shared between two hands as statics shares it. It is set as an operator sets `setLoad` on each grasp, and a policy can neither set nor clear it. **This is an assumption in the robot's favour.** With gravity carried beside the cap, a 12 N m wrist keeps all 12 N m for pushing while it also holds a 3.2 kg hammer, and a stretched arm's shoulder supplies

47 N m of its own weight for free. A real joint has a ceiling that includes gravity: libfranka reports an error when the *measured* torque leaves its safe range (`tau_j_range_violation`). No number is published for that range, so the ceiling cannot be grounded here and is absent; Section 7 reports one wall run with the tool's weight counted inside the cap, to show what the assumption is worth. An arm that carries more than its rated payload beyond its hand (2.27 kg, reading the FR3's flange rating of 3 kg as including the 0.73 kg hand) is not refused, as the real arm is not; the world logs it and the evaluator fails the episode once any arm has done so for more than 0.5 s ("no arm carried more than its rated payload"). The trajectory generator runs at 0.95 of the constant joint velocity limits; the FR3's real limits depend on joint position, which is not modelled. Table 5 measures the limits; Table 6 the one cap and the configured load. The gripper has a width mode and a force mode of up to 35 N per finger: the hand's continuous grasping force is 30–70 N (its product manual), and Franka's own simulation of the hand applies half of the commanded force to each finger. The force mode presses with the commanded force wherever the fingers are stopped, as the real hand's grasp does (a reference that stops at the commanded position gives a part narrower than 20 mm less, 12 N per finger on an M8 bolt for 35 N commanded, Section 8.4). Every robot record carries the force the fingers applied beside the force commanded. The two fingers are one mechanism, as in the real hand (Franka's description declares the second finger joint a mimic of the first). At the engine's default softness for that tie a pinched part sits as on a spring of 2.7 N/mm along the closing axis, so the tie is as hard as the press's rack (Section 8.4). The pads keep the rubber's anchored normal stiffness, and their friction rows remove a slip within one tick; what a pinched part still moves in a swinging hand is the time step's own error, 0.12 mm on the fastest 0.4 rad swing of the first joint, and is stated in the same section. The FR3's collision reflex is modelled: contact forces mapped to joint torques, less the configured load's weight and inertia, stop an arm when any joint exceeds 95% of its limit, and end its feed-forward. A policy acts through joint targets, Cartesian targets, gripper commands and keyboard chords; `cartesian_wrench` is an owner action, not on GAR's policy allowlist.

Table 5: E18a: a target step (0.5 rad on right/fr3_joint1, 0.8 rad on right/fr3_joint5) is followed by a jerk-limited trajectory: the commanded trajectory is within Franka's FR3 limits. The joint's own acceleration, measured from its positions, is the servo following that trajectory and is not bounded by them: it peaks at 10.97 and 10.49 rad/s^2 here, above the 10 the command keeps to.

joint	cmd. vel. (rad/s)	limit	cmd. acc. (rad/s^2)	limit	cmd. jerk (rad/s^3)	limit	measured acc. (rad/s^2)
right joint1	1.10	2.62	10.00	10.0	5000	5000	10.97
right joint5	2.82	5.26	10.00	10.0	5000	5000	10.49

Table 6: E18b: one torque cap for the servo and the wrench feed-forward, and the configured load.

case	measured
wrist twist 20 N m, wrench 10 N m helping the servo	joint 7 motor torque peak 12.00 N m (rating 12); the wrist turned -3.48 rad
drill (1.68 kg) held 100 mm up, load compensated	hand -0.18 mm below its command; servo torque on joints 2 and 4 0.11 and 0.13 N m
drill (1.68 kg) held 100 mm up, load not compensated	hand 11.97 mm below its command; servo torque on joints 2 and 4 7.62 and 7.46 N m

How firmly such a pinch holds a tool is measured in E19. A drill pinched at the hand's force does not creep at rest. Under the rotary hammer's recoil pulses it slips: along the bit's axis by less than a point mass in a fixed hand with Coulomb friction would at the full pulse, and in other rows by more than that model, because the drill also turns in the pinch, which a point mass cannot (Table 8; the model is a comparison, not a bound). The thin and light parts of the tasks are in Table 7: the force they get, what a swing of the arm moves them, and what a push along the closing axis moves them.

Table 7: E19, the thin and light parts of the tasks in the right hand, 35 N per finger commanded (placed between the fingers and steadied by a phantom spring while they close: ASSISTED; then only the fingers touch the part). The swing is 0.4 rad of the first joint, one target step out and one back; the shift is along the part's own axis, outwards, both times, and the time step's share is $a_c\,\Delta t^2/2$ summed over both swings with the lever arm from the first joint only. Without the force mode's run-on reference, the pads' direct contact law and the hard finger tie, the same parts get 8.5 to 24 N, shift 0.33 to 0.45 mm per swing and move 1.8 mm under the 5 N push (Section 8.4). Not corrected: MuJoCo's soft contact gives way per contact point, not per area, so a round part sinks deeper into the pads when it touches them at fewer points, and the count changes when it shifts; the M8 bolt's width between the pads and its place along the closing axis follow the count.

part	mass (g)	pinched width (mm)	force applied (N per finger)	pads press (N per finger)	shift after the swing out (mm)	after the swing back (mm)	time step's share (mm)	moved by 5 N along the closing axis (mm)	contact points per finger: still / after the swings / under the push	width under the push (mm)	held
centre punch	70	13.08	35.0	35.0	0.117	0.235	0.191	0.022	3+5 / 3+5 / 3+5	13.08	yes
m8 bolt	18	5.76	35.0	35.0	0.097	0.194	0.191	-0.155	3+3 / 2+3 / 2+2	4.63	yes
angle bracket	448	4.41	35.0	35.0	0.106	0.201	0.192	0.023	4+4 / 4+4 / 4+4	4.42	yes
carriage bolt	137	11.94	35.0	35.0	0.090	0.179	0.191	0.061	4+4 / 4+4 / 3+4	11.86	yes
wedge anchor	114	11.72	35.0	35.0	0.083	0.166	0.191	0.026	2+4 / 3+3 / 3+3	11.77	yes

Table 8: E19: the drill pinched at 35 N per finger (friction 56 N at μ 0.8), statically and under the rotary hammer's recoil pulse (3.9 ms half-sine at 76.7 Hz). The model is a point mass in a fixed hand with Coulomb friction. It is a comparison, not a bound: the hand yields, which lowers the slip, and the drill turns in the pinch, which the point mass cannot, and 2 of the 3 pulse rows slip further than the model.

case	impulse (N s)	peak (N)	slip along (mm)	slip down (mm)	turn (°)	model along (mm)	model down (mm)
static, 30 s	–	–	0.000	0.000	0.00	–	–
pulses, 2 s	0.447	158	39.72	3.00	33.2	94.5	20.1
pulses, 2 s	0.223	79	3.93	-6.28	7.5	9.9	2.6
pulses, 2 s	0.067	24	0.01	-0.16	0.2	0.0	-0.0

Box shapes have no contact gap; cylinders, capsules, spheres, hulls, meshes and the boxes of the holes' walls (which meet only round items) keep a 2 mm gap (Section 8.4 gives the measurements behind both). Three other rigs are qualified: two Pandas, and the right FR3 or the right Panda alone for punch-layout, press-drill and press-blind. A single arm is refused for the hand-drill and wall tasks because one Franka Hand pinch does not react a hand drill's cutting torque (in the robot's one-hand timber bore the drill turned in the pinch, Section 7) and the 3.1 kg rotary hammer exceeds one FR3's 2.27 kg tool payload. The G1 and the Mobile FR3 Duo are refused with their reasons (Table 9). The hand is the same Franka Hand on every rig, 35 N per finger; the Panda rigs take it from the Panda's `drilling` controller profile (the Panda's own profile from the barcode environment limits the hand to 20 N per finger). Table 10 shows a two-FR3 rig and a two-Panda rig lifting and holding the cordless drill and the rotary hammer at that force.

Table 9: Robot configurations, all through the public backend: 10 qualified ones built, left alone for 1 s and moved; 10 others asked for and refused with the reason shown. All as expected: yes.

robot	preset / task	result
dual-fr3	wall / clean-and-set	builds (rig drill-fr3; left, right, 18 joints). 1 s hands-off: largest object motion 0.0002 mm, largest joint drift 7.2e-14 rad; a 30 mm Cartesian move up went 30.06 mm; 0 reflexes, fault: none
dual-fr3	bench / press-drill	builds (rig drill-fr3; left, right, 18 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 3.1e-14 rad; a 30 mm Cartesian move up went 30.06 mm; 0 reflexes, fault: none
dual-panda	wall / anchor-holes	builds (rig drill-panda; left, right, 18 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 1.5e-08 rad; a 30 mm Cartesian move up went 30.00 mm; 0 reflexes, fault: none
dual-panda	bench / press-drill	builds (rig drill-panda; left, right, 18 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 3.7e-09 rad; a 30 mm Cartesian move up went 30.00 mm; 0 reflexes, fault: none
single-fr3	bench / punch-layout	builds (rig drill-fr3-single; right, 9 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 1.5e-14 rad; a 30 mm Cartesian move up went 30.06 mm; 0 reflexes, fault: none
single-fr3	bench / press-drill	builds (rig drill-fr3-single; right, 9 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 1.5e-14 rad; a 30 mm Cartesian move up went 30.06 mm; 0 reflexes, fault: none
single-fr3	bench / press-blind	builds (rig drill-fr3-single; right, 9 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 2.3e-14 rad; a 30 mm Cartesian move up went 30.06 mm; 0 reflexes, fault: none

robot	preset / task	result
single-panda	bench / punch-layout	builds (rig drill-panda-single; right, 9 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 7.5e-12 rad; a 30 mm Cartesian move up went 30.00 mm; 0 reflexes, fault: none
single-panda	bench / press-drill	builds (rig drill-panda-single; right, 9 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 7.5e-12 rad; a 30 mm Cartesian move up went 30.00 mm; 0 reflexes, fault: none
single-panda	bench / press-blind	builds (rig drill-panda-single; right, 9 joints). 1 s hands-off: largest object motion 0.0000 mm, largest joint drift 3.7e-09 rad; a 30 mm Cartesian move up went 30.00 mm; 0 reflexes, fault: none
single-fr3	wall / anchor-holes	refused (UnsupportedConfigurationError): drilling task wall/anchor-holes needs two arms: the 3.1 kg rotary hammer exceeds one FR3's 2.27 kg tool payload (drilling/paper/drilling_report.pdf, robot configurations table in section "Contact, solver and the robot"); single-fr3 is qualified for punch-layout, press-drill, press-blind
single-fr3	bench / match-bolt	refused (UnsupportedConfigurationError): drilling task bench/match-bolt needs two arms: one Franka Hand pinch does not react a hand drill's cutting torque: in the robot's one-hand timber bore the drill turned in the pinch and the hole came out of square, and take-drill asks for both hands (drilling/paper/drilling_report.pdf, robot configurations table in section "Contact, solver and the robot"); single-fr3 is qualified for punch-layout, press-drill, press-blind
single-panda	wall / anchor-holes	refused (UnsupportedConfigurationError): drilling task wall/anchor-holes needs two arms: the 3.1 kg rotary hammer exceeds one Panda's 2.27 kg tool payload (drilling/paper/drilling_report.pdf, robot configurations table in section "Contact, solver and the robot"); single-panda is qualified for punch-layout, press-drill, press-blind
single-panda	bench / match-bolt	refused (UnsupportedConfigurationError): drilling task bench/match-bolt needs two arms: one Franka Hand pinch does not react a hand drill's cutting torque: in the robot's one-hand timber bore the drill turned in the pinch and the hole came out of square, and take-drill asks for both hands (drilling/paper/drilling_report.pdf, robot configurations table in section "Contact, solver and the robot"); single-panda is qualified for punch-layout, press-drill, press-blind
g1	bench / press-drill	refused (UnsupportedConfigurationError): drilling does not support 'g1': the G1's wrist pitch and yaw joints are rated 5 N m (effort_limit in robots/models/g1/robot.json) and its Dex3-1 hand holds at most 500 g of a 5 cm object (Unitree's Dex3-1 product page, https://www.unitree.com/Dex3-1/, read 2026-10-07), against 1.7-3.1 kg drills and 50-150 N drilling pushes
mobile-fr3-duo	wall / anchor-holes	refused (UnsupportedConfigurationError): drilling does not support 'mobile-fr3-duo': the Mobile FR3 Duo is not laid out or qualified at the drilling stations: the layouts put the arms on the docked cart's fixed mounts, and how the base's passive rocker suspension takes a 50-150 N drilling push has not been measured (planned as an optional wall override after that measurement)
mobile-fr3-single	bench / press-drill	refused (UnsupportedConfigurationError): drilling does not support 'mobile-fr3-single': the Mobile FR3 with one arm is not laid out or qualified at the drilling stations: the layouts put the arm on the docked cart's fixed right mount, and how the base's passive rocker suspension takes a 50-150 N drilling push has not been measured (the same base as the Mobile FR3 Duo); the one-arm configurations qualified here are single-fr3 and single-panda
panda	bench / press-drill	refused (UnsupportedConfigurationError): drilling does not support 'panda': in drilling one Panda arm is the choice single-panda (rig drill-panda-single: the right arm at the drill-press bench, for punch-layout, press-drill and press-blind) and two Panda arms are dual-panda; the bare name panda is not offered because it does not say which
dual-panda-wsg50	bench / press-drill	refused (UnsupportedConfigurationError): drilling does not support 'dual-panda-wsg50': drilling's finger pads, grasps and force-mode grip are built and measured for the Franka Hand only (the pads are attached from that hand's catalogue data, drilling/scene.py; no WSG-50 has held a drilling tool here); use dual-panda
yam	bench / press-drill	refused (UnsupportedConfigurationError): drilling does not support 'yam': the YAM arm is not a catalogue robot model (robots/models has the FR3, Panda, G1 and the Mobile FR3 platforms); drilling builds its robots from the catalogue

Table 10: The two heaviest hand tools lifted 100 mm and held 10 s hands-off by a two-FR3 rig and a two-Panda rig with the Franka Hand's force-mode grip at 35 N per finger (ASSISTED scripted grasps of the demonstrations, no policy). Checks: lifted: the tool rose at least 80 % of the commanded 100 mm; held: after 10 s hands-off the tool is less than 5 mm lower than when the hold began; slip: the tool slipped at most 1 mm in each holding hand during the 10 s; force: each holding finger applied at least 95 % of the commanded 35 N at the end of the hold; quiet: no physics fault and no protective stop. All checks passed: yes.

robot	tool	hands	lifted (mm)	dropped in 10 s (mm)	largest slip in the hand (mm)	finger force at the end (N)	checks
dual-fr3	cordless drill (1.68 kg, 16.5 N)	left	96.5	0.00	left 0.000	left 35.0/35.0	lifted yes, held yes, slip yes, force yes, quiet yes
dual-panda	cordless drill (1.68 kg, 16.5 N)	left	90.2	0.00	left 0.000	left 35.0/35.0	lifted yes, held yes, slip yes, force yes, quiet yes
dual-fr3	rotary hammer (3.23 kg, 31.7 N)	left + right	99.5	0.00	left 0.000; right 0.001	left 35.0/35.0; right 35.0/35.0	lifted yes, held yes, slip yes, force yes, quiet yes
dual-panda	rotary hammer (3.23 kg, 31.7 N)	left + right	99.7	0.00	left 0.000; right 0.006	left 35.0/35.0; right 35.0/35.0	lifted yes, held yes, slip yes, force yes, quiet yes

6 Component experiments against published data

Before a robot touches a tool, each mechanism is tested in isolation against a published measurement or an exact expectation. Where a hand is needed, an ASSISTED fixture supplies it: a phantom spindle that turns a bit at a set speed under a set thrust, a phantom holder that keeps a tool on a line at the 24 N/mm stiffness of KIT's hand-arm model, a phantom finger on a trigger. These fixtures apply forces to the tool, not to the workpiece, and they are never part of the policy interface. Each experiment records its results as JSON and a film; the numbers here are read from those records.

6.1 Engine audit (E1a)

Exact expectations for the machinery the experiments rely on: the hole proxies park and unpark, a bolt's shank reaches the plug, the motor line (free speed, the loaded speed at 60 N m on the drive's torque–speed line), the press's rack and pinion, and momentum conservation through a two-body exchange. Table 11.

Table 11: E1a engine audit (Newton 1.5.0, MuJoCo 3.11.0, Warp 1.16.0). Checks: parked_pass yes, plug_rest yes, cage_centres yes, plug_force yes, reparked yes, motor_free yes, motor_line yes, motor_cap yes, gear2 yes, rack yes, momentum yes, mass_hammer yes, mass_hammer_bit yes, mass_drill yes, mass_press_quill yes.

quantity	measured	expected
parked hole proxy head height (m)	0.1037	0.1037 (re-parked)
bolt shank on the plug: force (N)	0.460	0.468 (its weight)
drill free speed (rad/s)	68.07	68.07
drill speed at 60 N m (rad/s)	22.69	22.69
drive torque capped (N m)	20.00	the clutch setting
gear 2 speed at 20 N m (rad/s)	65.00	65.00
press quill force (N)	519.90	519.57
two-body angular momentum (each)	0.15, -0.15	0.15

6.2 Metal cutting (E2, E3)

A phantom spindle holds the law's thrust for a published feed on a 100 mm S235 block and lets the bit advance under it; the measured feed and torque are compared with the law and with Rubenstein [2]'s AISI 1020 data (Figure 4). Below the chisel threshold the bit does not advance; a pilot removes the threshold. A motor that cannot give the torque the cut asks for stalls: the bit stops, the motor sits at its torque limit and the hole does not deepen (E3b, Table 14). The cut resists the spin as friction does: it applies the torque that leaves the bit standing at the end of the tick, up to the cutting torque. A cutting torque bounded by what stops the spin within a tick would leave the motor's own drive to turn the bit on at about a fifth of its no-load speed whatever the load, and a bit caught on a reinforcing bar would grind on in the same way.

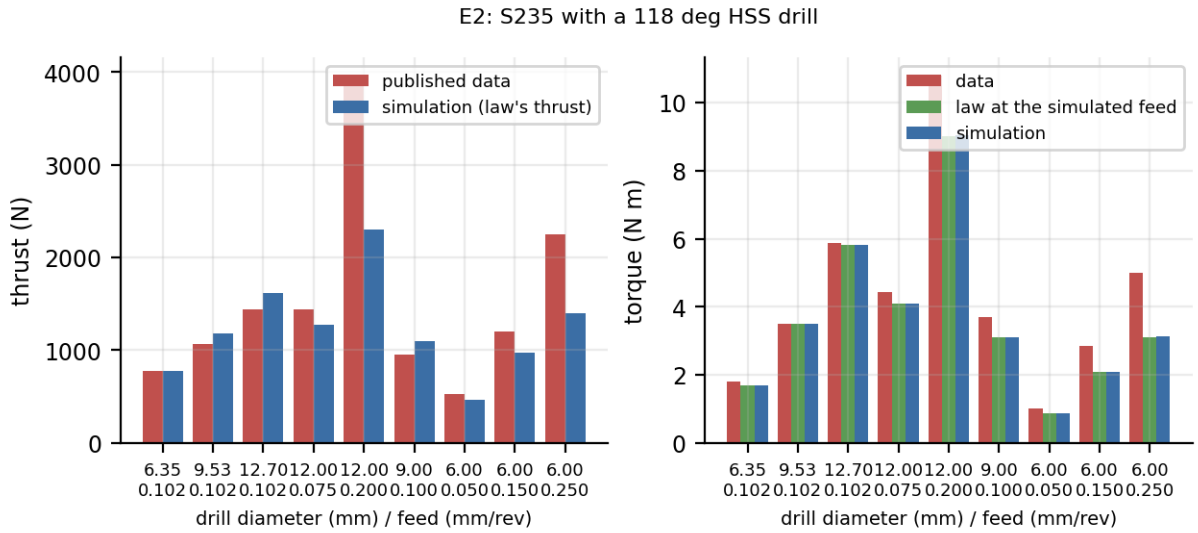


Figure 4: E2: thrust and torque, published data against the simulation, for six drills and feeds.

Table 12: E2: thrust and torque in steel against published data. The bench commands the law's own thrust at the data's feed, so the thrust column is the closed-form law and the simulated results are the feed and torque that thrust produces: E2 tests the implementation. Rows marked fit are the data K_d was fitted to; the rows marked independent are the only test of the law against data.

d (mm)	feed (mm/rev)	source	role	thrust data (N)	thrust law (N)	feed sim	torque data (N m)	torque sim (N m)	thrust vs data (%)	torque vs data (%)
6.35	0.102	Rubenstein AISI 1020 (Strenkowski 2004 Table 1)	fit	775	779	0.101	1.81	1.69	0.4	-6.7
9.53	0.102	Rubenstein AISI 1020	fit	1070	1179	0.100	3.51	3.50	10.0	-0.4
12.70	0.102	Rubenstein AISI 1020	fit	1440	1614	0.098	5.88	5.81	12.1	-1.1
12.00	0.075	Micheletti & Levi, mild steel	independent	1440	1273	0.072	4.43	4.10	-11.6	-7.4
12.00	0.200	Micheletti & Levi, mild steel	independent	3960	2304	0.192	10.79	9.04	-41.9	-16.2
9.00	0.100	Ema, 450 MPa steel, web 1.5 mm	independent	950	1095	0.098	3.70	3.10	15.1	-16.1
6.00	0.050	Bordinassi et al. 2003, ABNT 1020, read off Fig. 7	independent	530	461	0.050	1.00	0.86	-13.0	-14.1
6.00	0.150	Bordinassi et al. 2003, ABNT 1020, read off Fig. 7	independent	1200	967	0.149	2.85	2.08	-19.5	-27.2
6.00	0.250	Bordinassi et al. 2003, ABNT 1020, read off Fig. 7	independent	2250	1403	0.248	5.00	3.12	-37.7	-37.6

Table 13: E3: the chisel threshold and pilot holes.

case	feed sim (mm/rev)	feed law	torque sim (N m)	torque law
9 mm, no pilot, 200 N (threshold 231 N)	0.0000	0.0000	–	–
9 mm over 2 mm pilot, 200 N	0.0218	0.0219	0.881	0.881
9 mm over 2 mm pilot, 1094 N	0.1828	0.1831	4.819	4.812
9 mm over 4 mm pilot, 200 N	0.0333	0.0333	1.041	1.041
9 mm over 4 mm pilot, 1094 N	0.2778	0.2781	5.692	5.686

Table 14: E3b: an overloaded drill stalls. The bench spindle with a motor of limited torque and a 9 mm bit in steel at 650 rpm. Where the law's cutting torque at the thrust exceeds what the motor can give, the bit stops, the motor sits at its cap and the hole does not deepen; under the cap it cuts at the law's feed.

thrust (N)	cutting torque the law asks (N m)	motor torque cap (N m)	speed (rpm)	motor torque (N m)	feed sim (mm/rev)	feed law	stalled
1094	3.14	4.0	649	3.10	0.0983	0.0984	no
1094	3.13	2.5	0	2.50	0.0000	0.0992	yes
728	1.80	2.5	650	1.78	0.0492	0.0493	no

6.3 Percussion in concrete (E8)

A phantom holder pushes the rotary hammer into a concrete block at fixed forces. With no push there are no blows, and the rate of penetration rises with the push until every blow lands, then stays flat, as KIT measured between 110 and 170 N. What decides how many blows land is the housing's momentum budget. In steady drilling the push must cover three things: the mean recoil of the blows, the bit's return spring at the working position, and the force the housing passes to the bit when it bottoms on the chuck's stop. Below that sum the hammer delivers the share of its blows that the remaining push can balance, and Table 15 lists the measured share beside the budget's. A striker sized as if the whole of the maker's "gentle pressure (approx. 5 kg)" were recoil lands 63% of its blows at 49 N and all of them only from 75 N. A trace of every scheduled blow shows the delivered recoil capped at 34 N whatever the striker mass, with 9.7 N in the return spring (5 N with the bit out and 9.8 N with it pushed back) and 5 N through the stop. The striker is the largest that closes the budget at the maker's push, 0.047 kg at 9.4 m/s. That is a calibration to one sentence of the manual, so the check at 49 N confirms the arithmetic and does not test the model. The operator's stiffness does not enter: the same hammer pushed through the 49.29 N/mm spring of KIT's hand-arm rig delivered the same share at the same mean force. The recoil reaches the housing as a half-sine over 0.30 of the blow period, from the air-spring pressure trace of an electro-pneumatic hammer (Wang 2014 [42]). The idle-stroke window (2 mm) and the return spring are estimates without a source. The blow law treats the carbide head as one rigid face, so a face partly on steel advances at the steel's rate, and the bar's damage is measured over the whole face. A bit landing on rebar stops and trips the clutch (Figure 5).

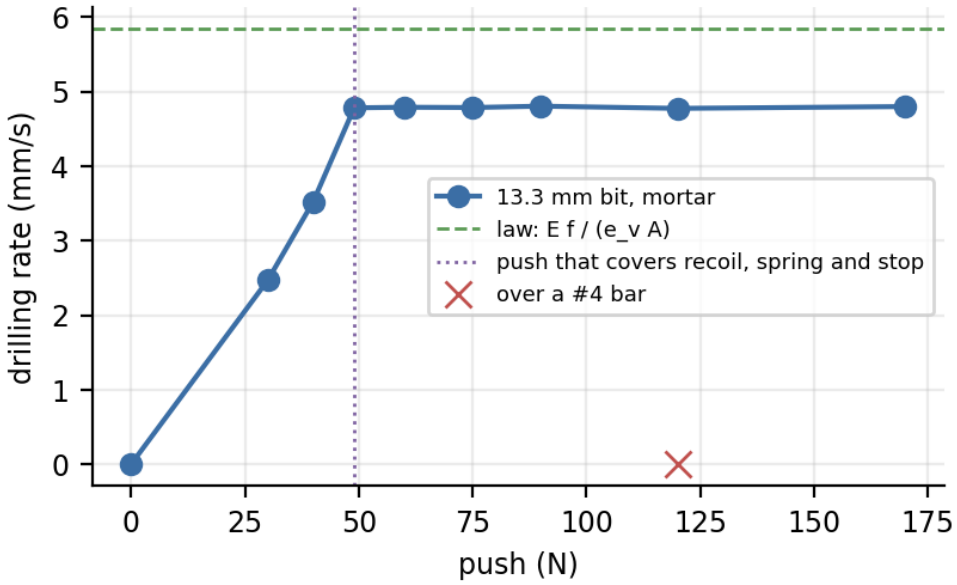


Figure 5: E8: penetration rate against push in C25/30, and the rebar stop.

Table 15: E8: percussion in C25/30. The push covers the mean recoil (34.2 N at the full blow rate), the bit return spring (9.8 N) and what the bottomed housing passes to the bit (5.0 N): every blow lands from 49.0 N, where the energy law predicts 5.84 mm/s in mortar without aggregate (the rows are measured in the wall, which has aggregate and a dust column). Checks: no blows without push yes, plateau flat yes, the maker's push keeps the blows yes (the striker is calibrated to it), rebar stops the bit yes.

bit (mm)	push (N)	rebar	rate (mm/s)	depth (mm)	blows delivered	share: measured / momentum budget	time (s)	clutch
13.3	0	no	0.00	0.0	0/321	0.00 / 0.00	4	no
13.3	30	no	2.47	10.7	152/321	0.47 / 0.44	4	no
13.3	40	no	3.52	16.0	234/321	0.73 / 0.74	4	no
13.3	49	no	4.78	20.5	309/321	0.96 / 1.00	4	no
13.3	60	no	4.79	20.5	309/321	0.96 / 1.00	4	no
13.3	75	no	4.78	20.5	309/321	0.96 / 1.00	4	no
13.3	90	no	4.80	20.6	310/321	0.97 / 1.00	4	no
13.3	120	no	4.77	20.5	310/321	0.97 / 1.00	4	no
13.3	170	no	4.80	20.5	310/321	0.97 / 1.00	4	no
10.0	120	no	6.19	27.0	310/321	–	4	no
13.3	120	yes	0.00	26.7	413/424	–	14	yes

6.4 The hand drill (E4–E6) and heat (E7)

Timber tear-out begins at the Hocheng–Dharan [12] force and not when the exit is backed; an unseated tilted drill walks and a pilot holds it (Figure 6); a backed sheet drills through. A brad point's centre spur holds its tip on the mark until the bit's full diameter is in the wood, as a short dowel in timber up to its embedment capacity (EN 1995-1-1 [43]); under a side load beyond that it ploughs out (Table 16). The wood law is MDF data extrapolated to a Douglas-fir auger, labelled so in the ledger; its sensitivity is in Section 10. Heat at the bit tip is the hotter of a lumped node fed by the cutting power and the cutting corners' tool–chip interface, Cook's rise, which grows with the cutting speed. So a dry 9 mm bit at 3000 rpm burns while 650 rpm does not, and a bit drilled on without withdrawing burns from depth (Table 17). Oil does not save the overspeeding bit, because Cook's rise has no coolant term. The burn is an instantaneous threshold on that temperature, where real tempering [7] takes time at temperature.

Table 16: E5b: the 9/16 in brad point under a steady side load, held by a phantom with no lateral stiffness. The spur (2.9 mm, leading the lips by 3 mm) holds the tip up to its embedment capacity, 368 N when fully embedded (EN 1995-1-1, $f_h = 42.9$ N/mm^2).

spur	side load (N)	slide at seating (mm)	largest slide before full diameter (mm)	depth after 8 s (mm)	fault
off	4	1.87	33.38	4.3	none
on	4	0.68	1.28	7.8	none
on	20	1.42	3.87	3.1	none

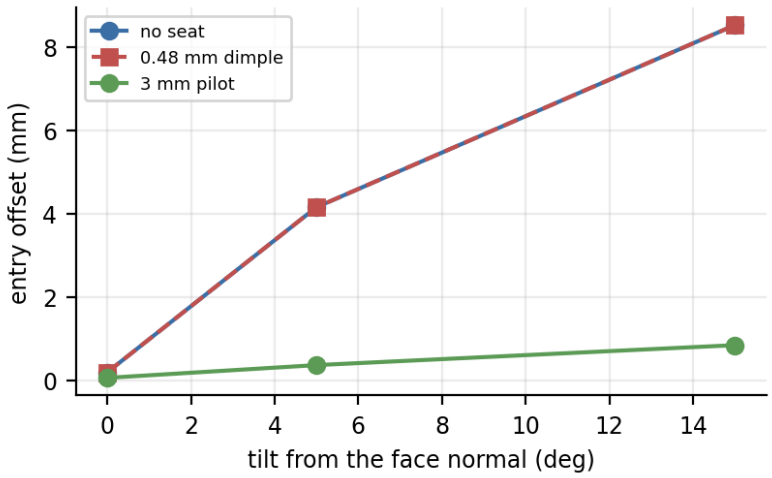


Figure 6: E5: how far an unseated drill walks at three tilts, with and without a pilot.

Table 17: E7: heat at the 9 mm bit tip in steel: the lumped tip node, or the cutting corners' tool-chip interface (Cook's rise, growing with cutting speed) while a chip is cut, whichever is hotter. Speed burns the bit: yes; oil prevents it: no (the interface rise has no coolant term). to 30 mm: max 410.9 C at 650 rpm dry (burnt False, at None mm), 651.4 C at 3000 rpm dry (burnt True, at 1.29 mm), 651.4 C at 3000 rpm oiled (burnt True, at 1.29 mm); 650 rpm dry without withdrawing to 80 mm: burnt True at 40.35 mm, 45.31 mm reached after 150.05 s (wear 4.541).

case	rpm	oiled	depth (mm)	time (s)	at 30 mm (°C)	peak (°C)	burnt	wear
650 rpm dry, 30 mm	650	no	30.0	28.2	376.6	410.9	no	1.090
3000 rpm dry, 30 mm	3000	no	30.0	11.8	568.4	651.4	yes	1.819
3000 rpm oiled, 30 mm	3000	yes	30.0	11.8	557.6	651.4	yes	1.803
650 rpm dry, on to 80 mm without withdrawing	650	no	45.3	150.1	376.6	805.7	yes	4.541

6.5 Setting a wedge anchor (E9, E20) and holes as geometry (E10)

E20 puts a ½ in anchor in a recorded hole on a test stand, places the impact wrench's socket on its nut and pulls the trigger with a phantom finger (Table 18). In a sound hole the wrench runs the nut down, impacts, and its module stops it with a green light when the torque it senses reaches the installation torque. The nut's torque is therefore close to that threshold by construction; the preload and the stud's rise follow from it through the nut factor and the published stiffness, which are inputs (Section 3.1). What the table shows that is not an input: the number of blows a setting takes, and what bad holes and bad handling do. In a hole wider than the clip's free diameter the clip holds nothing: the stud turns with the nut and is drawn out, no torque builds, and the wrench never finishes. An anchor that stands short of its embedment on a dust column is set there at full torque (green, 26.4 kN, its tip at 90.2 mm), which the evaluator's embedment check fails. The same wrench without its module, trigger held in its top gear, takes the nut past 1.7 times the installation torque and then breaks the stud at its steel strength; and pulling the trigger again and again on a set anchor adds a blow each time until the same happens. The evaluator's upper torque bound can therefore be reached, and it is scored. Two cases start with the socket off the nut. Pushed on with the hexes in line, the wrench sets the anchor. Pushed on 20° out of line with the trigger already pulled, the turning socket rides on top of the nut and never drops over it, however hard it is pushed; an operator who rolls the wrench until the socket drops on, and then pulls the trigger, sets the anchor. That failure is the operator's, and the tool's light stays off to say so. A ledger held by one anchor is then pulled straight off the wall, torqued and only hammered in (Table 21).

A blow written as a torque on the anvil's small inertia lets the socket's hex pass the nut's corners while the nut still runs freely (three times in the first nine blows of the ledger case, in which the nut first draws the ledger into its seat). The blow therefore passes through a hammer's inertia (Section 3.1). The probe reads the angle between the two hexes on every tick: in every case in which the socket is over the nut its largest value is between 3.5 and 6.9°, the play of a ¾ in socket on its nut (a corner is at 30°). In the two cases in which the socket is pushed on out of line it never drops over the nut, and the 30° the table shows there is the socket turning on top of it.

E9 runs the whole sequence on one hole: a phantom holder drills with the rotary hammer (116.0 mm in 23.0 s with the diameter's own energy per volume, Section 4), a phantom hand passes the nozzle and the brush or does not, the anchor is driven with the kit's hammer and torqued by the wrench (Figure 7, Table 22). The nozzle and brush strokes of this rig are its own choice; the anchor's maker asks for no cleaning sequence in a hammer-drilled hole. In this hole the uncleaned anchor also reaches its embedment: the hole is 21 mm deeper than the embedment and the 20.2 mm dust column leaves the anchor room to 94.5 mm, against 95.4 mm in the cleaned hole, and both are set (green; 26.7 and 27.2 kN). What dust does to an anchor is in E20's dusty case above and in clean-and-set, whose predrilled holes are no deeper than they must be.

The hammering. Every anchor in the evidence is driven the same way: the kit's 32 oz ball-peen hammer (1.04 kg with its handle), swung by a phantom hand at 5 m/s, 13.0 J a blow, strikes the anchor's end through contact, and the clip drags at 500 N in a nominal hole. The hand's force is taken off 5 mm before contact, so that the blow passes through the contact alone; it keeps the hammer's attitude through the blow with a torque, as a hand on the handle does (let go altogether, the hammer spins on the anchor's end and cocks the anchor in its hole: Section 8.4). E20's hammered case takes 10 blows, of which seven advance the anchor (the largest by 19.2 mm, where the blow's energy would allow 26.1 mm against the drag alone); the predrilled holes of clean-and-set take 9 to 10; the holes ledger-anchor drills itself take 7 to 11 over three seeds; E9's own hole takes 6, of which the first three seat the

anchor. The holes drilled in the same run are quicker because their walls wore under the bit's side load (0.10 mm in E9) and the clip drags about 300 N in them. The counts include the last blows, which no longer advance the anchor; the probe stops after three blows without 0.5 mm of advance. In no blow is the work done against the clip more than the blow's energy. The speed of the swing is the operator's; it was chosen while the hammer still flew free (runs at 3 and 4 m/s then stalled or took twelve blows, which measured the spinning hammer and not the swing) and was kept once the hand held the hammer's attitude. *What the last blow leaves is wrong and open*: in E20's hammered case the stud stands in 38.2 kN of tension with the cone caught after hammering alone, and in E9's cleaned hole in 20.6 kN, where a hammered anchor should hold no more than its clip's drag; the wrench then finds the nut tight (it stops after one blow in E20, after 18 in E9's cleaned hole against 42 in the dusty one). The cause and what was tried are in Section 8.4; both experiments carry the check and fail it. E9 also has a model of a hammer head, a mass meeting the anchor through a spring integrated with it; against the 500 N clip that model over-drives the anchor and the wrench phase then fails inside the solver, a defect that is open. It is kept behind a switch to reproduce that defect and is not evidence.

An M8 bolt pushed through two drilled 9 mm holes passes up to its clearance and jams beyond it (**E10**, Figure 8).

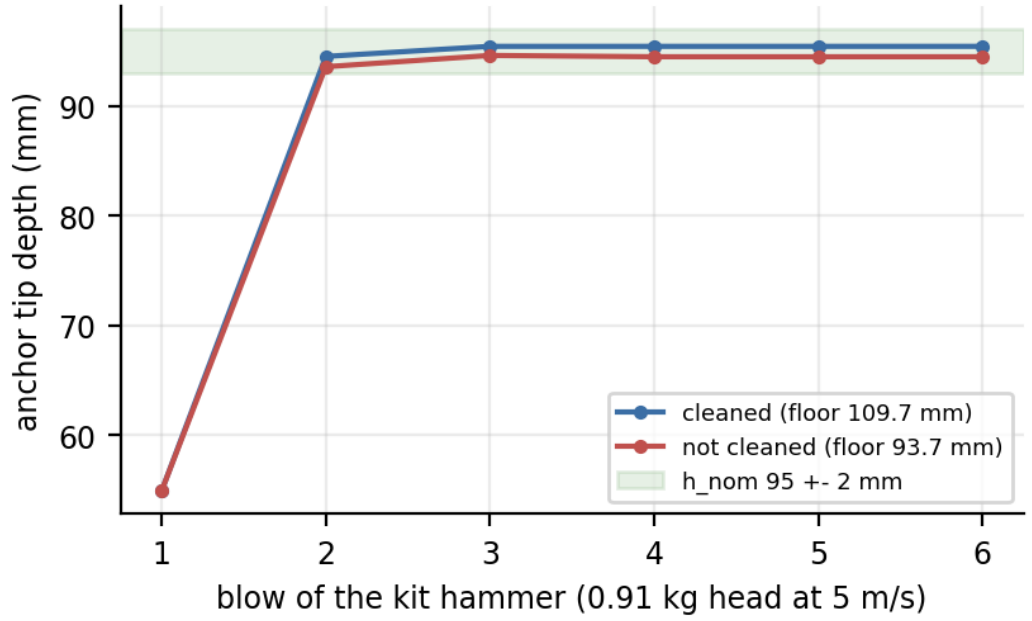


Figure 7: E9: the anchor's tip depth after each hammer blow, in the cleaned and the uncleaned hole, before its nut is torqued.

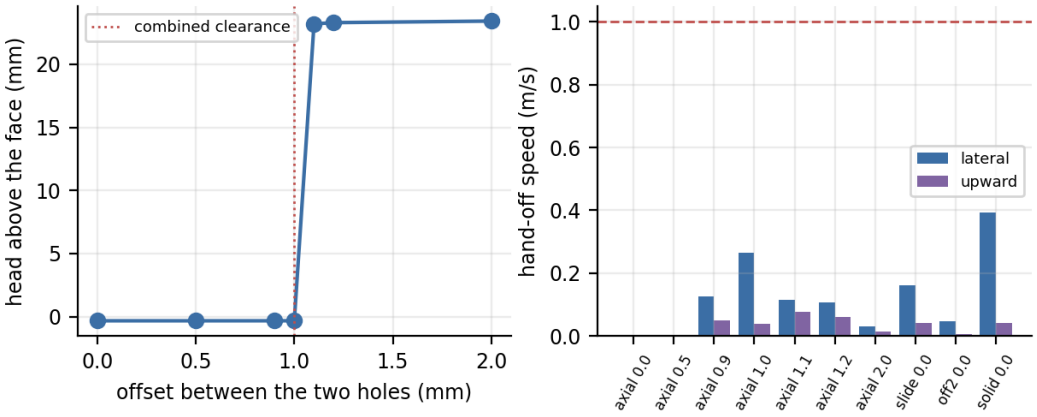


Figure 8: E10: outcome of a bolt through two holes against their misalignment.

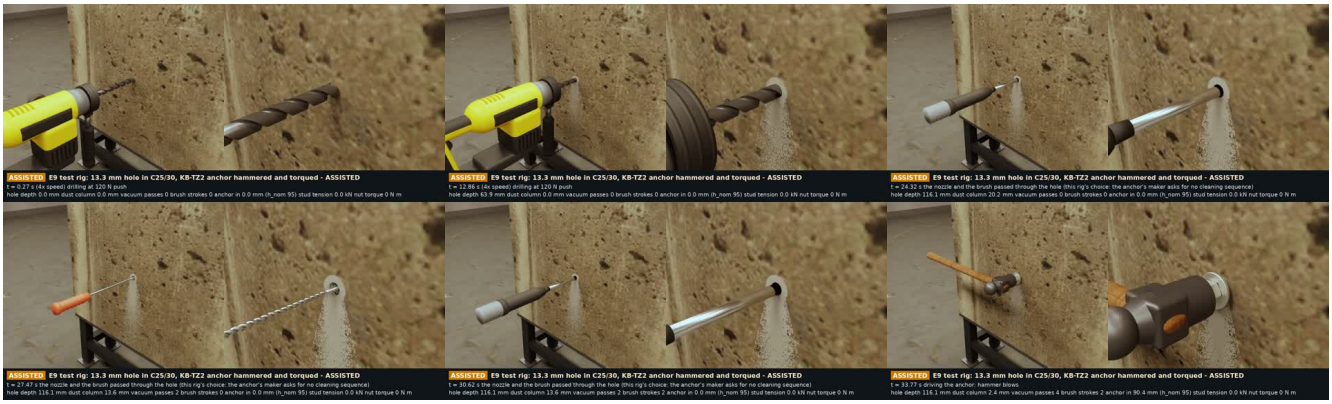


Figure 9: E9 as filmed (no robot; phantom tools): six frames, evenly spaced, of the hole drilled, cleaned, the anchor hammered and its nut tightened in the cleaned hole.

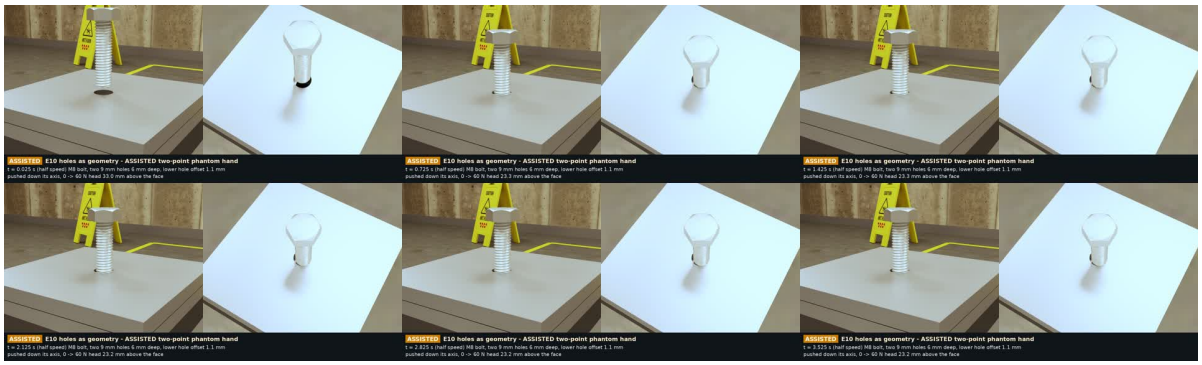


Figure 10: E10 as filmed: the bolt pushed into two holes 1.1 mm out of line, where it jams.

Table 18: E20: a 1/2 in wedge anchor standing in a recorded hole, its nut tightened by the impact wrench (ASSISTED: phantom holder, 20 N push, a phantom finger on the trigger). The module stops the wrench when the torque it senses reaches the installation torque (green), or gives up (off or red); *no module* is the same wrench in its top gear with the trigger held, and *retrigger* pulls the trigger again and again on a set anchor.

case	hole (mm)	tip at start (mm)	tip at end (mm)	stud rise (mm)	preload (N)	nut torque (N m)	blows	light	socket off the hex at the end (°)	fault
sound	13.3	95.00	93.56	1.44	27021	68.6	41	green	0.8	none
oversize	14.0	95.00	82.11	12.89	3	0.0	0	off	1.9	none
snug	13.6	95.00	93.53	1.48	27327	69.4	42	green	0.9	none
dusty	13.3	91.21	90.24	0.97	26393	67.0	34	green	-0.1	none
pushon	13.3	95.00	93.58	1.42	27146	69.0	41	green	0.6	none
pushon turned	13.3	94.99	92.92	2.07	3001	7.6	0	off	-29.6	none
pushon turned rolled	13.3	95.03	93.55	1.49	26889	68.3	40	green	1.3	none
pushon turned 50n	13.3	94.99	93.48	1.50	2568	6.5	0	off	-6.5	none
retrigger	13.3	95.00	92.90	2.10	45379	115.3	1	green	1.0	none
no module	13.3	95.00	90.97	4.04	603	1.5	155	off	-0.0	none

Table 19: E20: the anchor hammered in through contact with the kit's 1.042 kg ball-peen hammer (13.03 J a blow), then set with the wrench. Hammering alone must leave the stud holding no more than its clip's drag and the cone out of the clip. It does not: the return from the seating blow is caught by the cone and the stud is left in tension, an open defect (Section 8.4); the wrench then finds the nut tight and stops at once.

hammer blows	largest advance (mm)	energy allows (mm)	hammered to (mm)	stud tension after hammering (N)	cone free	light	wrench blows	preload (N)	socket off the hex, largest (°)	fault
10	19.2	26.1	95.73	38200	**no**	green	1	37651	3.2	none

Table 20: E20: the blow that seats the washer on the wall, in the wall station, from three distances short of seating and at two hammer speeds. On MuJoCo's default line search, with the solve unfinished, the blow from 0.9 mm at 3 m/s left 12.7 kN of tension the stud had not been given and threw the washer at 35–58 m/s (Section 8.4).

short of seating (mm)	blow (m/s)	largest washer load (N)	washer gives (mm)	washer fastest (m/s)	nut slips on its thread (mm)	stud tension left (N)	cone free	fault
0.1	5.00	6082	1.49	1.84	0.0188	0	yes	none
0.1	1.00	0	0.00	0.11	0.0030	0	yes	none
0.4	5.00	6260	1.19	1.85	0.0189	0	yes	none
0.4	1.00	0	0.00	0.12	0.0030	0	yes	none
0.9	5.00	5710	0.69	1.85	0.0166	0	yes	none
0.9	1.00	0	0.00	0.12	0.0030	0	yes	none

Table 21: E20: a 38 mm fir ledger held by one anchor and pulled straight off the wall by a phantom force ramped to 8 kN: with the anchor torqued, and with an anchor that was only hammered in.

anchor through the ledger	preload after setting (N)	pull at which the ledger leaves the wall by 0.1 mm (N)	ledger off the wall at the largest pull (mm)
torqued	26560	3998	0.21
hammered only	–	198	598.04

Table 22: E9: drill a 13.3 mm hole, clean it or not, hammer a 1/2 in wedge anchor in with the kit's hammer and tighten its nut with the impact wrench. Set = driven to the embedment, the cone drawn into the clip, the torque reached and the stud not turned.

hole	depth (mm)	dust left by drilling (mm)	vacuum / brush passes	dust after cleaning (mm)	hammer blows	driven to (mm)	wrench light	nut torque (N m)	preload (N)	tip at the end (mm)	driven to 95 ± 2	set
clean	116.0	20.22	4 / 2	2.42	6	95.44	green	67.7	26658	95.19	yes	yes
dusty	116.0	20.22	0 / 0	20.22	6	94.49	green	69.0	27158	93.59	yes	yes

Table 23: E10: an M8 bolt pushed through two 9 mm holes (0.5 mm radial clearance each, 1.0 mm combined). Passes within clearance yes; jams beyond yes; undrilled steel not entered yes. Two bolts 30 mm apart in the same parts: both seated yes, contacts between one bolt's hole and the other bolt 0, the first bolt moved 0.0000 mm while the second was handled.

misalignment (mm)	mode	seated	head above face (mm)	peak axial (N)	peak lateral (m/s)	peak tilt (°)	no pop-out
0.0	axial	yes	-0.34	0.0	0.000	0.00	yes
0.5	axial	yes	-0.34	0.0	0.000	0.00	yes
0.9	axial	yes	-0.33	3.8	0.124	0.96	yes
1.0	axial	yes	-0.33	6.3	0.265	1.62	yes
1.1	axial	no	23.20	58.0	0.114	1.00	yes
1.2	axial	no	23.30	57.4	0.107	0.98	yes
2.0	axial	no	23.42	0.0	0.030	4.02	yes
0.0	slide	yes	-0.07	5.0	0.161	1.65	yes
0.0	off2	no	29.62	0.0	0.045	4.02	yes

misalignment (mm)	mode	seated	head above face (mm)	peak axial (N)	peak lateral (m/s)	peak tilt (°)	no pop-out
0.0	solid	no	27.79	0.0	0.394	1.39	yes

6.6 The hold-down clamps (E21)

E21 builds the two clamped stacks without a robot, tightens the four nuts to the registered preload, lets each stack rest for 30 s, and then loads the top part along the straps, across them and in torque about the vertical until its corner has moved 0.1 mm (Table 24). The slip loads are compared with a hand calculation from the normal forces measured at rest and the faces' friction coefficients. A held load at half and at 0.8 of each measured capacity then looks for creep. The loads a drill puts on these parts (a few hundred newtons of push, at most the cordless drill's 27 N m stall torque) are under a tenth of the capacities. Every check of E21 passes. On an earlier version of the clamp hardware's contact two did not (a held load at 0.8 of capacity crept 0.03–0.04 mm in 2 s, and a face slipped below the measured capacity); the contact as it is set for the sliding bolt (Section 8.4) passes both.

Table 24: E21: the strap clamps that hold the pre-clamped workpieces (no robot in the scene). Each nut is tightened to the registered preload (5783 N, 10.8 N m at a nut factor of 0.20), the stack rests for 30 s, and the top part is then loaded along and across the straps and in torque about the vertical until its corner has moved 0.1 mm. The hand calculation is friction at the faces from the measured normal forces at rest. Creep is the corner's motion in 2 s under a held load at the given fraction of the measured capacity; bold fails the experiment's own limit. Every check of the experiment passes.

task	rest, 30 s: largest excursion (mm)	stud tension after rest (N)	load	slips at	hand calculation	creep in 2 s at 0.5 (mm)	creep in 2 s at 0.8 (mm)
panel-holes	0.0003	5783, 5783	force x	3649 N	3629	0.0003	0.0000
			force y	3642 N	3629	0.0000	0.0001
			torque z	523 N m	521	0.0000	0.0004
timber-bolt	0.0017	5780, 5786	force x	3926 N	3735	0.0003	0.0004
			force y	4764 N	4746	0.0001	0.0001
			torque z	613 N m	572	0.0004	0.0002

6.7 Phantom operators on the machines

A phantom operator runs the drill press through its own controls: it lays the bar in the vise, turns the crank at a hand's speed (at most 5 N m) until the jaws clamp, presses START with 15 N, pulls the feed handle with a knob force ramped to 100 N and presses STOP. A 9 mm drill does not cut below the 231 N chisel threshold (Machinery's Handbook [1], $c/d = 0.18$). The exit cap collapses when the remaining thickness falls below $\sqrt{2T/(\pi\sigma_y)}$, and the burr is a quarter of the cap left then: the geometric mean of nine measured exit burrs in St37 against the cap our law gives at their thrust (Alrabii 2009 [31]; the individual ratios run from 0.10 to 0.77). An operator who eases over the last 2 mm leaves a smaller burr than one who keeps pushing; both stay inside the 1/16 in limit for bolted faying surfaces (RCSC [30] Specification section 3.4), so the burr check is reported but not scored (Table 25, Figure 11). A phantom hand also stands the centre punch on each mark and pushes it along its axis, after a control push that stays under the punch's setting; each strike's dimple takes the strike energy the struck part's support does not (Table 26). Finally, the robot's own finger runs the cordless drill through the policy's gripper semantics alone.

Table 25: The phantom press operator: 100 N at the feed handle's knob, eased or not before the exit. The burr may be at most 1/16 in (1.6 mm), the limit for bolted faying surfaces (RCSC Specification section 3.4).

operator's knob force	exit burr (mm)	through after (s)	peak thrust (N)	evaluator
full push to the end	0.423	4.3	1381	pass
eased to 46 N over the last 2 mm	0.269	15.7	1322	pass

Table 26: The phantom punch hand: the loose bar moved 0.00 mm; the evaluator passes.

press	peak push (N)	point travel (mm)	strike	released after (s)	dimple depth (mm)	re-cocked
negative control: push only to 60 N	60.0	1.98	none	–	–	yes
mark 0	96.0	3.01	at 94.1 N	0.97	0.4698	yes
mark 1	96.0	3.01	at 94.1 N	0.97	0.4698	yes
mark 2	96.0	3.01	at 94.1 N	0.97	0.4698	yes
mark 3	96.0	3.01	at 94.1 N	0.97	0.4698	yes

The robot's own finger runs the cordless drill: at 35 N per finger the trigger travels 8.01 mm and the chuck runs at 2000 rpm; at 8 N per finger it travels 5.83 mm and the chuck turns at 1611 rpm; opening releases it (-0.002 mm, 0 rpm). The drill moved 0.54 mm on the bench; 0 reflex stops.

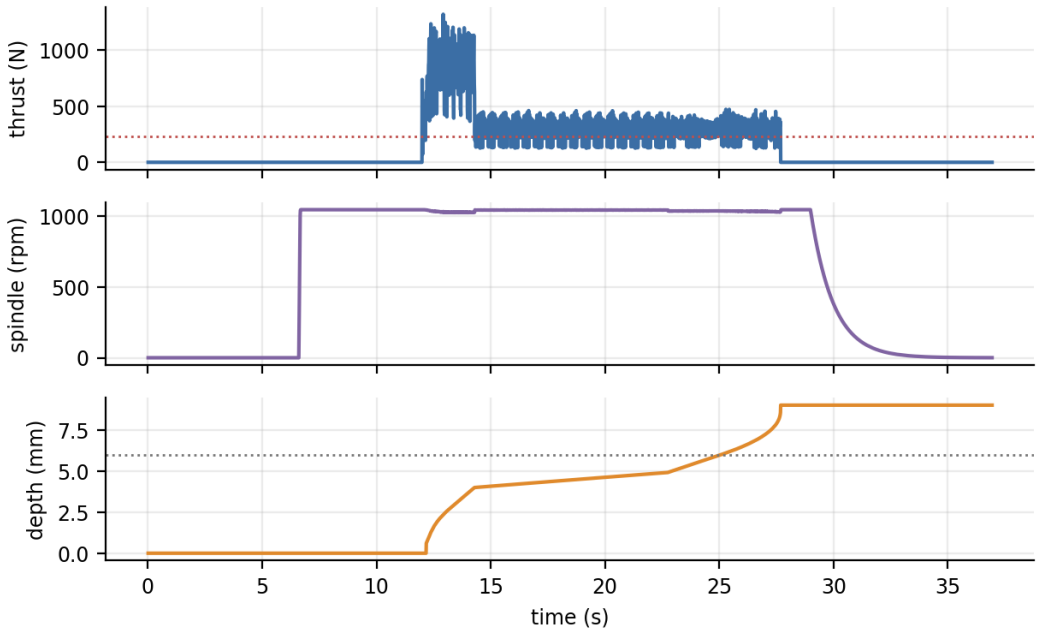


Figure 11: The phantom press operator eased to 46 N at the knob: thrust at the bit, spindle speed and depth over time.

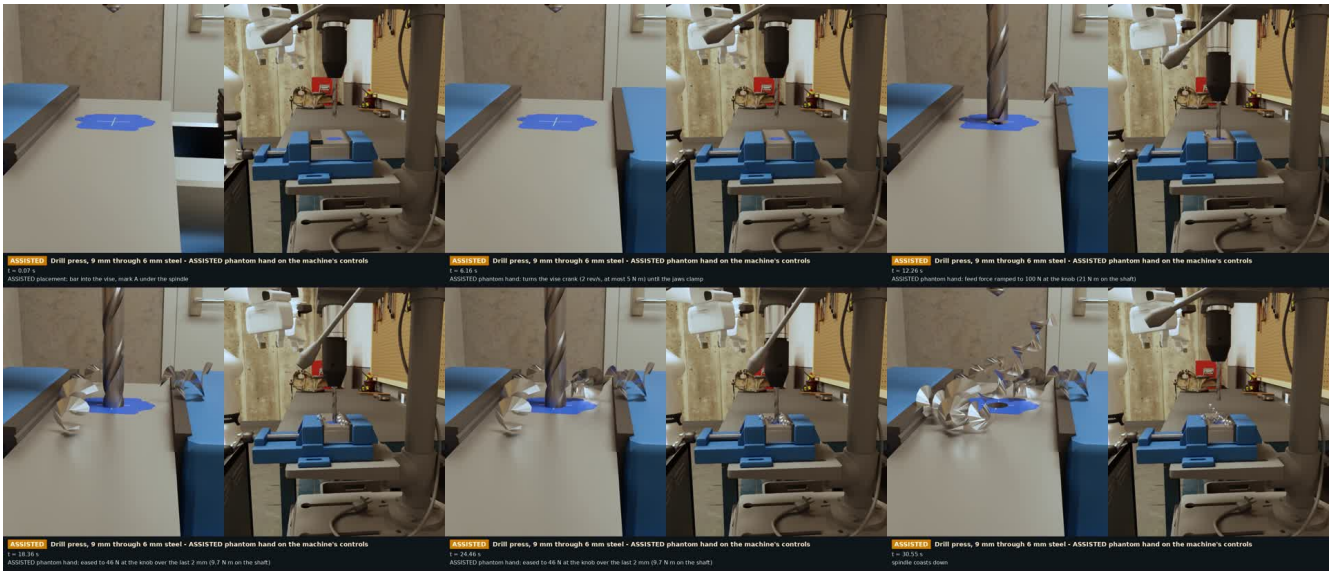


Figure 12: The same run as filmed (ASSISTED: phantom operator, no robot): the vise clamped by its crank, START, the feed handle pulled, the drill through the bar, STOP.

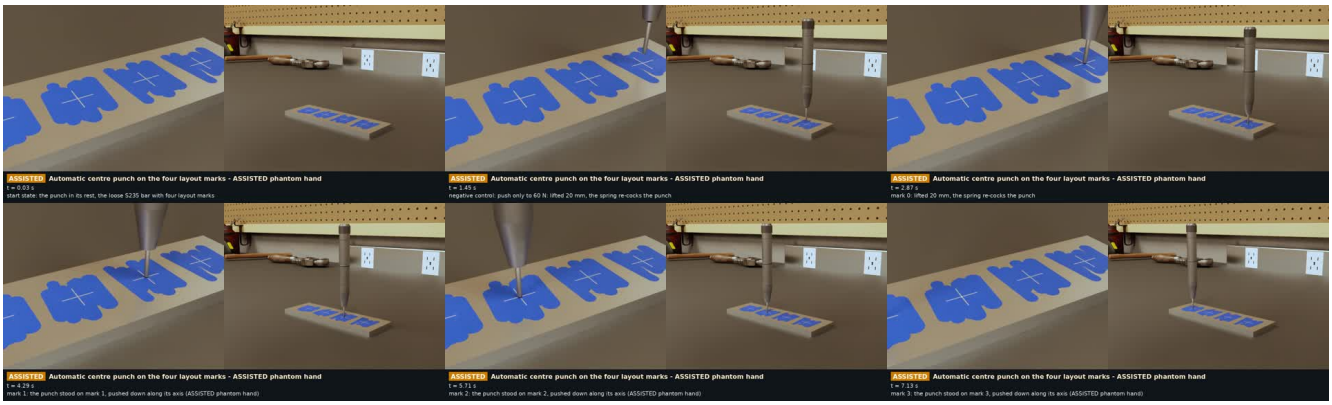


Figure 13: The phantom hand on the automatic centre punch (ASSISTED: phantom placement and push, no robot): four marks punched in the bar.



Figure 14: The robot's finger on the cordless drill's trigger (ASSISTED: scripted targets; the grip is the policy's own gripper command): the pinch closes, the lower pad pulls the trigger and the chuck turns.

6.8 Every task end to end (E12)

The component experiments do not show that a whole task can be completed in its own station: whole tasks find tasks that cannot be completed at all and mechanisms that defeat a later step, such as the second hole of every wall task (Section 8.4). E12 therefore runs each task once in its station, from the hand-over state to the task's own evaluator, with declared assistance: phantom hands on the tools' own handles, triggers, cranks and buttons, placement where a hand would carry a part or a tool, and privileged reading of the world where an operator would look. Nothing acts on a hole, a cut or a fastener's fit except through contact. The wall tasks are run without the robot, because a hammer placed by a phantom hand overlaps the parked arms at some marks. The anchors are driven by the kit's own ball-peen hammer, swung by a phantom force that is removed before the face meets the anchor (the phantom hand keeps the hammer's attitude through the blow with a torque), and tightened by the kit's impact wrench held on the nut by a phantom holder; the shop vacuum is switched on at its rocker by a phantom fingertip (Figure 15). The wrench is rolled about its anvil until the socket's hex meets the nut's, as an operator does: its anvil stops wherever the last setting left it, and a run that places the wrench at the nut's own angle sets a used wrench's socket up to 29° out of line, standing in the nut's corners. The probe records the largest angle between the two hexes on every tick of each setting. take-drill, punch-layout and press-drill are covered by the robot demonstrations of Section 7 and by the phantom press operator above. A run of this kind shows that the world lets the task be done. It says nothing about a robot doing it.

Four further things are part of E12. *Seeds:* the tasks are run at seed 0 and at other seeds (Table 27, one row per task and seed), and ledger-anchor at a seed of each mode the rotary hammer can be handed over in. *The mode selector:* where the hammer starts in *hammer* or *rotary*, a phantom finger holds its release button in and a phantom hand turns the knob to hammer-drill before the first hole, so that mechanism is operated too. *Rest after the work:* every run ends with all phantoms removed and 20 s with nothing acting, judged by E0's rule on the last 5 s (every free body below 0.1 mm/s and within 0.2 mm) and by the evaluator, which must say afterwards what it said before; match-bolt adds a second hold with the vise opened and the bolted stack lying loose, a state that can stay restless. *No phantom clamp:* match-bolt's bracket is not held on the bar by a phantom force, for which the scene has no tool. The bracket is a 50 × 50 × 5 angle, as wide as the 50 mm bar, and the vise's jaws grip both; nothing else holds it, and the run records how far the bracket turned on the bar. press-blind is run twice: once stopping at the depth read from the hole, and once (press-blind-stop) with the press's own depth-stop nut set after a touch-off and the quill fed against it, the hole never read.

Table 27: E12: each task end to end in its own station with declared assistance (phantom hands, placement, privileged reading where an operator would look; each JSON states it), one row per seed. The task's own evaluator judges the end state. World checks are the probe's own, on the world and not on the operator: that the scene is at rest through 20 s with every phantom removed (every free body below 0.1 mm/s from the second second on, read on every tick, and within 0.2 mm over the last 5 s; the column shows the fastest body); for the anchor tasks, that no hammer blow did more work against the expansion clip's drag than the blow's energy; for panel-holes, one hole record per mark in the sheet and in the backer; for match-bolt, rest again with the vise opened.

task	seed (hammer handed over in)	evaluator	checks	world checks	rest after: fastest body from 2 s on (mm/s)	failed checks	sim (s)
press-blind	0	pass	10/10	1/1	0.000	–	44
	3	pass	10/10	1/1	0.000	–	44
press-blind-stop	0	pass	10/10	1/1	0.000	–	72
match-bolt	0	pass	19/19	2/2	0.000	–	118
	3	pass	19/19	2/2	0.000	–	118
	777	pass	19/19	2/2	0.000	–	122
panel-holes	0	pass	19/19	2/2	0.000	–	58
	3	pass	19/19	2/2	0.000	–	57
	777	pass	19/19	2/2	0.000	–	56
timber-bolt	0	pass	10/10	1/1	0.031	–	70
	3	pass	10/10	1/1	0.000	–	70
	5	pass	10/10	1/1	0.000	–	70
	42	pass	10/10	1/1	0.000	–	70
	777	pass	10/10	1/1	0.070	–	70
	1234	pass	10/10	1/1	0.032	–	70
anchor-holes	0	pass	20/20	1/1	0.000	–	232
	3	pass	20/20	1/1	0.000	–	231
rebar-relocate	0	pass	19/19	1/1	0.000	–	270
	3	pass	19/19	1/1	0.000	–	269
clean-and-set	0	pass	23/23	5/5	0.040	–	60
	3	pass	23/23	5/5	0.040	–	60
ledger-anchor	0	pass	42/42	5/5	0.040	–	201
	1 (hammer)	pass	42/42	5/5	0.040	–	200
	2 (rotary)	pass	42/42	5/5	0.040	–	202



Figure 15: clean-and-set end to end in its station (ASSISTED: phantom hands, no robot): the nozzle in a hole with the vacuum running, the ball-peen hammer driving an anchor, the impact wrench on its nut.

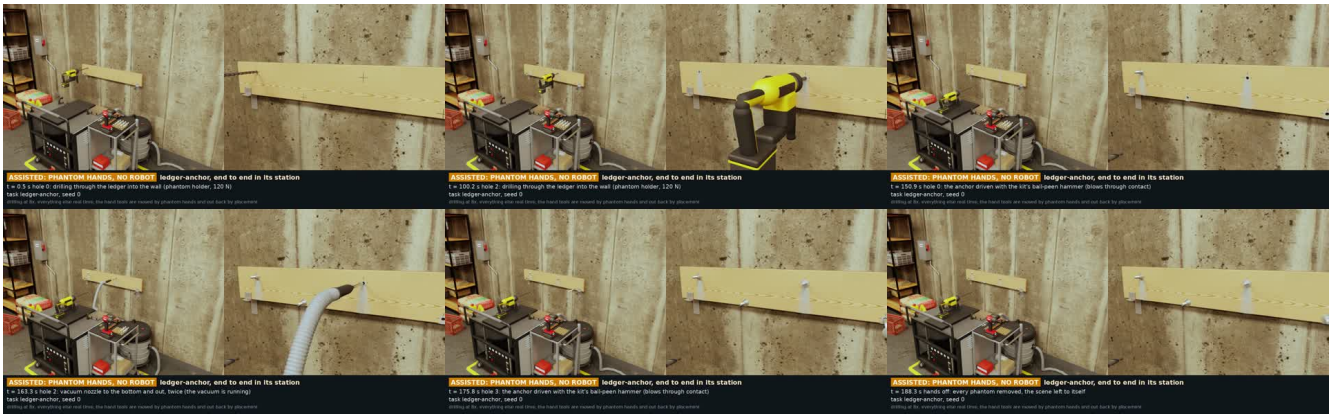


Figure 16: ledger-anchor end to end in its station (ASSISTED: phantom hands, no robot; the drilling is shown at 8×): four holes through the ledger into the wall, four anchors hammered in and their nuts tightened against the ledger.

6.9 The evaluator (E17)

Each task's evaluator is run on a correct end state (built by writing records, not by the robot), on an untouched scene, which must fail, and on deviations and fakes, each of which must fail the check it breaks (Table 28). This proves the checks, not that a task can be done; the robot demonstrations of Section 7 and the timber-bolt boards, which cannot be drilled through without their backer (Section 8.4), are why both are needed.

Table 28: E17: each evaluator on an correct end state, on an untouched scene and on deviations (states written directly). Doing nothing is its own column: it must fail, and is not one of the deviations. All as expected: yes.

task	correct end state passes	doing nothing	deviations caught	deviations tried
press-drill	yes	fails, score 0.00	8 of 8	hole 0.6 mm off the mark; hole not through; axis tilted 2 deg; wrong drill (10 mm); bar spun 3 deg; spindle left on; bit broken; fake: a dimple only
press-blind	yes	fails, score 0.00	3 of 3	too shallow (14.8 mm); too deep (15.7 mm); drilled through
punch-layout	yes	fails, score 0.00	4 of 4	one mark not punched; dimple 0.4 mm off; stray dimple near a mark; bar moved 3 mm
anchor-holes	yes	fails, score 0.00	8 of 8	one hole missing; hole 4 mm off; too shallow (100 mm); too deep (130 mm); axis 4 deg off; rebar cut 1 mm; bit chipped; one arm carried the hammer alone for 2 s
clean-and-set	yes	fails, score 0.00	10 of 10	dust left: anchor stopped short (88 mm); anchor driven 3 mm too deep (98 mm); anchor driven to 95 mm, then drawn 5 mm back out; anchor 0 crooked (5 deg); fake: anchor laid beside the hole; anchor 1 spun while torqued (90 deg); anchor 2 not torqued (40 N m); anchor 3 broken by over-torque (127 N m, the stud's steel strength); anchor 0 expansion not engaged (clip loose); anchor 3 over-torqued (120 N m)
timber-bolt	yes	fails, score 0.00	4 of 4	hole 2 deg off square (the robot's was 5.5 deg); charred wall; bolt forced (80 N); fake: bolt not through the lower board
match-bolt	yes	fails, score 0.00	4 of 4	holes 0.8 mm apart from drawing; bolt forced (70 N); bar not drilled; bolt beside it; angle spun 5 deg
panel-holes	yes	fails, score 0.00	4 of 4	hole 1.2 mm off; grabbed, out of round (lobed 0.4 mm); one hole missing; hole 2 left at its 3 mm pilot, the 6 mm drill only started
rebar-relocate	yes	fails, score 0.00	5 of 5	bar cut 1 mm; replacement 80 mm away (under twice the 45 mm stopped depth); replacement 200 mm away (beyond 150 mm); no replacement (the partner hole on the line does not count); stopped at 20 mm, short of the bar, and moved
take-drill	yes	fails, score 0.00	6 of 6	held only 2 s; both hands on the motor housing (none on the grip); trigger pressed 3 mm; bit touched the bench; bit 8 deg off vertical; tip 120 mm above the bench
ledger-anchor	yes	fails, score 0.00	9 of 9	one ledger hole missing; anchor short (90 mm); hole 0 drilled 135 mm deep (past 125 mm); ledger dropped 5 mm over its length; anchor 1 spun while torqued (90 deg); anchor 2 not torqued (40 N m); anchor 1 never brought above 55 N m; anchor 3 broken by over-torque (127 N m, the stud's steel strength); anchor 3 taken to 120 N m, 100 N m left

7 Robot demonstrations

The component experiments hold each tool with a phantom hand. The demonstrations remove it: the two FR3 arms take the tools from where they lie, hold them by finger contact and use them on the workpieces, through the joint servos, effort limits, force-mode gripper and collision reflexes a policy has. They are ASSISTED in one declared way. A script computes joint targets by inverse kinematics from privileged poses of the tools and workpieces, and a collision-checked planner (the arms, every carried body posed along the path, 1 mm clearance) chooses the paths. Where a run also feeds a held tool's weight forward, it is the arms' own joint torques ($J^\top F$, capped at each joint's effort limit, as the FR3 controller models a configured load), the `cartesian_wrench` action. No force acts on a tool or a workpiece except through the robot's fingers and hands. Each run ends with the task's own evaluator and is recorded, with a film captioned below the image. A scripted run proves that the world lets a robot do the task. It is not a policy and shows nothing about how often one would succeed.

Table 29: Robot demonstrations (ASSISTED: scripted IK targets, planned paths); the evaluator is the task's own.

task	the robot	evaluator	failed checks	reflex stops	sim (s)	wall (s)
take-drill	takes the drill with both hands, holds it bit-down over the bench	success (1.00)	none	0	36.9	274
punch-layout	hands the punch from one hand to the other, punches four marks	**fail** (0.25)	mark 0: dimple within 0.3 mm; mark 1: dimple within 0.3 mm; mark 3: dimple within 0.3 mm	2	73.2	427
anchor-holes	carries the rotary hammer to the wall and drills an anchor hole	**fail** (0.57)	no arm carried more than its rated payload (2.27 kg beyond its hand); hole 0: hole at the mark; hole 1: hole at the mark; hole 3: hole at the mark	0	97.9	794
press-drill	clamps the bar and drills it at the drill press	success (1.00)	none	0	177.5	923
timber-bolt	takes the drill with both hands, bores through both boards with one	**fail** (0.33)	square to the faces (axis within 1.19 deg of the face normal); no charred wall (< 200 C); through hole in the bottom board; carriage bolt through both boards	0	241.8	2646

7.1 Taking the drill (take-drill)

The left hand takes the cordless drill from above by its motor head over the centre of mass, lifts it, turns its grip towards the robot and stands it back on the bench. The right hand cannot reach behind the grip where the drill first stands, because the drill-press head and vise fill that space and every inverse-kinematics solution collides there. The right hand then takes the grip from the robot's side with the index finger over the trigger. Both hands lift the drill, turn it bit-down about its centre of mass and lower it until the tip is 50 mm above the bench (Figure 17).

Both grasps close at 35 N per finger, 2.6 mm and 0.6 mm from the planned grasp. The two-handed turn is planned so that no joint moves more than 0.12 rad from one waypoint to the next, and both arms' targets are interpolated between the waypoints every 20 ms. The drill then arrives 0.8 mm and 0.1° from its goal with no correction of the hands' targets; the grasps shift 0.14 and 1.35 mm; no reflex stops either arm. The evaluator passes all nine checks: held only by the two hands for 13.6 s, one hand on the grip and one on the motor housing, the bit 1.22° from vertical, the tip 49.3 mm above the bench, the trigger never pressed, the bit touching nothing, neither arm over its rated payload.

The waypoint spacing matters, and the failure without it is worth stating. A plan that lets an arm change its elbow's place by up to 0.35 rad between two waypoints makes the hand leave its path on the way, and the other hand, the drill and the wrist take the difference: the left wrist read 10.7 N m, its reflex stopped that arm eight times, and the right grasp tore by 44 mm and 23°. That defect is the operator's, and the script avoids it.

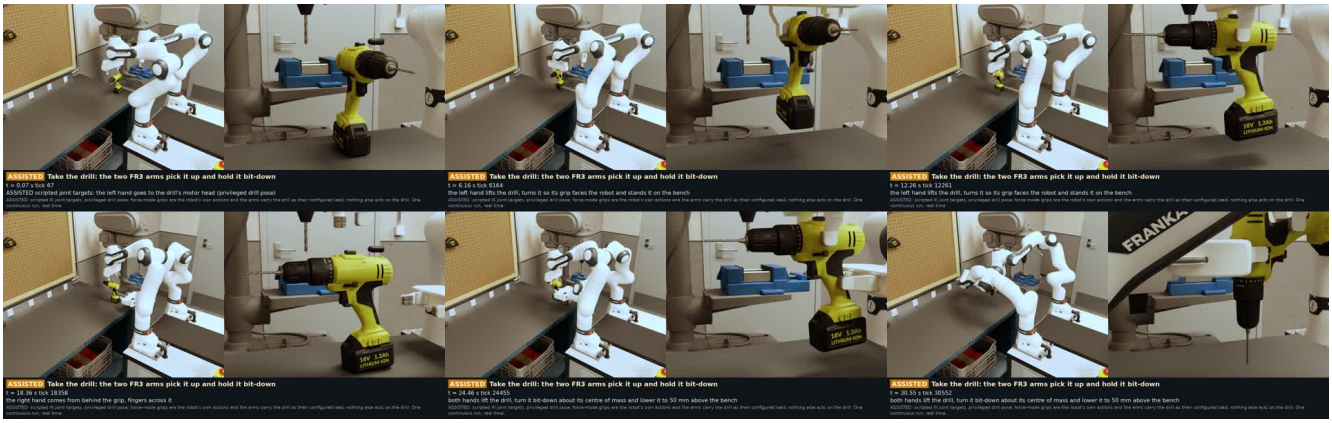


Figure 17: take-drill, performed by the robot (ASSISTED: scripted IK targets, planned paths). Wide and close views of the same instants.

7.2 Centre-punching the layout (punch-layout): fails

The right hand takes the automatic centre punch from its rest and turns it point-down. The left hand takes it from above along its axis with the cap against the palm, and the right hand lets go; the punch then stands 6.5° off vertical in the hand, and the hand is turned until the punch is vertical. The left arm pushes it down onto each of four layout marks on a loose steel flat bar, as a technician leans on the cap, until the punch's internal spring releases at this scene's cap setting, 94 N (Figure 18). The push is shaped through the servo: $q_{\text{target}} = q_{\text{hold}} + J^{\top} F / k_{\text{servo}}$, with F ramping at 12 N/s along the punch axis. This is the servo stiffness a policy also commands through joint targets. The punch is taken at 35 N per finger; for the push the operator lowers the grip to 24 N, because at the hand's full force the collision reflex stopped every push at 87 to 90 N.

The run fails: one mark of four. The point stands 0.18, 0.24, 0.05 and 0.17 mm from the four marks before the push. At marks 2 and 3 the punch fires at 94.1 N and leaves dimples 0.30 mm (0.298, inside the 0.3 mm allowed) and 0.43 mm from their marks: the point and the loose bar still move under the seconds of rising push. At marks 0 and 1 the collision reflex of the left arm's sixth joint stops the push at 92 and 93 N, short of the release, and the script abandons the mark. So this arm in these postures presses about as hard as this punch needs, and at a seed whose punch is set higher it would fire none. Both failures belong to the operator: a technician holds or clamps the bar and leans on the punch with a straighter arm.

The failure is not the world's: the punch gets the force commanded, does not walk outwards in the hand when the arm swings, and weighs what the maker's punch weighs (Section 8.4). The reflex stops at two marks are the price of pushing through a grip that does not give.

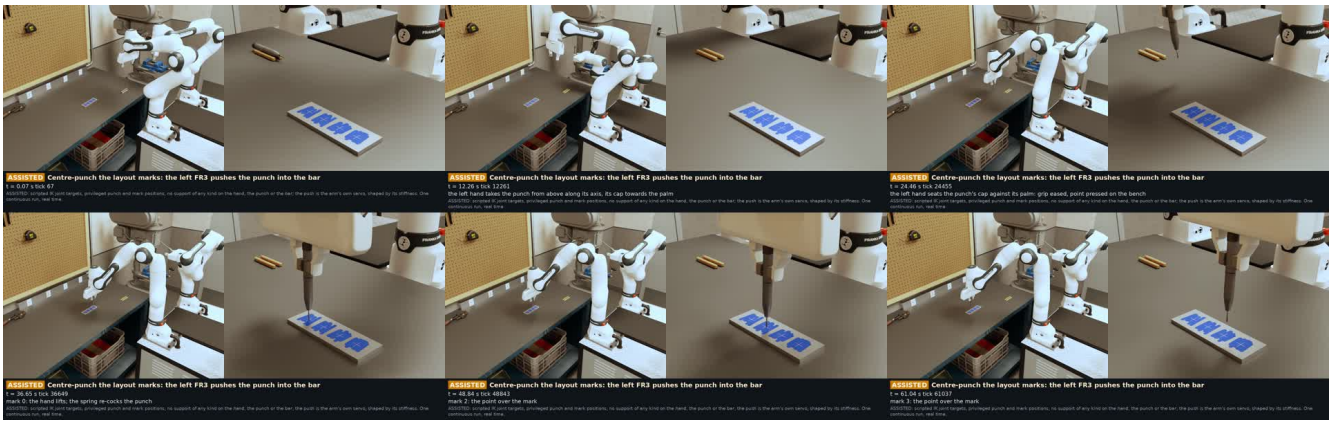


Figure 18: punch-layout, performed by the robot (ASSISTED: scripted IK targets, a shaped push through the joint servos). Two marks struck, one of them inside the tolerance; two pushes stopped by the arm's collision reflex; the evaluator fails the run.

7.3 Drilling an anchor hole into the wall (anchor-holes): one hole of four

Both hands take the rotary hammer (3.23 kg with its bit) from its cradle on the cart: the left hand pinches the barrel from above and the right hand pinches the D-handle front to back below the trigger. They carry it to mark 2 and bring the bit to the face 1.1 mm from the mark. The right hand's front pad presses the trigger through its 8 mm and the hammer starts. Both hands push until the pocket is begun; the left hand then lets go of the barrel, and the right hand alone pushes along the bit through its fingers, 28.7 N on average (Figure 19).

The hole reaches its 113 mm in 32.9 s at 3.44 mm/s. The hammer delivered 1528 of the 3327 blows it scheduled, 46 %: the share the housing's momentum budget gives for that push (Section 6.3); the concrete's energy per volume follows the bit's diameter (Section 4). No reflex stopped either arm in the whole run. The trigger is released, the left hand returns to the barrel and both hands draw the bit out (its tip 81 mm in front of the face at the end). The hole is 0.91 mm from its mark and 0.11° from the wall normal, 113.3 mm deep at full diameter, clear of the reinforcement. The evaluator fails the task (score 0.57): the other three marks were not attempted, and the rated-payload constraint fails (below). A variant with the hands mirrored drills mark 3 (Section 7.6); no run has tried marks 0 and 1. Static reach with the hammer in hand exists for all four (Table 3).

Two things in the record are not what a careful operator would accept. The right hand's grasp slid along the handle while the hammer struck (5.7 mm at the end of the run). And while one hand holds the hammer alone, which the right hand does for the whole of the drilling, that arm carries 3.23 kg against the 2.27 kg an FR3 is rated for beyond its hand: the world does not refuse it, and the evaluator fails the run on it (the left arm was over its rating for 45 s, the right for 46 s). The planner lets a tool lean on the palm of the hand that already holds it: the hammer slides back in the trigger hand's pinch and leans on that hand's palm, and a planner that counts that contact refuses every pose of the other hand at the barrel hand's return.

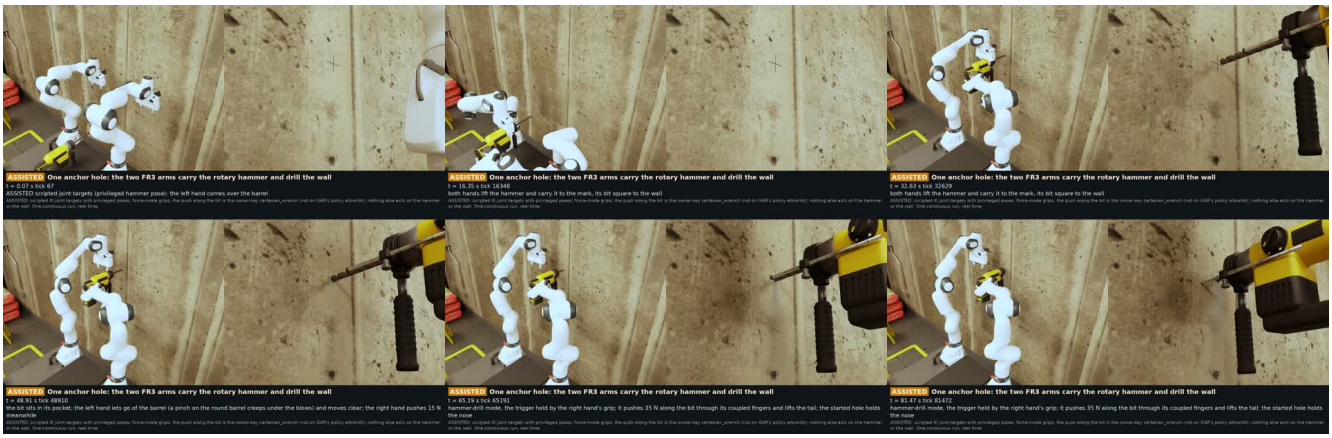


Figure 19: anchor-holes, performed by the robot (ASSISTED: scripted IK targets, planned paths, a push shaped through the right arm's servo). One hole drilled to specification at mark 2.

7.4 Drilling at the drill press (press-drill)

The two arms run the drill press through its own controls, and nothing else acts on the press, the vise, the handle, the buttons or the bar (Figure 20). The left hand picks the 6 mm steel bar off the bench across its width at 35 N per finger, carries it under the head 16 mm clear of the vise jaws and of the raised bit, lays it on the vise's parallels and slides it closed-loop on its measured pose until the punched mark is under the spindle (two fine steps; the bar slipped 0.18 mm in the fingers on the way). The right hand takes the vise crank's knob and turns the crank 10.1 turns. When the jaw meets the bar it tightens to 1.98 N m, and the self-locking screw holds a 788 N clamp. No reflex stopped either arm. A left fingertip presses START (2.33 mm of travel at 23 N; the spindle runs free at 1044 rpm). The right hand pinches the feed handle's top knob and pulls the handle round. The point touches after 0.5 s. The bulk of the cut is pushed with up to 68.0 N at the knob (peak thrust at the bit 586 N). Over the last 2 mm the push is eased to 40 N at the knob; the exit cap collapses at 308 N and the burr is 0.22 mm. At that push the thrust is only just above the 9 mm drill's 231 N chisel threshold, so the last two millimetres take most of the 68.3 s of drilling. The hand lets the handle return the quill to the top. A fingertip presses STOP and the spindle coasts down.

The evaluator passes all eleven checks: a through hole 0.066 mm from the mark, its axis 0.007° from the face normal, 9.002 mm in diameter, the bar unturned, the spindle off and stopped, the quill at the top and the bit clear of the work. The heaviest load on a wrist joint while drilling was 93 % of its reflex limit. The run lasts 178 s of simulated time. Beyond the scripted joint targets and planned paths it is ASSISTED in one more way: the point at which to ease is read from the hole record, as an operator reads the quill's depth scale. The film was rendered from the recording of this run.

The script carries the bar clear of the bit's point, and a bar carried against the stopped bit is not given a hole (Section 8.4).

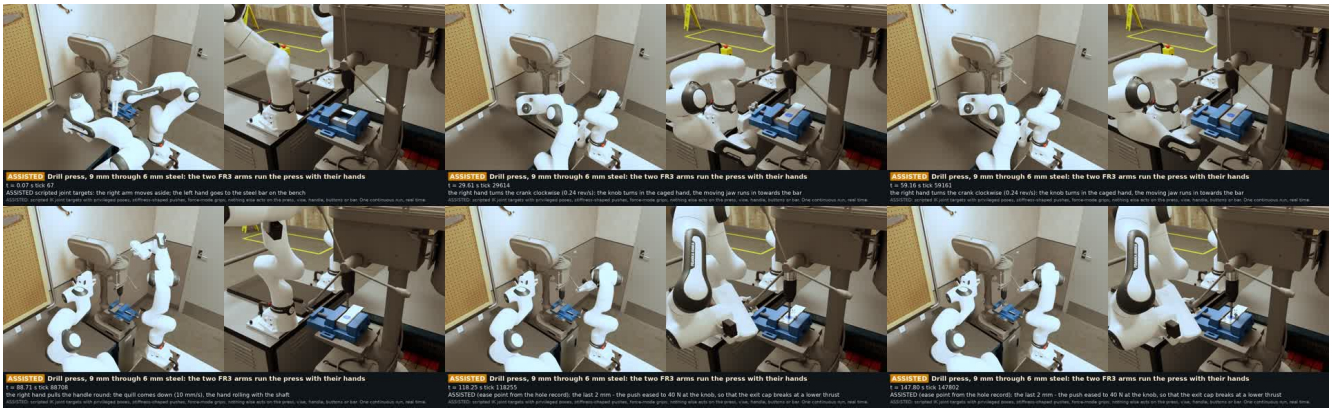


Figure 20: press-drill, performed by the robot (ASSISTED: scripted IK targets, planned paths, pushes shaped through the arms' servos, the ease point from the hole record). Bar placed, crank turned, START, feed, ease, through, STOP.

7.5 Boring through the timber (timber-bolt): fails

The right arm cannot reach the drill over the timber mark, so the left hand drills alone and the right hand helps it take the drill. The right hand lifts the drill by its motor head and turns it with the bit towards the robot. The left hand pinches the grip front to back, lower pad on the trigger, and closes at 35 N per finger, which pulls the trigger and starts the drill. Both hands turn it bit-down, the right hand lets go, and the left hand sets the brad point on the mark. The arm then pushes along the bit through joint targets shaped by its servo stiffness, up to 40 N. At breakthrough into the backer the arm eases off and opens the hand, the trigger springs back and the drill stands in its hole (Figure 21).

The run fails: the drill is lost before the bit is through. The 14.3 mm auger enters 0.14 mm from the mark (2 mm allowed) and goes through the top board; after 204 s of drilling at 644 rpm it stands 37.5 mm into the 38 mm bottom board when the drill, which has turned 25.6° about its bit in the one-hand pinch (26.6° at most) and tilted 2.2°, has turned far enough that the finger no longer holds the trigger (3.4 of 8 mm). The script then opens the hand; the trigger springs back, the brake stops the chuck and the drill stands in its hole. Four of the evaluator's ten checks fail: no through hole in the bottom board, the hole 1.91° off square against the 1.19° the instruction allows, the wall charred (the model's bit reaches 1336 °C against a 200 °C limit) and no bolt. No reflex stopped the arm.

One Franka Hand pinch does not react a hand drill's cutting torque, and that is the finding of this run: the cutting torque (0.25 N m) is about what the two pads' twisting friction can carry, and the drill creeps round under it for as long as the bore lasts. The bore is long: the script starts with 8 N and raises the push in steps, so the bit rubs for over a minute at pushes below what this auger needs to cut in solid wood, and only then pushes with 40 N; and the wood law (Section 4) feeds slowly in solid wood. The world was not changed for the script. The charring is the operator's technique meeting a heat model that has no ceiling from the wood's own charring. The boards are held by the strap clamps of Section 3.2 and did not move.



Figure 21: timber-bolt, performed by the robot (ASSISTED: scripted IK targets, planned paths, a push shaped through the left arm's servo). Through the top board and 37.5 mm into the bottom one; the drill turns in the one-hand pinch until the trigger is lost; the bolt was not attempted.

7.6 The wall again: variants of the anchor hole, and the anchor tools in the hand

Four further things are asked of the robot at the wall: whether the anchor hole is still drilled when the hammer's weight counts against the joint torque cap, what an arm over its rated payload does, whether the script reaches more than one mark, and whether the arm can do the anchor steps (a hammer blow, a vacuum pass, the wrench on a nut) that the end-to-end runs do with phantom hands. Each was run once, with the same declared assistance as the standard run; Tables 30 and 31 give what happened, failures included.

Table 30: The robot drilling one anchor hole (ASSISTED: scripted IK targets and planned paths; the push along the bit is the owner-key `cartesian_wrench`), the standard run and three variants. "Over the rated payload" is the time an arm carried more than 2.27 kg beyond its hand; the evaluator allows 0.5 s.

run	hole depth (mm)	drilling (s)	share of blows landed	reflex stops	over the rated payload	largest wall force on the bit (N)	script failure
standard run, mark 2	113.1	32.9	0.46	0	left 45.0 s, right 46.1 s	–	none
tool's weight inside the torque cap	113.1	32.9	0.46	0	left 44.8 s, right 46.3 s	2330	none
both hands on the hammer throughout	113.0	28.0	0.34	0	left 12.9 s, right 5.3 s	2831	none
mark 3, hands mirrored	113.1	32.0	0.47	0	left 44.1 s, right 44.5 s	2421	none

Table 31: The anchor tools in the robot's hand (ASSISTED: scripted IK targets; the anchor placed as the phantom runs leave it; where the trigger is pulled by a phantom finger the row says so).

the arm holds	what happened	script failure
impact wrench on a hammered anchor's nut	trigger: phantom; light green, 37 blows, preload 26226 N, 66.6 N m; wrist at most 11.08 of 11.4 N m; 0 reflex stops	none
ball-peen hammer on an anchor	3 blows; head speed at contact [0.39, 0.451, 0.33] m/s (the phantom run's: 5.0); anchor advanced [1.93, 0.73, -0.21] mm; wrist [12.19, 9.42, 11.38] N m against 11.4; 1 reflex stops; the hammer turned [7.73, 16.26, 41.55] degrees in the pinch; laid back on the tray: no	{'phase': 'put back', 'error': "Abort: right: no collision-free joint path found in 200 tries (clashes at the start: [(-0.0, 'hammer', 'right/fr3/fr3_link5')];
the vacuum's switch, then the wand in a predrilled hole	the closed fingertips press the rocker (21.1 N at most): unit running True; 2 passes, nozzle at the bottom [True, True], tip off the hole's axis [0.5, 0.25] mm; dust column 17.34 to 8.87 mm (loose dust 1205 to 27 mm3; the packed dust needs the brush); wand laid back 0.16 mm from its place; 0 reflex stops	none

- **The tool's weight inside the cap** (one run, a wrapper in the script; the world is unchanged): the hole is drilled as in the standard run, 113.1 mm in 32.9 s. The hammer's gravity torque is at most 43 % of a joint's

rating in this posture and was clipped on no tick. The assumption of Section 5 is therefore worth nothing for this hole; it is still an assumption, and a posture with the hammer far from the body would test it harder.

- **Rated payload.** One FR3 carries 2.27 kg beyond its hand; the hammer weighs 3.23 kg. In the standard run one hand holds the hammer alone while it drills and withdraws, and the evaluator's payload constraint fails (over the rating for 45 s and 46 s). With both hands kept on the hammer throughout, the hole is drilled (113.0 mm in 28.0 s, no reflex stop) and the constraint still fails (13 s and 5 s over the rating): the barrel hand's share grows past 2.27 kg as the bore deepens, and its pads lose the barrel in the second half of the bore (they read 0 N from 62.5 s). The run ends with the bit drawn 34 mm out of the wall, the hammer hanging 11° off in the trigger hand alone and the put-back refused by the planner. With these grasps the pair does not carry the rotary hammer inside its rating.

- **Other marks.** With the hands mirrored the script drills mark 3: 113.1 mm in 32.1 s, 0.85 mm from the mark, no reflex stop, and draws the bit out (95 mm in front of the face). Marks 0 and 1 were not attempted. The static reach probe finds every mark of seed 0 reachable, and at the extreme seeds (the left-most, right-most, lowest and highest hole of each wall task over seeds 0–999) every extreme hole is reachable for both two-arm robots.

- **The hand-back** after drilling still levers the bit: 2.3–2.4 kN on the bit as the second hand returns (1.8 kN at the hand-back and 2.8 kN in the withdrawal of the both-hands run). The cause is in the script (it sends the returning hand to the planned line, not to where the hammer is) and is not corrected.

- **The wrench on a nut.** With the maker's pack and module the wrench weighs 2.34 kg, over the 2.27 kg one arm is rated for: the arm takes it by its grip, carries it and puts it back, and the evaluator's payload constraint fails for the 37.5 s it is in the hand. It pushes the socket home on a hammered anchor's nut. With the trigger pulled the module sets the anchor: green after 37 blows, 26.2 kN, 66.6 N m, with the wrist at 11.1 N m against its 11.4 N m reflex (97 %: the heavier tool leaves no margin) and no reflex stop. The grasp slid 7.1 mm along the grip through the blows. The hand that carries the wrench cannot reach its trigger, so the trigger is pulled by a declared phantom finger: the robot holds the tool through the blows, it does not operate it. The arm then lays the wrench back in its place on the tray, 2.7 mm and 0.06° from where it took it, and nothing else on the tray moves.

- **A hammer blow.** Inside Franka's joint velocity, acceleration and jerk limits the arm brings the 1.04 kg hammer's head to 0.33–0.45 m/s, against the 5 m/s of the phantom runs: under 1 % of a blow's energy. The anchor, placed with its tip at the hole's mouth, advances 1.9 and 0.7 mm on the first two blows and is knocked back and askew by the third; it never enters its hole. The wrist reads 12.2, 9.4 and 11.4 N m and the first blow trips its collision reflex (11.4 N m). The hammer, pinched at its centre of mass, turns in the pinch with every blow, 7.7, 16.3 and 41.6° in all: a parallel pinch carries a moment about its closing axis only by the pads' twisting friction, and a blow on the face is a moment about it. After the third blow the handle lies against the forearm and no collision-free way back to the tray exists; the run ends with the hammer in the hand. An FR3 with a Franka Hand does not hammer an anchor in, and no robot of the catalogue does.

- **The vacuum.** The arm closes its fingers and presses the "I" half of the shop vacuum's rocker with the fingertips (21 N at most; the unit runs), takes the wand off the trolley by its handle, carries it to a predrilled dusty hole and passes the nozzle to the bottom and out twice with the unit running; the tip is 0.5 and 0.25 mm off the hole's axis as it enters. The loose dust goes (1205 to 27 mm^3) and the column falls from 17.3 to 8.9 mm: what is left is packed and needs the brush, which the robot is not shown using. The wand is laid back in its cradle 0.16 mm from where it was taken, with no reflex stop. The script keeps the wand in front of the wall's face on the carry. The shared planner counts only contacts the robot has a part in, and a held part meets the wall through stand-in colliders that the planner's copy of the world does not move, so a carry path that sweeps the nozzle along the wall (5 N for 0.6 s) turns the wand 14 to 15° in the hand and stops the run.

So of the anchor work the robot shows the dust step with the vacuum, holding the wrench through a setting, and that it cannot drive the anchor. The brush, and the wrench's trigger, are not shown.

8 Stability, abuse and runtime

8.1 Rest (E0)

Every task's start state and a loose-vise variant are handed over and left alone for 30 s with the robot holding its ready pose. Hand-over waits until every free body is below 0.1 mm/s at three checks 0.1 s apart. A free body passes if it moves less than 0.2 mm over the last 5 s, is slower than 0.1 mm/s there and never wanders more than 0.2 mm from where it was handed over (Table 32). Two 10 s films show the same at real time with the largest displacement of any body printed on every frame.

Table 32: E0: 30 s hands-off rest of every start state; all passed: yes.

preset	task	bodies	settle (s)	net (mm)	last 5 s (mm)	speed (mm/s)	largest excursion (mm, body)	reflexes	pass
bench	take-drill	13	1.2	0.000	0.000	0.000	0.000 (drill_press_quill)	0	yes
bench	punch-layout	12	1.2	0.000	0.000	0.000	0.000 (drill_press_quill)	0	yes
bench	press-drill	10	1.2	0.000	0.000	0.000	0.000 (drill_press_quill)	0	yes
bench	press-blind	10	1.7	0.000	0.000	0.000	0.000 (drill_press_quill)	0	yes
bench	match-bolt	17	1.2	0.000	0.000	0.000	0.000 (angle_bracket)	0	yes
bench	panel-holes	32	1.2	0.000	0.000	0.000	0.000 (cover_panel)	0	yes
bench	timber-bolt	29	1.2	0.001	0.000	0.001	0.001 (carriage_bolt_shank)	0	yes
wall	anchor-holes	7	1.2	0.000	0.000	0.000	0.000 (wall)	0	yes
wall	rebar-relocate	7	1.2	0.000	0.000	0.000	0.000 (wall)	0	yes
wall	clean-and-set	36	1.2	0.003	0.000	0.006	0.003 (anchor0_top)	0	yes
wall	ledger-anchor	38	1.2	0.003	0.001	0.005	0.003 (anchor1_top)	0	yes
bench	press-drill (loose-vise)	10	1.2	0.000	0.000	0.022	0.000 (press_vise_body)	0	yes

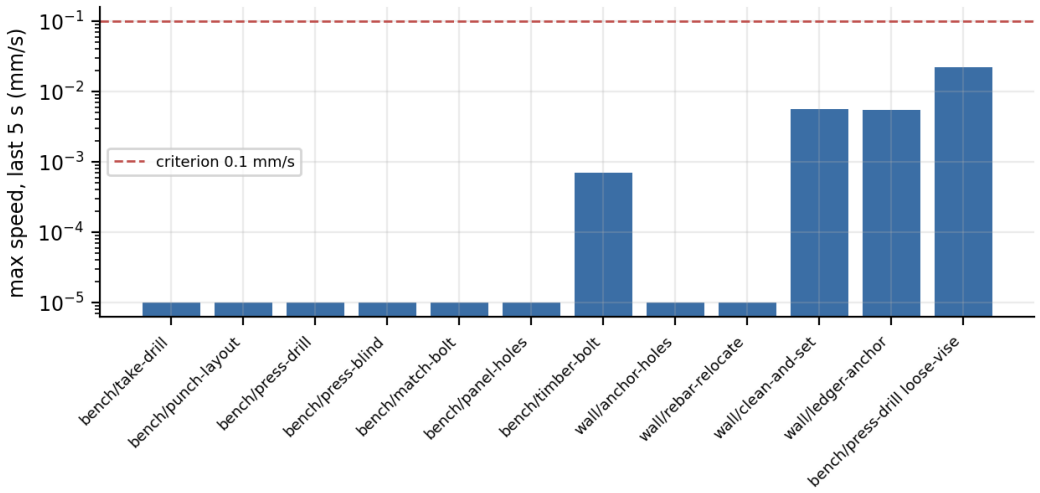


Figure 22: E0: the fastest free body in the last 5 s of the 30 s rest, per start state (log scale); the limit is 0.1 mm/s.

8.2 Drops and legal abuse (E14)

Every tool and part is dropped 1 m onto the concrete floor and onto the bench in several orientations, and a random sweep drops them from random poses with random spin; a drop passes if nothing faults and the energy never exceeds the release energy. Through the public backend, each arm in a wall and a bench task is sent legal but abusive commands: every joint to its upper and then lower limit, a pose 3 m away, the largest wrench and grip force the contract allows and beyond them, and 10 s keyboard chords that drive the hand up, down and diagonally at six times the jog speed. A command must be refused before it runs or must run without a fault and without moving anything the robot does not touch (Table 34).

Table 33: E14 drops from 1 m onto the floor and the bench. All passed: drops yes, sweep **no**.

set	object	drops	energy / release (max)	impact (m/s)	peak part speed after (m/s)	joint overtravel (mm or °)	faults	pass
drops	anchor	6	1.000	4.39	1.14	0.00	0	yes
drops	ballpeen	3	1.000	4.37	0.97	0.00	0	yes
drops	bolt	6	1.000	4.41	1.40	0.00	0	yes
drops	brush	3	1.000	4.39	1.48	0.00	0	yes
drops	carriage_bolt	6	1.000	4.39	1.52	0.00	0	yes
drops	drill	3	1.000	4.35	1.65	1.44	0	yes
drops	hammer	3	1.000	4.33	4.08	1.95	0	yes
drops	impact_wrench	3	1.000	4.42	2.91	0.04	0	yes
drops	nozzle	3	1.000	4.37	2.30	0.00	0	yes
drops	punch	3	1.000	4.40	2.55	0.04	0	yes
random sweep	anchor	8	1.000	4.42	5.07	0.00	0	yes
random sweep	ballpeen	8	1.000	4.09	2.89	0.00	0	**no**
random sweep	bolt	8	1.000	4.40	3.44	0.00	0	yes
random sweep	brush	8	1.000	4.00	3.51	0.00	0	yes
random sweep	carriage_bolt	8	1.000	4.39	3.45	0.00	0	yes
random sweep	drill	8	1.000	4.26	5.46	14.79	0	yes
random sweep	hammer	8	1.000	4.15	6.20	2.04	0	yes
random sweep	impact_wrench	8	1.000	4.41	3.67	0.07	0	yes
random sweep	nozzle	8	1.000	4.34	5.42	0.00	0	yes
random sweep	punch	8	1.000	4.40	5.77	0.04	0	yes

Table 34: E14 legal abuse through the backend, each command on both arms in a wall and a bench task. All as expected: yes.

command (each arm, wall and bench)	runs	ran	refused	reflex stops	why refused	object speed (m/s)	faults
all joints to upper limits	4	4	0	5	–	0.000	0
all joints to lower limits	4	4	0	22	–	0.000	0
back to ready	4	4	0	0	–	0.000	0
unreachable pose 3 m away	4	0	4	0	IK unreachable	0.000	0
wrench 150 N down 3 s	4	4	0	0	–	0.000	0
moment 25 N m 3 s	4	4	0	0	–	0.000	0
wrench beyond the contract (200 N)	4	0	4	0	Wrench beyond the accepted range	0.000	0
grip 70 N on nothing	4	4	0	0	–	0.000	0
grip beyond the contract (80 N)	4	0	4	0	inward_effort_n must be in [0, 70] N	0.000	0
open	4	4	0	0	–	0.000	0
keyboard SHIFT+R 10 s	4	4	0	0	–	0.000	0
keyboard SHIFT+F 10 s	4	4	0	0	–	0.000	0
keyboard SHIFT+W+D+I 10 s	4	4	0	3	–	0.000	0

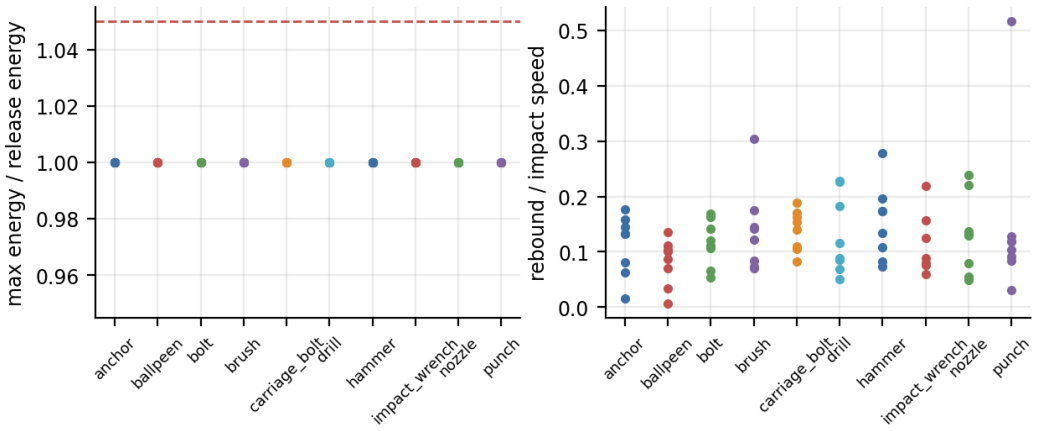


Figure 23: E14: for every drop, the largest energy over the energy at release (the dashed line is 1.05) and the rebound speed over the impact speed.

8.3 Timestep (E16)

The metal-cutting bench, the percussion bench and the bolt pass/jam pair are run at 1 ms and 0.5 ms; outcomes must not depend on the tick (Table 35).

Table 35: E16: the same benches at two ticks. Relative differences: feed 0.0001, torque 0.0005, percussion rate 0.0119; bolt outcomes the same yes.

observable	dt = 1 ms	dt = 0.5 ms
metal feed (mm/rev)	0.09822	0.09821
metal torque (N m)	3.1035	3.1021
percussion rate (mm/s)	4.774	4.831
percussion blows	310	309
bolt at 0.9 mm seated	True	True
bolt at 1.2 mm seated	False	False

8.4 Measured failures behind the design

Running whole tasks with a robot or a phantom operator exercises the world in ways the component benches do not. Each choice below answers a failure measured that way: the failure was reproduced, its cause found and corrected in the world, and the experiments of this report run on the corrected world. Two problems stay open and close the list. The world's fault guard (non-finite state, an object above 20 m/s or 1000 rad/s, MuJoCo's bad-acceleration warning) stops a run in which a body is thrown rather than letting a corrupted state continue. A slowly running mechanism does not trip it, so the rest test (E0) runs on every start state before any demonstration.

8.4.1 Contact and the constraint solve

- **Box–box contacts have no gap.** With a 2 mm contact gap on both boxes, MuJoCo's box–box narrowphase returns two bar–bench contacts 0.47 m and 1.73 m deep besides the four true ones, and the loose steel bar lying flat on the bench is thrown off at 37 m/s. A two-geom model at the captured poses reproduces it with the gap and not without it (E1b); a box dropped on a box rests at the same height with either gap.
- **Round shapes and hole walls keep a 2 mm gap** (`CONTACT_GAP_ROUND_M`). With no gap on a cylinder, the drill press's feed-shaft hub, flush against the head, gets a contact 136 mm deep at the first tick: the shaft spins at 90 rad/s and drives the quill 132 mm down, past its 82.6 mm travel, before any action. With the gap the hub rests with its contacts at 0.0 mm and the quill and shaft do not move. The boxes that form every hole's walls meet only round items, never another box, and keep the gap too: a pair's detection envelope is

the sum of both shapes' gaps, and with the walls at zero a 6 mm bit lunging at a thin sheet's breakthrough meets a wall within one tick and snaps (bending 41.7 N m against its 39.9 N m limit, E6), and a bolt handed into two holes pops out at 1.4 m/s (E10).

- **Thin workpieces have a softer face than the vise jaws.** With the bar's faces at the jaws' near-rigid impedance (0.99/0.999), the bar rocks on the bench under the robot's 100 N punch push at up to 15 rad/s and is thrown off at 22 m/s. A snapshot 90 ticks before, replayed with candidate settings, shows 0.97/0.99 holds it at 0.04 rad/s while light parts still rest within 0.01 mm; jaws and bolt heads keep the hard face.
- **The constraint solve is finished.** With MuJoCo's default line search inside each Newton iteration (20 iterations, tolerance 0.01), an anchor being hammered in leans from 0.6° to 15° in six ticks, and at the seating blow its washer leaves at 35 to 58 m/s. The state one tick before each fault was solved again under one changed setting at a time. With the default search, three wall boxes push a shank standing 0.00 to 0.08 mm inside them with 192, 96 and 108 N, and the washer takes 7.3 kN; with the conjugate-gradient solver, the projected Gauss–Seidel solver, a pyramidal cone, or a friction weight of 1 or 2 in place of 3, the same state gives 6 N and 1 N at the wall and 1.5 to 1.6 kN on the washer. The configured solve returns after one iteration with contact forces its accelerations do not contain, and the no-slip pass turns them into motion. With 200 iterations and 10^{-6} it agrees with the others at both states, with no change to the model, and that is the world's setting. In the station the anchor then advances at most 13.6 mm a blow (its energy allows 17.2 mm), leans at most 2.0°, and seats at 94.6 mm. A bolt handed into two holes 1.0 mm out of line leaves the hand with a one-tick jump of 1.09 m/s on the default search; on the finished solve it moves at 0.26 m/s and seats (Table 23).
- **The loose anchors rest on three rounded points per rail.** At its 10^{-6} m tolerance MuJoCo's convex collision routine places the contacts of a cylinder on a box anew each tick: on a two-rail rest each anchor rocks at 27 Hz with an amplitude of 10^{-5} rad and moves at up to 0.113 mm/s against the rest test's 0.1 mm/s limit. The anchor's own mechanism and the solver's line search are excluded by one run each. A tolerance of 10^{-9} m stops the rocking, but on that setting the hole brush rolls 26 mm at rest, the loose vise moves at 0.21 mm/s, and a hammered anchor seats with 12 kN of tension it has not been given. A rounded point's contact with a cylinder is analytic, and the anchors lie at 0.0001 mm/s in a single run (Table 32 has the experiment's figures). The hole brush has its own two-point rests.
- **The no-slip pass stays, and the match-bolt stack is built to rest under it.** With the bracket and both bolts loose on a part that carries the vise's clamp load, the no-slip pass can sustain an exact five-tick cycle, 4 µm of rocking at 200 Hz, with parts moving at up to 20 mm/s; with the vise opened everything lies still, more iterations of either solver change nothing (the main solve converges in three), and without the pass the motion decays to 0.02 mm/s in 2 s. The pass is what keeps a pinched tool from creeping in the fingers. match-bolt's bracket is a 50 mm angle, as wide as the bar, so the vise grips both and no force from outside the scene holds it, and the stack rests with the vise clamped and with it opened (Table 27).
- **A fastener set down on clamped work lies still.** Without the three settings below, a carriage bolt laid across the strap-clamped boards of timber-bolt shakes at 6 mm/s and slides 13 mm in the first minute and 32.5 mm in 150 s, towards the board's edge, and an M8 bolt on the bar clamped in the vise creeps 1.6 mm a minute. On the bench the same parts lie still. Measured on these cases:
  - The main solve is finished: four solvers give the same forces from one saved state, and the bolt's contact rows follow their reference accelerations. With the no-slip pass the head's acceleration is 12.4 m/s^2 against a reference of 6.9.
  - The bolt's own contact is not the lever. With its contact softened the bolt hops more; with the clamped parts made too heavy to answer within a tick it lies at 0.01 mm/s.
  - The cause is in the support. MuJoCo's friction holds a load only against a slip acceleration; the no-slip pass then removes the slip with the normal forces held fixed. A strap clamp keeps hundreds of newtons of friction locked in under its washer, toe and heel. Solved with and without the second pass from the same state, with nothing lying on the stack: before the pass the top board accelerates at 5.6 m/s^2 and 75 rad/s^2 and the straps at 1,200 rad/s^2; after it, at zero. The stack never shows this in a velocity. A light part on the board gets its normal forces for the accelerating board.
  - A second source: rolling and twisting friction on hardened faces, there so that round parts stop rolling on steel, is slack in the main solve and rigid in the no-slip pass. Inside the clamp the second pass carries moments on those rows that the first has not seen.

– A third, on the part itself: on a hardened face a 137 g bolt lies on a contact 0.02 µm deep, and with the stack quiet it still hops at 12 mm/s in bursts in five of six placements.

Three settings follow, each chosen by the quantity it acts on and not by whether a bolt happens to rest (a dozen settings each passed some placements and failed others, and one passed when switched on in a settled scene and failed when the scene was built with it): friction in the main solve 100 times harder than the normal contact instead of 3, which leaves the second pass less to change (the set-down cases behave alike from 30 to 300); the clamps and the wooden stack at impedance 0.999/0.9999 with no rolling rows on the faces that bear load; and a small loose part's own contact law deciding where it lies on a hardened face. The top board's acceleration before the second pass is then 0.002 m/s^2 and 0.06 rad/s^2, and the stack shortens 0.04 mm under its clamps (0.36 mm with the stiff clamping impedance; a real one about 0.14 mm). The harder friction alone, without the loose parts' own law, makes the centre punch laid on the loose flat bar chatter at 0.19 mm/s. Every start state rests (Table 32). Table 36 lays every loose part of the bench tasks on every workpiece of its scene, in three placements, read on every tick.

Table 36: E22: a loose part set down on a workpiece. One placement 2 mm above the face (seed 0 centred, the others up to 8 mm and 6° away), then nothing acts for 60 s. Read on every tick. Rest rule of E0: from the second second on every 5 s window below 0.1 mm/s for the part and for every other free body, the part within 0.2 mm of where it lay at 2 s. 42 of 42 runs at rest. ASSISTED: the placement; in the vise rows the phantom press operator.

part and what it is laid on	placements (seed)	at rest	part, fastest tick after 2 s (mm/s)	part, moved after 2 s (mm)	any other body, fastest tick (mm/s)
carriage bolt across the clamped boards	0, 1, 2	3 of 3	0.000	0.003	0.000
carriage bolt along the clamped boards	0, 1, 2	3 of 3	0.000	0.003	0.000
carriage bolt pushed through both boards (holes written)	0, 1, 2	3 of 3	0.000	0.000	0.000
carriage bolt on the bench (control)	0, 1, 2	3 of 3	0.000	0.004	0.000
centre punch across the clamped steel sheet	0, 1, 2	3 of 3	0.000	0.004	0.000
centre punch along the clamped steel sheet	0, 1, 2	3 of 3	0.000	0.006	0.000
centre punch on the bench (control)	0, 1, 2	3 of 3	0.000	0.003	0.000
centre punch on the loose flat bar	0, 1, 2	3 of 3	0.000	0.000	0.000
M8 bolt on the loose flat bar	0, 1, 2	3 of 3	0.039	0.000	0.000
M8 bolt on the loose angle bracket	0, 1, 2	3 of 3	0.050	0.001	0.000
M8 bolt on the bar in the open vise	0, 1, 2	3 of 3	0.000	0.000	0.000
M8 bolt on the bar clamped in the vise	0, 1, 2	3 of 3	0.000	0.000	0.000
M8 bolt on the bracket and bar clamped in the vise	0, 1, 2	3 of 3	0.002	0.000	0.000
wedge anchor on the ledger's top edge	0, 1, 2	3 of 3	0.001	0.000	0.040

- **A carriage bolt's end stands on three points.** The seated carriage bolt stands on its end in the backer's hole, its head 18 mm above the top board. As a flat face on the hole's flat floor, a contact whose points MuJoCo's convex collision routine places anew every tick, it moves at 0.56 to 0.62 mm/s for the whole minute at all three placements, over 7 µm, and at 0.24 mm/s after timber-bolt at seed 3; a tighter collision tolerance makes it worse (1.29 mm/s), and without the no-slip pass it moves at 0.55 mm/s. One crown in the middle of the end rests where the bolt stands on the bare backer and rocks in the dimple the auger's point leaves: 1.0 to 1.2 mm/s, decaying, over 0.18 to 0.30 mm, at all six seeds of timber-bolt. A bolt on one point is a pendulum in its hole. The flat end is therefore carried by three points on a circle inside its chamfer, 0.3 mm proud (ledger entry `BOLT_END_FEET`): a flat face stands on three asperities, and three spheres on a flat floor are three contacts the engine computes in closed form. The three points are authored; Table 36 and the timber-bolt holds of Table 27 give the result.

8.4.2 Robot and hand

- **Joint motion stays inside Franka's limits.** Targets follow a time-synchronized jerk-limited trajectory inside Franka's published limits, and the servo and the Cartesian wrench share one cap per joint (Section 5, Table 5). Targets slewed at the velocity limit alone are commanded at thousands of rad/s^2, and a wrench capped separately from the servo lets a wrist joint deliver the sum.
- **Only a body that can be carried is a held load.** The collision reflex subtracts the torques of a held load from each arm's external-torque estimate, as the FR3 does for a load set with `setLoad`. Counting the whole subtree of a touched body's root would make a pinched mechanism a load: the vise is welded to the world with a subtree of 9.315 kg, and with its crank's knob counted as the whole vise, 0.5 N of real knob contact

reads as 17.7 N m on joint 3 and trips the reflex. A load is a body whose root is free: of the press-drill bodies the flat bar counts, the vise and the press's feed shaft (a hinge) do not.

- **A hand holds a tool when its fingers touch any bodies of that tool.** A trigger is a body of its own. Requiring both fingers on the same body leaves the hand that drills with the rotary hammer with no configured load (880 of 880 samples) and charges the other hand the whole 3.23 kg while both carry it.
- **The Cartesian wrench acts at the fingertips.** `cartesian_wrench` feeds $J^\top F$ forward at the TCP, taken from the robot rig's own `tcp` frame, the frame the inverse kinematics checks itself against: 0.2104 m along link 7, 0.0 mm from the IK's TCP. Adding the 0.1034 m flange-to-TCP offset to the origin of link 7, which lies one flange length (0.107 m) behind the flange, would put the TCP 107 mm behind the fingertips, and every force not along the hand's axis would carry an extra moment of $0.107\,F$.
- **The grip applies the commanded force on thin parts.** The force mode's reference runs on past the commanded position by the travel the force needs. A reference that stops at the commanded finger position, which the gripper action sets to the closed end, gives a part of half-width q only 3,500 N/m $\times\ q$: 23.7 N per finger on the centre punch, 21.3 on a carriage bolt, 20.9 on an anchor, 12.1 on an M8 bolt and 8.5 on a 5 mm plate, for 35 N commanded. With the reference run on, all five read 35.0 N, and records carry the force applied.
- **A pinched part shifts in a swinging hand by a known law, and the fingers are tied hard.** Without the two measures below, each of six light parts shifts 0.33 to 0.45 mm in the hand during a 0.4 rad swing of the first joint, whatever its mass (18 to 448 g) and the grip force (8.5 to 35 N), in the same direction on the way out and on the way back, and a hammer carried through the swing moves 0.6 mm with both fingers, which go the same way. Neither is the grip.
  - *The shift.* Three swings of the same angle at three speeds move the centre punch 0.442, 0.118 and 0.039 mm, in proportion to $\int \omega^2\,dt$ (0.574, 0.160 and 0.053 rad^2/s) within 6 %, along the line from the first joint's axis outwards. A hand that turns carries its contact points on an arc; MuJoCo's constraints are written with the Jacobian of the tick's start, so each tick a held part is left with a slip velocity $a_c\,\Delta t$ outwards, a_c the centripetal acceleration. Friction rows then remove a slip at the damping rate of the contact's soft normal law, $b = 2/(d_{\max}\,\tau)$, and the no-slip pass does not change that reference. For the pad's $\tau = 5$ ms, $b = 421$/s, a slip is 42 % removed per tick and the part slides $a_c\,\Delta t\,(1/b - \Delta t/2)$ per tick. The law was checked by changing b alone with the pad's stiffness kept: 0.442 mm at $b = 421$/s, 0.118 mm at 1000/s, 0.106 mm at 1053/s; the three agree with it to 1 % with one common geometric factor (1.24 times the lever arm from the first joint to the grip point). The pads therefore carry MuJoCo's direct form of the same normal stiffness with $b = 1/\Delta t$, the largest rate the engine's own stability rule allows. What is left, $a_c\,\Delta t^2/2$ per tick, is the time stepping: a free body moves along a tangent for a tick and a joint along its arc. On the same swing the punch, the carriage bolt and the bracket shift 0.117, 0.090 and 0.106 mm. It is not zero and cannot be made zero at this tick; it halves with the tick and falls with the square of the hand's angular speed, and a policy that swings a pinched part at the arm's full acceleration should expect it.
  - *The spring.* The hand's description ties the second finger to the first, and the robot import carries that tie into MuJoCo as a joint equality, by default at the engine's default softness (time constant 20 ms, impedance 0.9 to 0.95). There, a 5 N push on the pinched punch along the closing axis moves the punch and both fingers 1.83 mm (6.54 and 6.54 mm become 8.35 and 4.73), and they return when it is removed: 2.7 N/mm, where the real hand has one mechanism. With the tie as hard as the press's rack the same push moves the punch 0.023 mm and 30 N move it 0.133 mm, the pads' own give. A finger that meets a part alone presses with the whole hand's force until the other arrives, as with one motor.
- **The phantom hammer keeps its attitude.** The phantom hand keeps the hammer's attitude through the swing and the blow with a torque and no force (declared with the assistance); this is part of the operator, not the physics. A blow of steel on steel lasts a fraction of a millisecond and the head does not turn in it; at this world's 1 ms tick the contact lasts four to five ticks, and a free hammer, whose centre of mass lies on the handle's side of the strike line, turns 20 to 33° while it touches the anchor's end. Let go 5 mm before contact, in a rig of one anchor through a ledger with the holes in line, it tilts the anchor up to 1.9° in a blow, presses it on the holes' walls with 1 to 3.5 kN and advances it 12 to 21 mm; in the task's own station it knocks the ledger 0.71 mm along the wall while the first anchor is driven, the other holes come out of line, and a later

anchor lying 1.0 to 1.6° askew across the ledger's hole and the wall's is wedged by 1.6 and 3.3 kN sideways from the two holes' walls (ledger-anchor stops one anchor of four 19.9 and 43.2 mm into the wall at seeds 1 and 2). With the attitude held the hammer turns at most 1.4°, the anchor 0.4 to 0.6° once it stands in both holes (1.6° on the first blow, with only the ledger's hole to guide it), the ledger moves 0.08 mm, and a blow advances the anchor about 31 mm against the 300 N the clip drags in those holes.

- **The end-to-end operator centre-punches each mark first,** as the trade does, with the kit's punch at its seeded setting. Piloted on a bare mark at seed 3 of the sheet task, the 3 mm drill walks 0.71 mm before it bites and the enlarged hole comes out oval (roundness 1.108 against 1.10).

8.4.3 Holes, tools and tasks

- **Push buttons stand proud of their bezels by more than their travel.** The caps start 3.3 mm proud for a 3 mm travel, and a steady 15 N push holds the cap at its 3 mm stop and switches the press on and off. With the cap 1.5 mm in front of the head's face, the robot pressing START with its fingertips stops at 1.59 mm of travel where the switch needs 2.1 mm. The switch point is where the button's spring pushes back with its rated 8 N, half the travel: at 0.7 of the travel a steady push needs 9.6 N. A travel audit (Table 37) sweeps every mechanism of every preset through its range and lists every collider it runs into; none is blocked.

- **The hammer's mode selector turns only while its release button is held in,** as on the real tool. Without that lock, a docked arm's link pressing the knob with up to 131 N while the hammer is pushed at the wall turns it against its 0.48 N m/rad detent between holes. Four holes in a row pass with one hammer (E12).

- **A concrete hole widens.** The mouth is the spall crater the hole record describes, the wall wears under the bit's side load, and an SDS shank bends within its limit. With the mouth drawn as a chamfer at the nominal radius and a hole that never widens, a hammer held 0.2 to 0.4° off its axis wedges the bit with 6.7 kN on the hole's walls; with them the same robot run carries 0.55 kN.

- **A bit's bending check follows the bit as a spring.** A real bit is a cantilever spring $k = 3EI/L^3$ between the wall and the mass that moves with the tool, and an impulse J loads it to $J\sqrt{k/m_{\text{eff}}}$. The check follows the wall forces with that time constant, so a sustained wedge load still breaks a bit and a one-tick impact counts for what it is. Compared with the bit's static yield moment instead, the rotary hammer drilling through the ledger into the wall breaks its 13.3 mm bit at 138.8 N m against the body's 136.7, from rare single-tick spikes on an average of 4 N m: the tool rattles sideways in its hole and the rigid contact stops it within one tick (640 N for a millisecond).

- **A cutting bit's walls stand at its radius plus its own wander.** A drill enlarging a pilot through the cover panel into its backer has a hole record in each part, each centred where the pilot seated. Held at its nominal radius about two axes 0.57 mm apart, the bit breaks at 1.7 kN.

- **Each part of a stack is cut by its own law** (Section 4). If the sheet's stage followed the tip's depth once the drill's point is in the plywood backer, four 6 mm holes in 1.5 mm steel would take 0.2 s each, and a 6 mm drill that is not turning would sink through a 3 mm pilot; with its own law they take 1.9 to 2.0 s at 80 N. The evaluator asks a through hole for its diameter, so a 3 mm pilot under a just-started 6 mm stage does not pass.

- **An enlarging drill stays with the pilot until its own stage is through.** It is captured by a pilot hole when its point falls in the pilot's bore, and a cutting bit is never handed to a record whose floor is above its tip. Compared with the hole's largest stage, its own from the tick after capture, a drill 0.49 mm off the pilot's axis would cut the backer as a bare face, and withdrawn it would be handed to the pilot's record, whose floor stands 7.8 mm above its tip: 26 kN of side load.

- **A cut handed from one clamped part to the part beneath starts at the bit's tip only while the bit stands in a hole of another part.** Applied to any tip below any face, the rule gives a bar carried against the drill press's stopped bit a 0.87 mm hole record with nothing removed.

- **A wandering start stays one hole.** A tip that seats again within its own radius of a start it has not yet entered at full diameter goes on with that start. Otherwise, at panel-holes' first mark the 3 mm pilot

meets the backer 0.38 mm off as it breaks through the 1.5 mm sheet, cuts 0.3 mm, leaves that start and seats again 0.62 mm away, and the hole module opens a second record for it. The 6 mm drill then enlarges both, is pushed about between them, and leaves the sheet's hole lobed (roundness 1.12 against the 1.10 allowed). With the rule panel-holes passes all 19 checks at seeds 0, 3 and 777 (roundness at most 1.099; Table 27).

- **A fast item is handed from the mouth to the hole.** An item that moves more than 3 mm in a tick through a hole's mouth, 0.3 mm off the axis, fails the test for a fresh capture and is beyond the rim's reach, and without the hand-over it has no stand-in collider at all: an anchor struck at 4 m/s goes 93 mm into the wall on one blow of 8.6 J against a clip that drags 500 N. The hammer probe's check that no blow does more work on the clip than its energy catches it.
- **The centre punch's support is placed on the nearest face.** On the far face of a 6 mm bar the support under the punch's point appears and vanishes once per cycle and the spring force swings between 3.6 and 97.8 N at a steady 60 N push; on the nearest face it stays at 61 to 62 N. A round part's shank support is placed from its surface, not its axis, and round parts lying on their sides come to rest in about a second.
- **The anchor procedure is the wedge anchor's own.** Its instructions ask for no cleaning in a hammer-drilled hole, only for the embedment (the sequence vacuum twice, brush twice, vacuum twice belongs to an adhesive anchor), and they end with the nut tightened to the installation torque. The anchor has a nut on a thread, a washer, and a cone that the nut draws into the expansion clip; the kit holds an impact wrench with an adaptive-torque module (Section 6.5).
- **An anchor cannot be pushed in without a hammer.** At a clip drag of 150 N a steady 160 N push seats an anchor in 0.4 s, where the approval document requires hammering. The drag is 500 N, still without a published value (Section 3), and the kit hammer is a 32 oz ball-peen. The embedment is judged where the anchor was driven to, before its nut draws the stud out, and as the deepest place the tip has stayed (within 0.05 mm for 20 ms): in the blow that seats the washer the stud goes 2.2 to 2.4 mm past its place for a few milliseconds and springs back (the washer's contact gives 1.2 to 1.5 mm at a 6 kN peak, Table 20), so the deepest point reached would score four correctly driven and set anchors of clean-and-set at 97.05 to 97.21 mm, outside 95 ± 2, while they stand at 94.6 to 94.9 mm. E9 and E20 print that reading beside the depth they measure themselves.
- **Every task can be completed at every seed.** The ledger takes the 7 in anchor the maker's fixture table gives, so the anchors reach their embedment with the washer on the ledger; the marks are laid out clear of the bars by more than the evaluator's own tolerances; rebar-relocate has an admissible site at every seed; and every seed of every task is checked (Section 1.1). The timber-bolt boards lie on a sacrificial plywood backer, as a carpenter backs a through hole: on the bare steel bench a phantom-held drill goes through the top board and stalls with its tip on the bench, and on the backer the same drill goes through both with a backed exit (tear-out factor 1.0).
- **Checks that nothing can fail are not scored.** Where the mechanism could be built it is (a bit burns at excessive speed, Section 6.4). Checks that nothing a policy does can fail (the sheet does not grab, the burr does not reach its limit, the bolt is pushed with less than 50 N while untouched) are reported but taken out of the score, and each task's table row lists goals and constraints separately.
- **Tools weigh what their data sheets say.** Tool masses come from the ledger's data-sheet masses (rotary hammer 3.12 kg with battery, cordless drill 1.68 kg, impact wrench 2.33 kg with the maker's pack and module, more than one arm's 2.27 kg, centre punch 70 g), not from shape densities, and the hammer's recoil pulse and striker from the maker's blow energy and the manual's contact force. Tool figures such as the punch's release force, the drill's stall torque and the wood law's slope are the tools' own (Section 3, Appendix A). Concrete is drilled at a maker's rated volume per minute by diameter (0.198 J/mm^3 at 13.3 mm), and the wood law's datum is the paper's air-dry row, not its average over all moisture contents.
- **Dust needs the vacuum.** A drum vacuum with a rocker switch stands by the kit trolley; a nozzle pass removes dust only while it runs, and dust left in a hole is compacted by an anchor driven onto it.
- **Interfaces.** A checkpoint restores the arm controllers' internal state and a broken bit's geometry; the render worker is given back 600 s after a pause; the runtime experiment goes through the public commands. A refused reset by the session's owner and a batch with a command under the floor do not lock a session; the Panda rigs grip with the same 70 N as the FR3 rigs; an inverse-kinematics solve takes 0.07 s at worst; the command floors are sized from several sessions running side by side (Section 11).

Table 37: Travel audit: every non-robot hinge and slide swept through its range; contacts deeper than 0.1 mm with anything it does not touch at its start position. A joint that ends on its own seat (a clamp's nut on the strap it tightens, the nut of an anchor lying on its rest meeting that rest) is counted apart from a blocked one; the seats are declared in the probe.

preset / task	joints	end on a seat	which seat	blocked	blocked by
bench/press-drill	8	0	none	0	none
bench/take-drill	13	0	none	0	none
bench/timber-bolt	17	4	2 x board_clamp_nut_slide against board_clamp_rail (up to 5.6 mm); 2 x board_clamp_seat against board_clamp_rail (up to 0.5 mm)	0	none
bench/punch-layout	9	0	none	0	none
wall/anchor-holes	6	0	none	0	none
wall/clean-and-set	21	4	4 x anchor_nut_slide against anchor_rest, anchor_rest_point (up to 8.8 mm)	0	none

8.4.4 Open problems

- **A dropped hammer that does not lie down.** Dropped from 1 m in one orientation of eight, the 32 oz kit hammer goes on rolling on its head, about the handle's end, at 3 to 5 cm/s for the whole 10 s (0.85 m), and the sweep fails that drop (Table 33). The head is one plain cylinder without rolling resistance; drawn, it has a ball and a flat-sided eye section. Rolling resistance on the head stops the roll (at 4.9 s) and jams an anchor: the blow of the face on an anchor's end carries the same rows, and in clean-and-set's first hole the eighth blow leaves the anchor 3.5° askew with the clip gripping at 7 kN. A head built as it is drawn (a cylinder from the face, a ball at the peen end, a flat-sided eye section) falls onto its side when dropped on its ball and lies still within 1.2 s in seven drops of eight; in the eighth it stands on its face and the handle's butt and rolls on the face's rim for more than 10 s, and lying on the kit tray it begins to tremble after half a minute (0.9 mm/s, E22's anchor on the ledger). The plain cylinder is kept because its behaviour is known in every scene; what it does wrong when dropped is this item.

- **A hammered anchor is left in tension by its last blow.** The blow that seats the washer leaves the cone caught in the clip and the stud in tension, where a hammered anchor should hold no more than its clip's drag. Measured after hammering alone: 38.2 kN in E20's hammered case (the wrench's module then stops after one blow, the nut has not turned and no installation torque is recorded), 20.6 kN in E9's cleaned hole (the wrench takes 18 blows to its 26.7 kN, against 42 in the dusty hole, where the hammer leaves none); in the tasks the wrench needs between 1 and 48 blows for the same reason, and one ledger anchor of seed 1 stands near 28 kN before the wrench touches it. Followed tick by tick in E20: in the tick in which the washer meets the wall it is already 2.1 mm inside the face under 8 kN, and the cone is caught in that same tick, with 0.96 mm of cone travel booked while the stud is still going in. The probable cause, not confirmed: the washer's give is read from the contact pass at the start of a tick and the stud's position after it, one tick apart, and at the 1 to 2 m/s of a seating blow that is about a millimetre, more than the clip's 0.5 mm play. Over the next fifteen ticks the contact pushes the washer back out to 0.3 mm and draws the tied stud 1.4 mm out with it: 41 kN falling to 38. With the catch held back for 20 ms the washer instead comes to rest 1.66 mm inside the wall's face, pressed on it by 340 N over sixteen contact points and held there by the clip's drag, and catches 28 to 33 kN when the cone does take hold; leaving the thread tie's stretch out of the cone's travel leaves 10.8 kN. Neither is used: behind the booking stands a contact that does not push a part held by friction rows back out of the face it has entered, so removing the catch alone would leave hammered anchors standing 1.7 mm too deep. The washer's give is not counted as cone travel, which covers the washer's seat and not the thread tie. The evaluator's torque check reads only the torque a turning nut was turned against, which is zero after hammering alone, so no task passes without the wrench; its expansion check alone is met by a hammered anchor, and the wrench then has less to do.

8.5 Runtime (E15)

Wall-clock over simulated time of policy commands through the path a policy uses: a `gar run` session, HTTP, the policy key, the action history and the physics trace recording, every camera image downloaded (Table 38). Each configuration is timed at its command floor (21 to 68 ms, Section 9), at 0.1 s and at 0.5 s, with six kinds of command in equal shares at the floor: holds, joint moves, gripper commands, keyboard chords, Cartesian moves of 4 mm and Cartesian moves to poses up to 0.30 m and 0.8 rad from the hand's start pose (the long inverse-kinematics solve; a pose out of reach is refused before anything runs and counted apart); 1,200 commands at the floor without a camera and 300 with each camera count, enough to see a tail of one command in a hundred. The wall station (anchor-holes) carries the camera counts; the bench default (press-drill) and the three start scenes with the costliest physics (timber-bolt, ledger-anchor, clean-and-set) are timed with the default cameras; the other six tasks are not timed one session at a time. Two groups then run several sessions at once, as a benchmark would: six camera-less sessions on six tasks and four sessions with the two default cameras, all streams started together. The table's last rows are the lifecycle cases (a long gap between commands on a live session, and the owner's pause and resume) and the physics alone while cutting. The limit is 200×. The machine was shared with other work during the measurement: the 1-minute load average and the GPU's utilisation are recorded with every command, and the evidence file keeps every command's row.

The slowest part of a command without a camera is not physics. A timer around each stage inside the session server (5,000 commands of 5 ms, not part of E15) puts 84% of the server's time in the shared action history, which commits each command to disk three times; one commit took 33 ms at the median, 0.34 s at the 99th percentile and 1.21 s at worst, in bursts while other jobs wrote to the same disk and the container sat at its CPU quota. The slowest of those commands took 1.69 s, of 400 earlier ones 1.76 s, and of 1,200 later ones 2.02 s. At a 5 ms floor 37 of the 5,000 commands ran over 200×, a tail that ten hold commands per configuration do not show. The floors are sized to it (Section 9); the stall itself is in the shared action history, outside this environment.

With several sessions the stall grows: seven camera-less sessions side by side produced a 2.61 s command and six produced a 2.79 s command (2.70 s without its physics), the slowest commands of all sessions falling in the same moments; the same six-session case had found nothing over 1.01 s forty-five minutes earlier. The floors are sized to the largest of these.

Table 38: E15 runtime through the path a policy uses (`gar run`, HTTP, policy key, recording on, images downloaded). Limit 200×; worst one session at a time 125.2×; commands over the limit 0; within the limit (every timed command, one session at a time and several at once) yes. One session at a time: 3329 commands in 13 sessions, 0 over the limit, 0 over the 150× design ratio, 0 over the cost model's worst case, 75 far targets refused as unreachable (no simulated time). Several sessions at once (10 sessions in 2 groups, load 5.9–21.5): 4003 commands, 0 over the limit, 0 over the design ratio, 0 over the cost model's worst case; slowest command 2.77 s (131.0×). Load average at start 12.7, 14.5, 13 (16 CPUs shared with other jobs).

station/task	policy cameras	floor (ms)	command (ms)	n	physics (s)	rest (s)	median wall/sim	worst
wall/anchor-holes	none	21	21	1177	0.066	0.057	6.2	52.7
wall/anchor-holes	none	21	100	12	0.237	0.101	3.5	6.0
wall/anchor-holes	none	21	500	12	0.951	0.060	2.0	2.6
wall/anchor-holes	1: station_rgb	28	28	293	0.078	1.030	39.9	125.2
wall/anchor-holes	1: station_rgb	28	100	12	0.211	1.003	12.1	16.2
wall/anchor-holes	1: station_rgb	28	500	12	0.971	1.153	4.2	4.6
wall/anchor-holes	2: station_rgb, overview_rgb	35	35	293	0.094	1.304	40.5	107.0
wall/anchor-holes	2: station_rgb, overview_rgb	35	100	12	0.209	1.136	13.6	16.5
wall/anchor-holes	2: station_rgb, overview_rgb	35	500	12	0.982	1.033	4.1	4.5
wall/anchor-holes	3: station_rgb, overview_rgb, right_wrist_rgb	40	40	293	0.104	2.222	58.8	87.8
wall/anchor-holes	3: station_rgb, overview_rgb, right_wrist_rgb	40	100	12	0.217	2.128	23.4	33.3
wall/anchor-holes	3: station_rgb, overview_rgb, right_wrist_rgb	40	500	12	0.958	1.660	5.2	6.8
wall/anchor-holes	4: station_rgb, overview_rgb, left_wrist_rgb, right_wrist_rgb	45	45	293	0.116	1.923	45.4	84.4

station/task	policy cameras	floor (ms)	command (ms)	n	physics (s)	rest (s)	median wall/sim	worst
wall/anchor-holes	4: station_rgb, overview_rgb, left_wrist_rgb, right_wrist_rgb	45	100	12	0.200	1.387	16.1	17.9
wall/anchor-holes	4: station_rgb, overview_rgb, left_wrist_rgb, right_wrist_rgb	45	500	12	1.002	1.925	5.8	6.5
bench/press-drill	2: station_rgb, overview_rgb	35	35	293	0.089	0.995	30.9	61.0
bench/press-drill	2: station_rgb, overview_rgb	35	100	12	0.202	0.731	9.4	10.0
bench/press-drill	2: station_rgb, overview_rgb	35	500	12	0.941	1.129	4.2	5.5
bench/timber-bolt	2: station_rgb, overview_rgb	35	35	117	0.138	0.828	27.5	39.4
wall/ledger-anchor	2: station_rgb, overview_rgb	35	35	116	0.199	0.937	33.3	48.6
wall/clean-and-set	2: station_rgb, overview_rgb	35	35	116	0.193	0.883	31.2	45.9
wall/anchor-holes	1: station_rgb (width 1920, height 1080)	46	46	58	0.121	2.114	48.6	78.4
wall/anchor-holes	4: station_rgb, overview_rgb, left_wrist_rgb, right_wrist_rgb (width 1280, height 720)	68	68	58	0.159	2.754	42.7	61.8
wall/anchor-holes	1: station_rgb (samples 32)	33	33	58	0.094	0.773	26.6	79.9
wall/anchor-holes	1: station_rgb (idle_release_s 5.0)	28	28	12	0.073	0.462	19.5	57.5
6 sessions at once: anchor-holes, timber-bolt, ledger-anchor, press-drill, clean-and-set, match-bolt	none	21	21	3537			10.9–14.5	131.0
4 sessions at once: anchor-holes, timber-bolt, ledger-anchor, press-drill	2 each: station_rgb, overview_rgb	35	35	466			52.1–55.5	79.1
wall/anchor-holes	after 75 s idle, policy key		28	1	0.088		18.6	
wall/anchor-holes	next, policy key		28	1	0.203		24.2	
wall/anchor-holes	after 75 s idle, owner key		10	1	0.035		48.4	
wall/anchor-holes	next, owner key		10	1	0.035		47.3	
wall/anchor-holes	after 75 s idle, policy key		500	1	1.146		3.3	
wall/anchor-holes	next, policy key		500	1	0.997		3.2	
	rotary hammer under a phantom holder at the wall (3 s), trace on		3000	1			11.19	
	the whole assisted press-drill demo (clamp, start, feed, cut through, stop), build included		36940	1			4.09	

Table 39: E15: the two fixed terms of the command floor's cost model (`drilling/runtime_cost.py`) against what this run needed: the slowest floor command's wall time less its cameras and its physics, and the slowest accepted inverse-kinematics solve. The values in the code are the largest of the kept measurements with several sessions side by side (`runtime_cost.py` lists them), rounded up to 0.1 s.

fixed term of the cost model	in the code (s)	needed in this run (s)	covers this run
per-command overhead (`OVERHEAD_S`)	2.7	one session at a time 2.49; several sessions at once 2.58	yes
inverse-kinematics solve (`IK_S`)	0.1	slowest accepted solve alone 0.184; beside six sessions 0.070	**no**

Table 40: E15: the inverse-kinematics solve inside a Cartesian policy command, in process, over targets up to 0.6 m and 1.5 rad from the hand. Solver: `drilling/kinematics.py` (analytic Jacobian); reference: the shared solver (2-point numerical Jacobian), same problem, bounds, tolerances and cap of 300 evaluations. Load 5.2–7.2. Beside the six camera-less sessions (load 6.4–8.2) the same 900 targets took 0.015 / 0.065 / 0.070 s (accepted) and at most 0.073 s (refused).

scene	targets	accepted: solver / reference	accepted by one only: reference / solver	solver, accepted: median / 99th / worst (s)	reference, accepted: median / 99th / worst (s)	same solution within 10^{-6} / 10^{-3} rad	largest joint difference (rad)
dual-fr3 bench/press-drill	300	260 / 260	0 / 0	0.015 / 0.063 / 0.070	0.112 / 0.503 / 0.555	160 / 247	0.0278
dual-fr3 wall/anchor-holes	300	263 / 263	0 / 0	0.016 / 0.065 / 0.184	0.113 / 0.503 / 0.544	174 / 255	0.1087
dual-panda bench/press-drill	300	259 / 259	0 / 0	0.016 / 0.064 / 0.067	0.114 / 0.500 / 0.530	162 / 247	0.1599
all	900	782 / 782	0 / 0	0.015 / 0.064 / 0.184	0.113 / 0.503 / 0.555	496 / 749	0.1599

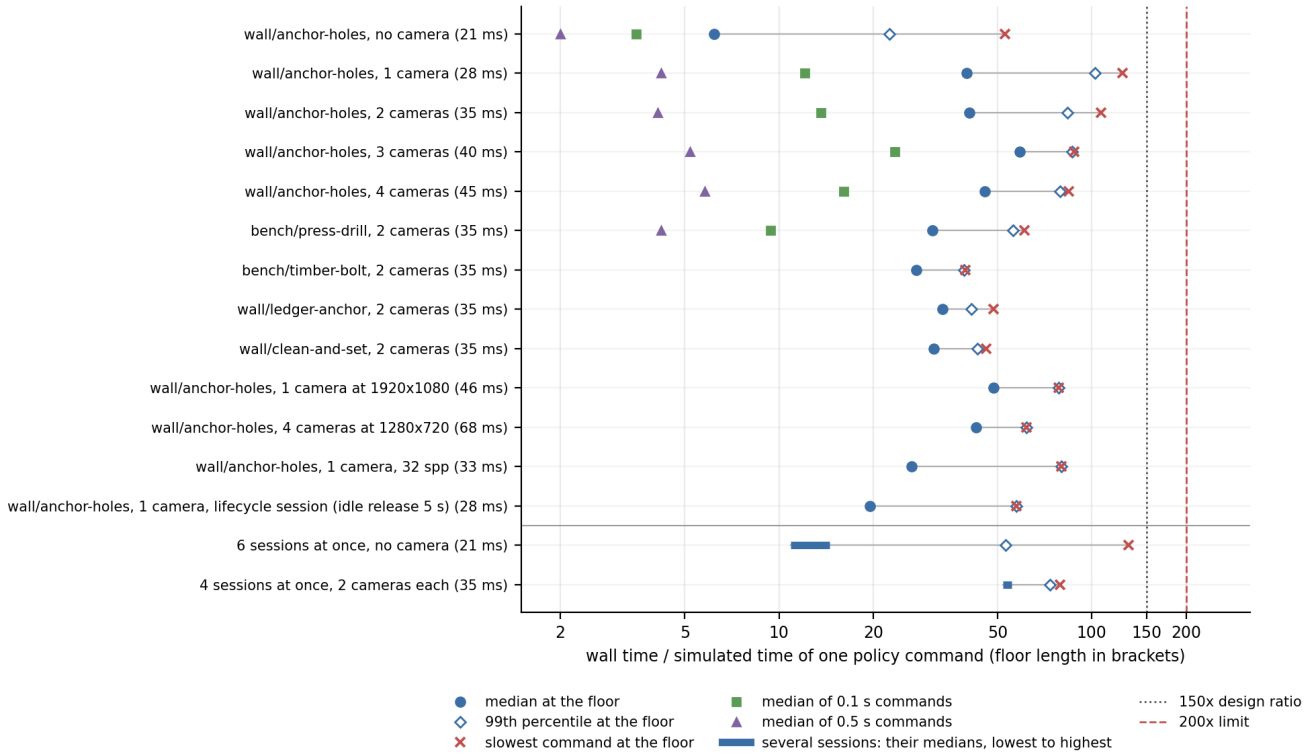


Figure 24: E15: wall time over simulated time of policy commands, one row per timed configuration (floor length in brackets): the median, the 99th percentile and the slowest command at the floor, the medians of the 0.1 s and 0.5 s commands, and below the line the groups of several sessions at once. Logarithmic axis; the 150× design ratio and the 200× limit are marked.

9 Interfaces

The environment is registered with GAR as `drilling` (aliases `drill-hole`, `drill`) with its catalogue `drilling/benchmark.json` (eleven tasks, seeds 0–9999, selectors preset, task and seed) and the backend `gym_anything_robotics/backends/newton_drilling.py`. The unified contract holds. The policy sees `/v1/observation` (joints, protective-stop timers, external joint torques, gripper widths, the policy cameras) and the instruction. The owner also sees `/v1/state`, `/v1/evaluation`, `/v1/geometry`, `/v1/history` and the diagnostic render. Every request, one command or a batch, is checked whole before its first command runs for everything that does not depend on where the arms will be (durations against the floor, arm and joint names, joint limits, finger widths, grip forces, keyboard

chords), and a refused request or a refused reset by the owner (an unknown task, a seed out of range) leaves the session usable. One thing is still decided when its turn comes: the reach of a Cartesian or keyboard-move command that is not the first of its batch. If it is out of reach then, the earlier commands have run and the gateway asks for the owner's recovery; such commands are best sent one per request. A keyboard chord is accepted only as a command to the robot, mapped to joint or Cartesian targets, and never moves anything in the world directly.

A session test starts a real background server with `gar run drilling --task press-drill --port 0 --policy-key`, drives it over HTTP with both keys, opens the dashboard in a browser with each key and ends with `gar stop` (Table 41).

Table 41: The GAR session test, task press-drill: `gar run` started a background server in 42.8 s; every request answered as expected: yes. Keys are read from the credential file and never recorded.

step	key	request	answer	wall (s)
health (no key)	none	GET `/v1/health`	200	0.003
help	privileged	GET `/v1/help`	200	0.001
capabilities	privileged	GET `/v1/capabilities`	200	0.001
task (policy)	policy	GET `/v1/task`	200	0.001
privileged route refused to policy	policy	GET `/v1/state`	403 AccessDenied	0.001
privileged route refused to policy	policy	GET `/v1/evaluation`	403 AccessDenied	0.001
privileged route refused to policy	policy	GET `/v1/geometry`	403 AccessDenied	0.001
privileged route refused to policy	policy	GET `/v1/history`	403 AccessDenied	0.001
observation (policy)	policy	GET `/v1/observation`	200	1.239
camera image (policy)	owner	camera image	200	0.002
named_joint_target 0.5 s (policy)	policy	POST `/v1/action`	200	2.739
out-of-limit target refused (policy)	policy	POST `/v1/action`	400 ActionRejectedError	0.002
shorter than min_command_s refused	policy	POST `/v1/action`	400 ActionRejectedError	0.001
keyboard chord (policy)	policy	POST `/v1/action`	200	3.184
gripper force mode 35 N (policy)	policy	POST `/v1/action`	200	2.605
cartesian_wrench refused to policy (not in the shared allowlist)	policy	POST `/v1/action`	403 AccessDenied	0.002
cartesian_wrench 20 N 0.2 s (owner)	privileged	POST `/v1/action`	200	1.511
evaluation (owner)	privileged	GET `/v1/evaluation`	200	0.003
state (owner)	privileged	GET `/v1/state`	200	0.002
geometry index (owner)	privileged	GET `/v1/geometry`	200	0.001
history (owner)	privileged	GET `/v1/history?after_seq=0&limit=20`	200	0.001
checkpoint save	privileged	POST `/v1/checkpoints/save`	200	0.001
checkpoint list	privileged	GET `/v1/checkpoints`	200	0.002
checkpoint restore	privileged	POST `/v1/checkpoints/restore`	200	1.526
variations	privileged	GET `/v1/variations`	200	0.001
variation select (preview)	privileged	POST `/v1/variations/select`	200	0.001
reset to an unknown task refused (owner)	privileged	POST `/v1/reset`	400 ActionRejectedError	0.044
policy command after that reset attempt	policy	POST `/v1/action`	200	2.211
batch with a command under min_command_s refused whole (policy)	policy	POST `/v1/action`	400 ActionRejectedError	0.001
policy command after that batch	policy	POST `/v1/action`	200	2.075
health afterwards (no key)	none	GET `/v1/health`	200	0.001
pause	privileged	POST `/v1/session/pause`	200	0.001
action while paused refused	privileged	POST `/v1/action`	400 ActionRejectedError	0.001
resume	privileged	POST `/v1/session/resume`	200	20.892
dashboard	owner	dashboard	200	0.002
gar observe	owner	gar observe	exit 0	1.760
gar verify	owner	gar verify	exit 1	0.190
gar status	owner	gar status	exit 0	0.200
reset to the other preset (owner)	privileged	POST `/v1/reset`	200	41.679
policy command after the reset to the other preset	policy	POST `/v1/action`	200	2.254

`gar test --suite contract`: Contract tests PASSED: 99 run, 0 failures, 0 errors.

A warning every session prints. Each world build prints MuJoCo Warp's warning that MULTICCD is enabled but some pairs (capsule, cylinder, mesh) lack multi-contact support, "at most 1 contact will be generated for these pairs". The warning comes from MuJoCo Warp's model conversion, which Newton runs while building the solver. The world then steps on CPU MuJoCo, and that does generate several contacts per pair: a bolt shank against a hole's wall boxes 2–5, the hammer's bit against the wall boxes up to 5, a drill point against its floor plug up to 4 (measured). The warning does not apply to the drilling contacts.

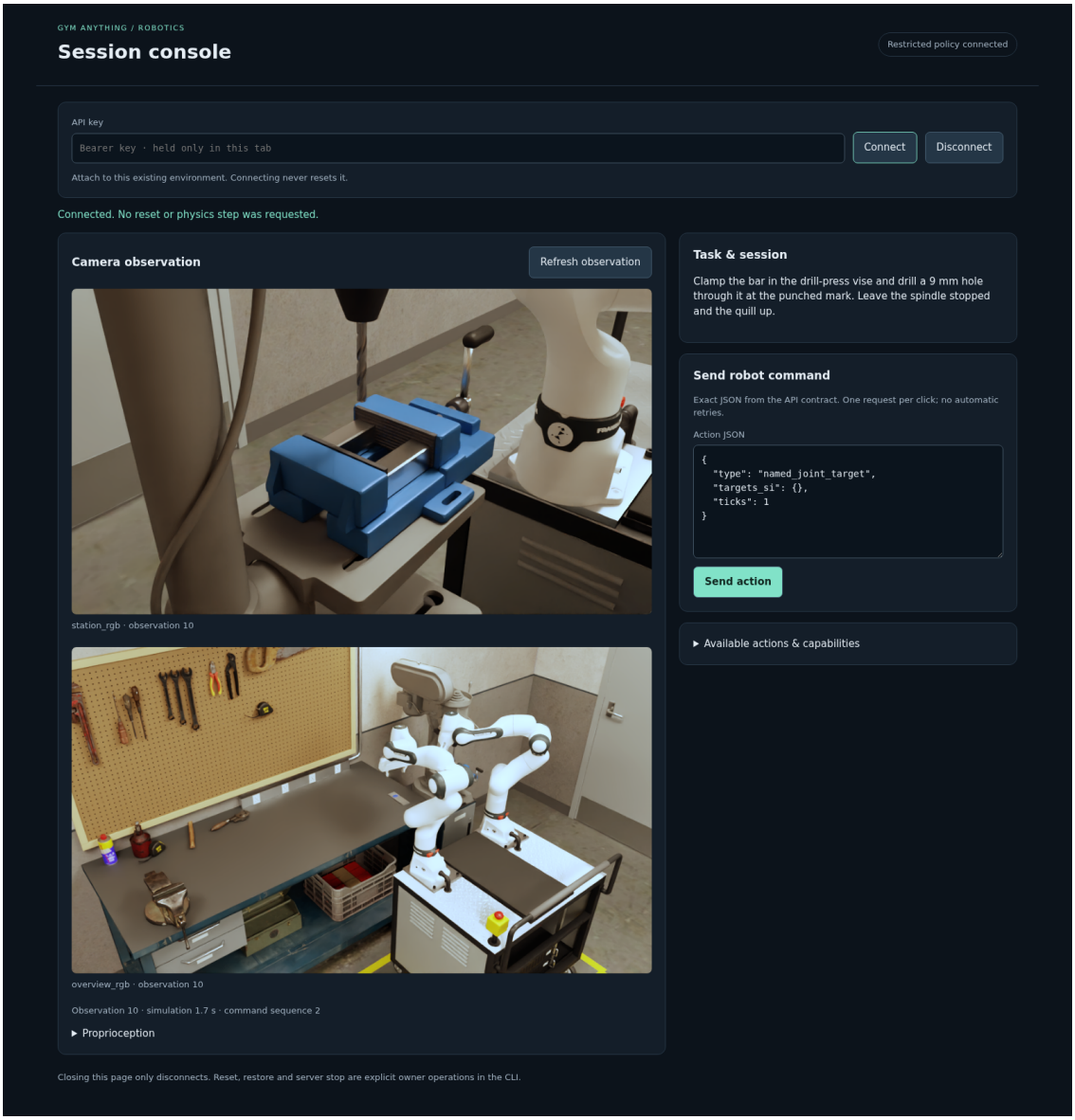


Figure 25: The GAR dashboard with the restricted policy key: the task instruction, the policy camera and no privileged panel.

The wrench and the policy. The backend advertises `cartesian_wrench`. GAR's shared policy allowlist (`policy_access.ROBOT_ACTIONS`) does not contain it, so the policy key is refused it (403) while the owner key may use it. It is part of the arm's desired joint torque, capped with the servo at each joint's rating, so it adds no strength the joint servos lack; it only shapes a push without position targets. The punch, press and timber pushes in Section 7 are shaped through joint targets alone, which the policy has. Carrying needs neither: the pinched tool's weight is the arm's configured load. The wall push in Section 7 uses the wrench, so that demonstration shows what the owner key can do. Whether a force-mode action belongs in the shared allowlist is a decision for the shared contract, not for one environment.

Policy cost model. Each policy command returns its camera observation and is recorded in the session's action history before and after it runs. The capabilities publish `min_command_s`, computed from a measured worst-case model (`drilling/runtime_cost.py`): two fixed terms per command (2.7 s of overhead and 0.1 s for the inverse-kinematics solve), each camera's frame cost by its pixel count and samples (38 ms plus 0.40 µs per pixel at 16 samples, times 3.9 for the GPU contention of other jobs on this machine) and physics at up to 14× (the rotary hammer cutting, measured in process with the container at its CPU quota). The overhead is the slowest non-

camera, non-physics part of a command measured with several sessions running side by side, as a benchmark uses the machine, rounded up to 0.1 s: 2.70 s with six camera-less sessions, 2.57 s with seven, and 0.86 s in the same six-session case forty-five minutes before the 2.70; 1.99 s with one session at a time. It is the action history's three disk commits per command stalling together, in every session at the same moment (Section 8); it is the largest stall seen, not a bound, and it has grown with every sample taken. The solve term is the slowest accepted inverse-kinematics solve over 900 targets up to 0.6 m and 1.5 rad from the hand, 0.071 s rounded up; the shared solver, which this environment does not use, takes 0.55 s on the same targets and 1.79 s on a loaded machine. The floor is the shortest whole-millisecond command whose modelled worst case stays within the design ratio of 150×, a quarter under the 200× limit, so that a stall longer than the longest one measured still fits: 21, 28, 35, 40 and 45 ms for zero to four of the native cameras (35 ms for the default pair, station and overview); 46 ms for one 1920×1080 camera; 68 ms for four at 1280×720; 33 ms for one camera at 32 samples per pixel. At the 21 ms floor a command reaches 200× only when its non-physics part reaches 3.91 s. A policy command of any kind shorter than its floor is refused before anything runs. Table 38 measures every configuration at its floor through `gar run` and HTTP.

Render worker and the exceptions to the guarantee. A session keeps its render worker for as long as it is live and not paused, whatever the time between commands, so no policy command waits for a worker to start; the launch option `idle_release_s` cannot change that and the floor does not depend on it. The price is GPU memory: the worker holds about 4.6 GB with the two default cameras (measured) for as long as the session is live. The owner's pause is the idle signal: a paused session keeps the worker for `idle_release_s` more seconds (600 by default; 0 gives it back at the pause) and then gives the GPU back, and the owner's resume request restarts it and returns only after its first frame (11 to 17 s measured, the longer with another job on the GPU). That wait is the owner's: the first policy command after such a resume ran at 23.1× in E15, the resume itself taking 13.7 s. Outside the guarantee are: commands sent with the owner key, which are not floored; a camera command that arrives while a new worker is still starting after a crash of the worker process or a frame error (the session notices within a second and starts one by itself, so only the 11 to 17 s of that start are exposed: in E15 the worker was killed and the first policy command 30 s later ran at 23.6×); and the wall-over-sim figure that `gar benchmark` prints for a whole run, which includes the owner's resume when the run is started with `--resume` (266× for a two-command run resumed after the paused worker was released, while each of its commands ran at 22×).

10 Limitations

Each item is unfinished work against the real job, not a property of it.

- **The robot does not complete every task.** Under the FR3's own limits (Section 5) the robot takes the drill and drills at the press, and both pass their evaluators. At the wall it drills one anchor hole to specification and does not attempt the other three. It punches the layout but fails on the dimples' offsets and on its wrist reflex short of the release. It bores through the timber, but the drill turns in its one-hand pinch and the hole ends off square. The remaining tasks are shown end to end with phantom hands (Table 27). Whether a policy can do them is the benchmark's question.
- **Nothing catches the work when a drill breaks through metal.** A twist drill breaking through metal lunges and its lips catch the remaining ring: a hand-held drill is wrenched round, and work that is not held down is drawn up the flutes or spun. Here the thrust on a bit is a contact force that can only push the work away, and at breakthrough the cap under the point collapses and the lips cut the remaining ring at the law's thrust, with no lunge and no catch (E6). This holds for every metal through hole, not for thin sheet only. In press-drill and match-bolt it means that clamping matters only against the cutting torque: a bar laid against the vise's fixed jaw as a fence, the vise left open, is drilled through without lifting 0.1 mm and passes every hole check, where a bar clear of the jaw is turned by the 5.9 N m of the cut and fails "bar turned". Nothing here rewards tightening the vise on a bar that already bears on the jaw. Published sources agree on the mechanism: below a critical remaining thickness the material under the lips deforms plastically and no longer carries the cutting force (Segonds et al., 2013 [32]), the drill pushes through leaving two remnants (Schneider, *Cutting Tool Applications* [33]), and the burrs catch the flutes with a rapid rise of torque that can turn the tool in the operator's grasp (US 7,410,006 [34]). The collapse load can be derived from the yield stress without a new parameter. The torque while the lips take the collapsed ring cannot: the cutting law

at a feed of one helix lead gives a jam no motor leaves, a shear estimate gives 1–2 N m, and no published torque or lunge trace for a hand-fed drill in thin sheet was found to decide. The motor's stall and the clutch would bound the jam, at 8.8 or 27 N m, still an order of magnitude above the shear estimate, and which of the two a real sheet follows is the missing measurement. The mechanism is therefore not built, the panel's "did not grab" check stays out of the score, and panel-holes is described as what it tests: placing, piloting and enlarging holes in a sheet that is strapped on a backer, which is how the trade avoids the grab.

- **Checks out of scoring, because the physics behind them is missing or cannot vary here.** The grab above; entry spall depending on push and angle ("spall at most 2 d"); the plywood backer under the boards always backing the exit (timber exit factor); a burr over the 1/16 in limit (the model's largest burr is about 0.5 mm); a bit burning at the press, whose belt step is fixed ("never above 550 °C"); the dimple depth, which follows the strike energy but cannot fail on its own. Each switch and its reason are in `task.py`.
- **Two of the machinist's decisions are in no task.** The press's spindle speed is a start state: changing a belt step is a mechanism that was not built, so no task lets the operator choose a speed and a bit cannot burn at the press. Easing the feed at breakthrough leaves a smaller burr (Table 25), but with no deburring step and no burr limit a shop would apply, nothing rewards it.
- **Bit heat** is the hotter of a lumped node and Cook's tool–chip interface rise. The interface rise has no coolant term, so oil does not save an overspeeding bit, and the burn is an instantaneous threshold where real tempering takes time at temperature.
- **Holes as geometry are proxies.** Contact uses a 12-gon, a rim ring and a 45° cone (a concrete hole's mouth is its spall crater); a bit's lateral compliance in its hole is contact softness, not the bit's bending stiffness; items enter a hole only from its entry face. A part drilled through into another part has one hole record in each, centred where its first bit seated there. A cutting bit's walls follow its own stage's wander, and a bolt through two holes passes or jams at 1.0–1.1 mm against the geometric 1.0 mm (Table 23). The pool of proxies is finite: the ledger has six slots (four anchors and one item in each hand), and a seventh item touching it raises a fault rather than passing through. An item more than 40° off a hole's axis is not captured by it.
- **Impacts resolve at the 1 ms step.** A stop struck at v is penetrated by up to about $v\,\Delta t$ in that tick (a dropped hammer's trigger 2.7 mm past its stop at 4 m/s). Energy is never gained (Table 33).
- **A hammered anchor can be left in tension by its last blow (open defect).** The blow that seats the washer carries it up to 2 mm into the face for a tick, the cone is booked into the clip in that tick, and the washer's way back out loads it: 38.2 kN after hammering alone in E20's hammered case, 20.6 kN in E9's cleaned hole, where the stud should hold no more than its clip's 500 N. The wrench then finds the nut tight. No task passes without the wrench, because the installation torque is read only from a nut that a tool turns (Section 8.4).
- **A phantom hammer needs a phantom hand.** A simulated blow lasts four to five ticks where a real one lasts a fraction of a millisecond, and a hammer left free in that time turns on the anchor's end. The phantom hand that swings the kit hammer therefore keeps its attitude through the blow with a torque; a robot's hand has to do the same with its grip, and the Franka Hand's pinch does not (Section 7.6).
- **A pinched part and the time step.** A part held in a hand that turns moves in it by the time step's own error, $a_c\,\Delta t^2/2$ per tick outwards (a_c the centripetal acceleration): 0.08 to 0.12 mm on a 0.4 rad swing of the first joint at the arm's full acceleration, and as much again on the way back (Table 7, Section 8.4). It is not friction and no grip force changes it; a policy that needs a tenth of a millimetre at the tip of a held part should move slowly or look at the part.
- **Soft contact gives way per point.** A round part between the flat pads sinks deeper when it touches them at fewer points (an M8 bolt's shank 1.13 mm into each pad on three points, 1.70 mm on two, at 35 N), and the count changes when the part shifts: the bolt's place along the closing axis moved 0.27 mm in a swing (Table 7). This is not corrected.
- **Contact devices that are authored, not measured:** the carriage bolt's flat end stands on three points 0.3 mm proud of it; the anchor's driven depth is the deepest place its tip has stayed for 20 ms; the clip's drag (500 N), the kit hammer's swing (5 m/s) and the rigid tie between the hand's fingers have no published values.

- **Open and estimated parameters** (Appendix A).
  - Dust: the residual in the hole and the vacuum pass and brush stroke efficiencies have no published data; they are open, and their effect is measured in E9.
  - Percussion: the striker's momentum is calibrated to one sentence of the maker's manual (the push that keeps the blows); the 2 mm idle-stroke window, the bit's return spring and the electronic clutch's trip level [22] are estimates without a source.
  - Wood: the wood law's form is an MDF fit. Its two slopes are fixed by the one solid-wood measurement found (a 25.4 mm spur machine bit in southern pine); its intercepts, the thrust and torque at zero feed, stay MDF's, transferred to a brad point in Douglas fir without a measurement.
  - Tools: the press's feed pinion is transferred from another press (no figure for the modelled one was found; a smaller pinion would lower every handle force by 35%); the punch's dimensions and force range come from a retailer's listing read at search level.
  - Friction: the friction coefficients are chosen, not measured. Steel on steel (0.2) and steel on other metals (0.3) lie between Machinery's Handbook's oiled range (up to 0.16) and its clean range (from 0.4), inside neither; steel on concrete (0.45) is below a published 0.57 to 0.70. The rolling lever arm (0.1–0.51 mm) is likewise an estimate.
  - Exit burr: the burr fraction of the collapsed cap has a spread of 0.10–0.77 across its nine data points.
- **The robot carries its own weight and its tool outside the torque cap.** Each arm's configured load is the carried tool both its fingers pinch, as an operator sets `setLoad` on each grasp; a policy can neither set nor clear it. Its gravity, and the arm's own, are compensated beside the joint torque cap. That follows Franka's definition of the commanded torque and is an assumption in the robot's favour: a real joint also has a ceiling on its measured torque, with no published value (Section 5). The FR3's velocity limits depend on joint position; constants are used. The desired-torque rate is limited only for the wrench feed-forward, not inside the servo.
- **Not modelled:** noise and hand-arm vibration; chip and dust particles as bodies (drawn only, without a collider, as is the vacuum hose); coolant flow; the bending of the 1.5 mm panel; an anchor's preload relaxing over days, and the fir ledger crushing under the anchor's washer.
- **Anchors through one rigid ledger share its seat.** The ledger is one rigid body on a soft contact, so tightening a later anchor draws the whole ledger in and unloads the earlier ones (Section 6.8); a real ledger crushes locally under each washer. The evaluator therefore reads the torque each nut was set to, not the torque left at the end.
- **Contact settings chosen on an unfinished solve.** The contact impedances of metal faces, hole walls and clamping faces were chosen with MuJoCo's default line search, on which the constraint solve is unfinished (Section 8.4); each passes the experiment it was chosen on with the finished solve, and none was derived again on it.
- **Equal boxes exactly flush.** MuJoCo's box–box collision returns a 29 mm deep contact for one captured pose of two flush boards of equal footprint (a replay of the collision routine alone reproduces it). The timber boards are therefore laid 2–3 mm out of flush; a fault raised by the world would follow such a contact.
- **Runtime** is measured on a machine shared with other work (Table 38), and the command floors are sized to the stalls of that machine's disk and CPU quota: on a quieter machine they are longer than needed, and on a slower disk a stall beyond 3.9 s would put a zero-camera command over 200×. The stall is in the shared action history (three full-sync commits per command), which this environment does not own.

11 Running the environment

The environment ships as the package `drilling` and is started and driven with the `gar` command. The scripts of the experiments and demonstrations are not part of it; the numbers in this report come from their recorded results.

```
gar inspect drilling ; gar doctor drilling ; gar tasks inspect drilling
gar tasks resolve drilling --select preset=bench --select task=press-drill --select seed=3
gar run drilling --task press-drill --port 0 --policy-key
    # then gar status / observe / action / verify / dashboard / stop
gar run drilling --task anchor-holes --robot dual-panda --port 0
gar test --suite contract
```

A Parameter ledger

Every physical parameter is registered with its value, unit, a provenance label and its basis (`params.LEDGER`). Labels: *read* (a datasheet or standard), *literature* (a published measurement), *derived* (computed from read values), *estimate* (an engineering estimate, its basis stated), *open* (a value no source pins down; its effect is reported), *authored* (a task or scene choice), *repo* (validated in another environment of this repository), *engine* (a numerical setting chosen by a measured engine behaviour), *measured* (measured on this environment's own models). The ledger holds 346 parameters (estimate 97, read 84, derived 43, authored 40, engine 29, literature 28, repo 15, open 9, measured 1).

name	value	unit	label	basis
GRAVITY	9.80665	m/s2	literature	standard gravity
DT	0.001	s	authored	the FR3's 1 kHz control rate (barcode_scanning, wine_pouring); E16 compares 0.5 ms
SOLVER_ITERATIONS	50	-	repo	barcode_scanning / wine_pouring Config
SOLVER_IMPRATIO	100	-	engine	how much harder the main solve's friction is than its normal contact (MuJoCo impratio, elliptic cones). MuJoCo's friction carries a load only with a slip acceleration, (1 - d)/d x inverse inertia / impratio x the load, which the no-slip pass then removes with the normal forces held fixed. With 3 the main solve has the clamped timber stack accelerating before that pass (top board 5.8 m/s2 and 79 rad/s2, the straps 1,200 rad/s2: same state solved with and without the pass), so a light part lying on the stack gets normal forces for a support that then does not move, and hops: a carriage bolt on the clamped boards slides 13 mm a minute, an M8 bolt on the bar clamped in the vise creeps 1.6 mm a minute. With 100 the top board's residual is 0.03 m/s2 and 0.7 rad/s2 (together with HOLD_DOWN_BEARING_ROLLS). Chosen inside a measured plateau, not at an edge: the same six set-down cases rest at 30, 50, 100 and 300; at 10 nothing changes and at 1000 the boards creep 0.23 mm in 6 s. A numerical setting with no material meaning: real shear and normal contact stiffness are of one order, and what the larger ratio removes is slip under a held load
NOSLIP_ITERATIONS	100	-	repo	barcode_scanning
NOSLIP_TOLERANCE	1e-10	-	repo	barcode_scanning: the default tolerance leaves grip creep
SOLVER_LS_ITERATIONS	200	-	engine	line-search iterations of MuJoCo's Newton solve. With the engine's default search (20 iterations, tolerance 0.01) the main solve returns after one iteration with contact forces its own accelerations do not contain, and the no-slip pass turns them into motion: a hammered anchor sliding at 1.4 m/s along its hole got 192, 96 and 108 N from three wall boxes it was 0.00-0.08 mm into, where the converged solve gives 6 N, and was thrown 4.6 mm sideways in one tick; its washer meeting the wall got 7.3 kN where the converged solve gives 1.6 kN (the same state solved 22 ways: the converged answer also comes from the CG and PGS solvers, from impratio 1 or 2 and from the pyramidal cone). Only the solve is changed, not the model
SOLVER_LS_TOLERANCE	1e-06	-	engine	line-search tolerance of the Newton solve (the default is 0.01): see SOLVER_LS_ITERATIONS

name	value	unit	label	basis
SOLVER_CCD_TOLERANCE	1e-06	m	engine	tolerance of MuJoCo's convex collision routine, which places the contacts of every cylinder with a box, a cylinder or a hull: the engine's default, kept. Its contact points move by about this much from tick to tick, and a round shank lying across two box rails rocked on them at 27 Hz (four loose anchors in the wall kit, 0.116 mm/s against the rest test's 0.1 mm/s). A tighter tolerance is not used: at 1e-9 m with 200 iterations the anchors lie at 0.0003 mm/s, but a hammered anchor then seats with 12 kN of tension it has not been given (E9 clean), the hole brush rolls 26 mm in its 30 s of rest, and the loose vise moves at 0.21 mm/s (rest test): flat faces meeting through this routine have no unique contact point, and a finer tolerance moves it elsewhere. The rests are shaped instead (stations.py: rounded bearing points, whose contact with a cylinder is analytic)
SOLVER_CCD_ITERATIONS	35	-	engine	iterations allowed to the convex collision routine (the engine's default): see SOLVER_CCD_TOLERANCE
CONTACT_TIMECONST	0.004	s	repo	rigid contact of every non-pad geom (barcode E17)
CONTACT_SOLIMP	0.95, 0.99, 0.001	-	repo	MuJoCo contact impedance (d0, dmax, width) of every geom without its own entry: the engine's numerical softness, not a material property (barcode_scanning's setting)
STOP_SOLIMP	0.99, 0.999, 0.5, 2	-	repo	hard joint stops (barcode_scanning: d0, dmax, midpoint, power; width per joint)
SOLID_SOLIMP	0.97, 0.99, 0.0005	-	engine	metal workpiece faces: with STIFF_SOLIMP a loose 6 mm bar under a 100 N punch push rocks at 15 rad/s on the bench and is thrown off at 22 m/s; 0.97/0.99 holds it (0.04 rad/s) and a 13 g shank or a 0.35 kg block still sinks under 0.01 mm at rest
STIFF_SOLIMP	0.99, 0.999, 0.0005	-	engine	hardened steel clamping faces (vise jaws): with the default a clamped bar sinks 0.7 mm into the jaw under 3.8 kN. A numerical stiffness, chosen as the hardest the 1 ms step holds
WALL_SOLIMP	0.99, 0.999, 0.0005	-	engine	metal hole walls carrying a fastener. TUNED on E10, not grounded in a material property: of three values tried with the chamfer cone, d0 = 0.999 threw a bolt jammed between a wall and a chamfer at 5.7 m/s (a solver jump, 20 N to 0 N in one tick); 0.995 and 0.99 jam 1.1 and 1.2 mm calmly and pass 0.9 mm. The independent observable it is compared with is geometric: two 9 mm holes pass an 8 mm bolt up to 1.0 mm out of line, and E10's pass/jam boundary sits at 1.0-1.1 mm; the 0-0.1 mm band is the contact's softness
PAD_TIMECONST	0.005	s	repo	Franka fingertip rubber pad (label_application, anchored)
PAD_SLIP_RATE_DT	1	1/tick	engine	rate at which a pad's friction rows remove a slip, as a fraction per tick. MuJoCo gives the friction rows of a contact the damping rate of its soft normal contact, b = 2/(dmax x time constant) = 421 /s for the pad's 5 ms, and neither solve nor the no-slip pass knows that a rotating hand's contact points accelerate inwards (the Jacobian's rate of change is not in the constraint). At that rate a part pinched in a swinging hand slides outwards by a_c dt (1/b - dt/2) per tick, a_c the centripetal acceleration, whatever its mass and the grip force: the centre punch 0.442 mm on a 0.4 rad swing of joint 1 and as much again on the way back, 0.118 and 0.039 mm on swings of 1 and 3 s. The pads therefore carry MuJoCo's direct form: the normal stiffness of PAD_TIMECONST unchanged, the rate 1/dt, the largest MuJoCo's own stability rule allows (a time constant of two ticks). The same swing then moves the punch 0.118 mm: the remainder a_c dt^2/2 is the time stepping's (a free body moves along a tangent for a tick, a joint along its arc) and halves with the tick. The pad's normal contact is over-damped by this (ratio 2.4)
FINGER_GEAR_SOLREF	0.002, 1	s, -	engine	the tie between a Franka Hand's two fingers, with FINGER_GEAR_SOLIMP. The hand is one mechanism: Franka's description declares the second finger joint a mimic of the first (franka_description hand.xacro; fr3_franka_hand.urdf in Newton's assets, line 358) and the hand's interface has one width and one force. At MuJoCo's default softness for a joint equality (time constant 20 ms, impedance 0.9-0.95) a pinched part sits on a spring of 2.7 N/mm along the closing axis: 5 N on the pinched centre punch moves it and both fingers 1.83 mm the same way (6.54/6.54 mm become 8.35/4.73), and a hammer carried through a swing moves 0.6 mm with both fingers. The tie is therefore as hard as the drill press's rack and the vise's screw. The force a finger meets alone is then the whole hand's (2 x the per-finger figure), as with one motor

name	value	unit	label	basis
FINGER_GEAR_SOLIMP	0.9999, 0.9999, 0.001, 0.5, 2	-	engine	see FINGER_GEAR_SOLREF
DRIVEN_DWELL	seconds: 0.02; band_m: 5e-05	s, m	authored	how the world records where an anchor was driven to (the evaluator's embedment): the deepest its tip has stayed within band_m for seconds, not the deepest it has been. In the blow that seats the washer the stud goes 2.2-2.4 mm past its place for a few milliseconds and springs back (the washer's contact gives 1.2-1.5 mm at a 6 kN peak, E20's seating table): read as the deepest point, four correctly driven and set anchors of clean-and-set would score 97.05-97.21 mm, outside 95 +- 2, while they stand at 94.6-94.9 mm
BOLT_END_FEET	circle_r_m: 0.0045; foot_r_m: 0.001; proud_m: 0.0003	m	engine	the carriage bolt's flat end is carried by three points: spheres of foot_r_m on a circle of circle_r_m (inside the chamfered face), proud_m proud of it. A bolt dropped through the bored boards stands on its end in the backer's hole, its head 18 mm above the top board. Flat on the hole's flat floor it stands on contact points that MuJoCo's convex collision routine places anew every tick and never comes to rest (0.24 mm/s at seed 3 of timber-bolt after the task, 0.56-0.62 mm/s in E22's seated case); a tighter collision tolerance makes it worse (1.29 mm/s). A flat face stands on three asperities, and three spheres on a flat floor are three contacts computed in closed form (E12 and E22). One crown in the middle of the face rocks in the dimple the auger's point leaves (1.0-1.2 mm/s decaying, 0.18-0.30 mm of travel): a bolt on one point is a pendulum in its hole
PAD_TORSIONAL_M	0.0042	m	repo	wine_pouring: mu L/4 line-contact torsion
CONTACT_GAP_M	0	m	engine	box shapes of the world (not the hole proxies' boxes, which meet only round items): MuJoCo 3.11's box-box narrowphase with a 2 mm gap returned 0.47 m and 1.73 m deep contacts for a bar resting flat on the bench and threw it off at 37 m/s; gap 0 gives the face contacts and the same rest (E1b)
CONTACT_GAP_ROUND_M	0.002	m	engine	cylinders, capsules, spheres, convex hulls and meshes. With a zero gap on these shapes the drill press's feed-shaft hub, flush against the head, gets a 136 mm deep contact from MuJoCo 3.11's convex narrowphase: the shaft spins at 90 rad/s and drives the quill 132 mm down before any action
MAX_OBJECT_SPEED_M_S	20	m/s	repo	fault threshold (barcode_scanning)
MAX_OBJECT_SPIN_RAD_S	1000	rad/s	repo	fault threshold; spindles exempt
FINGER_EFFORT_N	35	N	read	Franka Hand product manual 1.2 section 5.1: grasping (continuous) force adjustable 30-70 N, a gripper-level figure (its travel speed is the one quoted per finger); Franka's own simulation of the hand applies half the commanded grasp force to each finger (franka_ros franka_gripper_sim.cpp)
GRIP_FORCE_BAND_M	0.0005	m	authored	force mode of the hand: a finger stopped at least this far short of its commanded position presses with the whole commanded force, and with a proportional part of it nearer (the hand-over between holding a force and holding a position; world._servo). Franka's grasp call holds its force on whatever stops the fingers and reports success within a 5 mm width tolerance by default (libfranka Gripper::grasp, epsilon_inner / epsilon_outer); 0.5 mm is chosen here so that every part of the tasks (thinnest pinched: the 4.9 mm bracket leg) gets the commanded force
FRANKA_HAND_FINGER_SPEED	0.05	m/s	read	Franka Hand datasheet 50 mm/s per finger
FR3_PAYLOAD_KG	3	kg	read	FR3 datasheet v2.6: 'Rated payload 3 kg', measured at the flange. The datasheet does not say that the hand counts against it; that it does is this environment's reading of a flange rating (the hand is bolted to the flange), not a sentence of Franka's
FRANKA_HAND_KG	0.73	kg	read	Franka Hand product manual: mass about 0.73 kg. With the reading above an arm's rated payload beyond its hand is 2.27 kg. The world does not refuse a heavier load (the real arm does not either: it leaves its specification); it logs one payload_over_rating event per episode and the evaluator fails the episode once an arm has carried more than that for PAYLOAD_OVER_S (world._payload_torques, task.common)
REFLEX_FILTER_S	0.02	s	repo	filtered external torque estimate (wine_pouring)
REFLEX_RECOVERY_S	1	s	repo	targets refused until recovery (barcode_scanning)

name	value	unit	label	basis
ROBOT_VELOCITY_FRACTION	0.95	-	authored	the arm's trajectory generator plans to this fraction of each joint's velocity limit. Planned at the limit itself, a full-speed move ended in the robot's own velocity reflex: the joint servo overshoots the generated velocity by 5 % at the end of its acceleration (2.752 rad/s against 2.62 in E14), and the reflex stands at 1.05 x the limit. Franka's FR3 velocity limits also depend on joint position (https://frankarobotics.github.io/docs/robot_specifications.html); the constants used here are their published maxima
PAYLOAD_OVER_S	0.5	s	authored	an arm counts as having carried more than its rated payload once its carried load has been above the rating for this long in total (the carried-load estimate switches with the fingers' contacts during a hand-over; evaluator constraint, task.py)
FR3_MAX_JOINT_ACC	10, 10, 10, 10, 10, 10, 10	rad/s2	read	FR3 limit table: 10 rad/s2 on every joint
FR3_MAX_JOINT_JERK	5000, 5000, 5000, 5000, 5000, 5000, 5000	rad/s3	read	FR3 limit table: 5000 rad/s3 on every joint
PANDA_MAX_JOINT_ACC	15, 7.5, 10, 12.5, 15, 20, 20	rad/s2	read	Franka Emika Robot (FER, Panda) limit table
PANDA_MAX_JOINT_JERK	7500, 3750, 5000, 6250, 7500, 10000, 10000	rad/s3	read	Franka Emika Robot (FER, Panda) limit table
FRANKA_MAX_TORQUE_RATE	1000	N m/s	read	FR3 and FER limit tables: desired joint torque rate on every joint
IK_MAX_EVALUATIONS	300	-	repo	cap on the least-squares function evaluations of one inverse-kinematics solve, as drilling/kinematics.py has always called the shared solver (max_evaluations=300); a count, so which targets are accepted does not depend on the machine's load
IK_ROTATION_WEIGHT	0.12	m/rad	repo	weight of the rotation-vector error against the position error in the solve's residual (robots/controllers/kinematics.py SerialKinematics.solve residual)
IK_SEED_WEIGHT	1e-05	m/rad	repo	pull towards the seed joints in the same residual: picks one of a 7-joint arm's solutions
WALL_FACE_Y	3	m	authored	north wall interior face
WALL_THICK_M	0.2	m	authored	cast-in-place basement wall
CART_DECK_Z	0.8	m	authored	robot cart deck height
ARM_SPACING_M	0.56	m	authored	arm bases at the cart's front corners
BENCH_TOP_Z	0.9	m	authored	steel workbench top (commercial 36 in)
DRILL_BARE_KG	1.28	kg	read	DCD800 'Weight (without battery pack) 1.28 kg' (https://service.blackanddecker.co.za/i/DEWALT/GLOBALBOM/B1/DCD800N/1/Instruction_Manual/EN/NA408973_DCD800_DCD805_T1_Asia.pdf)
BATTERY_2AH_KG	0.4	kg	read	DCB183/DCB203 2.0 Ah 0.40 kg, same manual (https://service.blackanddecker.co.za/i/DEWALT/GLOBALBOM/B1/DCD800N/1/Instruction_Manual/EN/NA408973_DCD800_DCD805_T1_Asia.pdf)
DRILL_MASS_KG	1.68	kg	derived	DCD800 bare + 2.0 Ah battery: the built drill without its bit (tools.CordlessDrill; printed by E1a)
DRILL_LENGTH_M	0.162	m	read	DCD800 161.8 mm; NOT built: the colliders follow the Drill_01 visual model, 183 mm from the battery's back to the chuck's front (tools.CordlessDrill), open
DRILL_GEAR_RPM	650, 2000	rpm	read	DCD800 no-load speeds gear 1 / gear 2
DRILL_SOFT_TORQUE_NM	27	N m	read	DCD800 'Max. torque (hard/soft) 90/27 Nm' (https://service.blackanddecker.co.za/i/DEWALT/GLOBALBOM/B1/DCD800N/1/Instruction_Manual/EN/NA408973_DCD800_DCD805_T1_Asia.pdf): the soft-joint figure, the torque the drill sustains into a yielding joint; the hard-joint 90 N m is a peak that includes the rotor's kinetic energy
DRILL_STALL_NM	27, 8.78	N m	derived	sustained stall torque at the chuck per gear: the manual's soft-joint torque in gear 1 (27 N m) and the same motor through gear 2's ratio (27 x 650/2000 = 8.78 N m). It decides whether a jammed bit in gear 2 can trip an FR3 wrist's 11.4 N m reflex by itself: 8.78 N m cannot
DRILL_POWER_W	459.458	W	derived	peak mechanical output of the linear motor line M = M_s (1 - w/w0): M_s w0/4 = 459 W. For comparison only (nothing reads it): DeWalt rates the DCD800 'up to 40% more UWO vs. the DCD791B' (https://www.dewalt.com/product/dcd800b/20v-max-xr-brushless-cordless-1-2-in-drill-driver-tool-only), whose output is listed as 460 W (RS 136-2870, search-level)
DRILL_CLUTCH_NM	1.7, 8.5	N m	read	Milwaukee 2903 clutch 15-75 in-lb over 16 settings, applied to this class (transferred)
DRILL_SOFT_START_S	0.2	s	estimate	electronic soft start ramp

name	value	unit	label	basis
DRILL_BRAKE_S	0.15	s	open	brake on release; no stop time published
TRIGGER_TRAVEL_M	0.008	m	estimate	variable-speed switch travel (no datasheet)
TRIGGER_DEADBAND_M	0.001	m	estimate	switch dead band
TRIGGER_PRELOAD_N	4	N	estimate	return spring at rest (5-15 N range, notes §1.6)
TRIGGER_FULL_N	10	N	estimate	return spring at full travel
TRIGGER_ARMATURE_KG	0.1	kg	estimate	reflected inertia of the trigger mechanism (switch slider, contacts, spring) on the trigger axis; sized by the drop sweep E14: a 6 g trigger between a 3 kg housing and its stop was thrown at 27.9 m/s (fault) and 14 mm past its 8 mm stop by 1 m drops; 0.1 kg: none of 32. Costs a finger 0.16 N at a 1.6 m/s^2 squeeze
SELECTOR_ARMATURE_KG	0.1	kg	estimate	reflected inertia of the drill's direction and gear sliders (detent, linkage); E14: a gear slider 9 mm past its stop in a 1 m drop at 0.002 kg
ROTOR_INERTIA_CHUCK	0.0002	kg m2	estimate	rotor reflected through gear 1 to the chuck
HAMMER_BARE_KG	2.5	kg	read	DCH273 technical data 'Weight (without battery pack) 2.5 kg' (https://service.blackanddecker.com.au/i/DEWALT/GLOBALBOM/XE/DCH273/2/Instruction_Manual/ZH/DCH273_T2_ASIA.pdf); the built tool without its bit (housing, chuck, trigger, mode knob and button, depth rod) is scaled to it
BATTERY_5AH_KG	0.62	kg	read	DCB184 5.0 Ah 'Weight 0,62 kg', same manual (https://service.blackanddecker.com.au/i/DEWALT/GLOBALBOM/XE/DCH273/2/Instruction_Manual/ZH/DCH273_T2_ASIA.pdf)
HAMMER_MASS_KG	3.12	kg	derived	DCH273 bare + DCB184 5 Ah, the built hammer without its bit (tools.RotaryHammer; printed by E1a)
SDS_BIT_KG_PER_M	0.4	kg/m at 10 mm	derived	KIT (Uhl et al. 2022, https://publikationen.bibliothek.kit.edu/1000151192/149581508) weighed 10 mm SDS-plus bits with 150 mm working length: Bosch SDS plus-7X 91 g, Hilti TE-CX 81 g; per metre of a 215 mm overall bit (Bosch 10/150/215 designation, inferred), scaled by (d/10 mm)^2: the 13.3 x 160 mm bit weighs 0.113 kg
MODE_BUTTON_N	8	N	estimate	mode selector release button force; DCH273 manual: 'Before attempting to rotate the mode selector depress the mode selector button' and 'There are no operable positions in between' (https://service.blackanddecker.com.au/i/DEWALT/GLOBALBOM/XE/DCH273/2/Instruction_Manual/ZH/DCH273_T2_ASIA.pdf); force not published, the push-button value (PRESS_BUTTON_N)
MODE_BUTTON_TRAVEL_M	0.002	m	estimate	mode selector button travel
MODE_UNLOCK_FRACTION	0.75	-	authored	the lock pin leaves its notch only in the last quarter of the button's stroke: the button stands 1 mm proud of the knob's pointer, so a flat body lying on the knob cannot unlock it
MODE_LOCK_PLAY_RAD	0.0349066	rad	estimate	angular play of the selector's lock pin in its notch; outside it the gears are not engaged (no rotation, no blows): 'no operable positions in between'. Without the lock the knob is turned from hammer-drill to rotary by 0.28-0.37 N m of contact from a robot arm link against a 0.48 N m/rad detent spring
HAMMER_BLOW_J	2.1	J	read	DCH273 EPTA 05/2009 single impact energy
HAMMER_BPM	4600	1/min	read	DCH273 0-4600 bpm
HAMMER_RPM	1100	rpm	read	DCH273 0-1100 rpm
HAMMER_ENERGY_EXP	2.4	-	derived	KIT 2026 rig: 0.59 J @24.6 Hz, 1.65 @36.9, 2.95 @50 Hz
HAMMER_PUSH_N	49.0332	N	read	DCH273 manual: 'Apply only a gentle pressure to the tool (approx. 5 kg). Excessive force does not speed up drilling but decreases tool performance' (https://service.blackanddecker.com.au/i/DEWALT/GLOBALBOM/XE/DCH273/2/Instruction_Manual/ZH/DCH273_T2_ASIA.pdf): the push that keeps every blow
IDLE_WINDOW_M	0.002	m	estimate	bit shank must be pushed back within 2 mm of the stop
SDS_PLAY_M	0.012	m	estimate	SDS-plus bit axial play in the chuck
SDS_SPRING_N	5	N	estimate	light bit return spring: its force with the bit fully out
SDS_SPRING_K_N_M	400	N/m	estimate	rate of the bit return spring: 9.8 N with the bit pushed fully back (tools.RotaryHammer)
HAMMER_BOTTOM_N	5	N	engine	read from this simulation, not from a tool: mean force the housing passes to the bit through the chuck's stop in steady drilling at the manual's push: 49.0 N push - 9.7 N return spring - 34.2 N mean recoil = 5.1 N, the same for striker masses of 0.045 to 0.075 kg (an E8 diagnostic: per scheduled blow, the bit's slide, the window and whether the blow landed). Used only to derive STRIKER_MASS_KG

name	value	unit	label	basis
STRIKER_MASS_KG	0.0474716	kg	derived	steady drilling returns each blow's momentum sqrt(2 m_s E_b) to the housing; the push that keeps every blow must cover that mean recoil f_b sqrt(2 m_s E_b) AND the bit return spring at the working position (9.8 N) AND what the bottomed housing passes to the bit (HAMMER_BOTTOM_N). The manual's 49 N then leaves 34.2 N for the recoil: m_s = 0.047 kg at 9.4 m/s. It is the LARGEST striker momentum with which the simulated hammer keeps its blows at the maker's push (a calibration to that one statement; E8's check at that push is therefore not an independent test). Taking the whole 49 N as recoil gives 0.097 kg, with which the hammer lands 63 % of its blows at 49 N and all of them only from 75 N, the delivered recoil capped at 34 N whatever the striker (97 of 153 blows at 0.097 kg, 119 at 0.075, 138 at 0.055, 153 at 0.045). No SDS-plus striker mass found in KIT's hammer-drill papers (Uhl 2022, Zimprich 2026)
HAMMER_POWER_W	400	W	read	DCH273 technical data 'Max power output 400 W' (https://service.blackanddecker.com.au/i/DEWALT/GLOBALBOM/XE/DCH273/2/Instruction_Manual/ZH/DCH273_T2_ASIA.pdf)
HAMMER_STALL_NM	13.8899	N m	derived	rotation stall torque at the chuck, from the linear motor line M = M_s (1 - w/w0) whose peak power M_s w0/4 is the manual's maximum output: 13.9 N m at 1100 rpm (an upper bound, since the hammer mechanism takes part of that output)
HAMMER_CLUTCH_FRACTION	0.9	x stall	estimate	the electronic overload clutch: 'In case of jamming of a drill bit, the drive to the drill spindle is interrupted ... the motor is repeatedly switched on and off for a few seconds ... Release and depress the trigger to re-engage drive' (DCH273 manual, Overload Clutch). How it detects a jam and at what level is not published; here: the motor at or above this share of its stall torque (the spindle slowed to a tenth of its commanded speed)
HAMMER_CLUTCH_S	0.05	s	estimate	time the overload must last before the electronic clutch interrupts the drive (not published)
HAMMER_CLUTCH_NM	12.5009	N m	derived	motor torque at which the electronic clutch counts an overload at full trigger (12.5 N m). The mechanical slip clutches of Milwaukee's corded 1-1/8 in SDS hammers slip at 21.7-48.8 N m (Product Support Bulletin #403, https://documents.milwaukeetool.com/TIY403.pdf:), a level this 400 W tool's motor cannot reach
PRESS_TRAVEL_M	0.0826	m	read	G7943 quill travel 3-1/4 in
PRESS_PINION_R	0.0202	m	open	TRANSFERRED from another press: the JET JDP-17 advertises 127 mm of quill travel per handle revolution (20.2 mm pitch radius). No figure for the Grizzly G7943 was found (its product page and manual could not be opened). If the G7943 covers its 82.6 mm in one turn the radius is 13.1 mm and the handle's leverage over the quill is 16 instead of 10.4: every handle force in this environment's press runs would be 35 % lower and every handle angle 54 % larger. Not changed without a figure for this press
PRESS_HANDLE_R	0.21	m	measured	knob distance from the hub on the drill_press_01 model that draws the press (0.20 estimated before measuring)
PRESS_QUILL_KG	2.5	kg	estimate	quill, spindle and chuck (without the bit); tools.DrillPress builds exactly this (E1a prints it)
PRESS_CHUCK_KG	1.1	kg	estimate	keyed 5/8 in chuck on its arbor, the turning part of PRESS_QUILL_KG (the value built before)
PRESS_SPRING_PRE_NM	1.5	N m	estimate	return spring preload at the feed shaft (open)
PRESS_SPRING_RATE	1	N m/rad	estimate	return spring rate (open)
PRESS_MOTOR_W	560	W	read	3/4 hp motor
PRESS_BREAKDOWN	2.25	x rated	literature	induction motor breakdown 2-2.5 x rated
VISE_PITCH_M	0.0025	m	estimate	drill-press vise screw pitch
VISE_THREAD_MU	0.15	-	literature	Shigley Table 8-5 (threaded pairs; reproduced at https://amesweb.info/Screws/screw-nut-collar-friction-table.aspx): steel screw in a steel or cast-iron nut 0.15-0.25 dry, 0.11-0.17 with machine oil; a vise screw is oiled: 0.15 (the jaw plates' friction is VISE_JAW_MU)
VISE_COLLAR_MU	0.12	-	literature	thrust-collar friction, Shigley Table 8-6 (same reproduction): soft steel on cast iron 0.12 running (0.17 starting); the screw's shoulder bears on the cast-iron body end without a thrust bearing
VISE_COLLAR_R_M	0.0095625	m	derived	mean radius of the collar face: from the screw's major radius (VISE_SCREW_RM + pitch/4) to the 12 mm screw hub (tools.Vise); with the collar the vise gives 396 N of clamp per N m at the crank (724 N per N m without it; a Kurt D688 with a roller thrust bearing: 1,129 lb at 10 ft lb = 370 N per N m, 7,968 lb at 80 ft lb = 327 N per N m, https://ctemag.com/node/17211)

name	value	unit	label	basis
HOLD_DOWN_STRAP_M	0.1016, 0.0254, 0.015875	m	read	strap length, width, height: TE-CO plain clamp 30402 (https://catalog.te-co.com/viewitems/machined-clamps/plain-clamps): total length 4.000 in, width 1.000 in, height 0.625 in, slot length 1.438 in, for 5/16 and 3/8 in studs; 1018 steel, case hardened, black oxide, tapered nose
HOLD_DOWN_SLOT_LEN_M	0.0365252	m	read	stud slot length: TE-CO plain clamp 30402 (https://catalog.te-co.com/viewitems/machined-clamps/plain-clamps): total length 4.000 in, width 1.000 in, height 0.625 in, slot length 1.438 in, for 5/16 and 3/8 in studs; 1018 steel, case hardened, black oxide, tapered nose
HOLD_DOWN_SLOT_W_M	0.0111125	m	estimate	stud slot width 7/16 in, a clearance slot for the 3/8 in stud (not in TE-CO's table); the slot is taken centred along the strap (not in the table either)
HOLD_DOWN_STUD_D_M	0.009525	m	read	3/8-16 clamping stud: Carr Lane, Standard Strap Clamp Forces (https://www.carrlane.com/engineering-resources/technical-design-information/clamp-strap): 3/8-16 stud, recommended torque 16 ft lb, clamping force 1300 lb, tensile force in stud 2600 lb; 'based on a clean, dry clamping stud that is torqued to approximately 33% of its 100,000 psi yield strength (2:1 lever ratio)'
HOLD_DOWN_STUD_AT_M2	4.99999e-05	m2	literature	tensile stress area of a 3/8-16 UNC thread, 0.0775 in2 (ASME B1.1; handbook value, not re-opened)
HOLD_DOWN_NUT_M	0.022225, 0.0127, 0.003175, 0.0174752	m	read	spherical flange nut: flange diameter, height, flange thickness, width across flats: TE-CO spherical flange nut 41803, 3/8-16, 12L14 case hardened, black oxide, 0.875 x 0.500 in, 0.688 in across flats (https://catalog.te-co.com/viewitems/nuts-washers/spherical-flange-nuts); the 0.125 in flange thickness is that of TE-CO's plain flange nut 41603 of the same envelope (https://catalog.te-co.com/viewitems/nuts-washers/flange-nuts)
HOLD_DOWN_WASHER_M	0.022225, 0.0039624, 0.03175	m	read	female spherical washer under the nut: outside diameter, thickness, seat radius: TE-CO 42742 (inside diameter 0.438 in, 12L14 case hardened, black oxide; 'compensates for slight misalignment between workpiece and strap clamp', https://catalog.te-co.com/viewitems/spherical/spherical-washers-steel-female)
HOLD_DOWN_SEAT_LIMIT_RAD	0.0523599	rad	estimate	largest tilt the spherical seat takes up (spherical washer pairs are made for about 3 degrees, DIN 6319; not re-opened)
HOLD_DOWN_REC_TORQUE_NM	21.6931	N m	read	the maker's recommended nut torque, 16 ft lb: Carr Lane, Standard Strap Clamp Forces (https://www.carrlane.com/engineering-resources/technical-design-information/clamp-strap): 3/8-16 stud, recommended torque 16 ft lb, clamping force 1300 lb, tensile force in stud 2600 lb; 'based on a clean, dry clamping stud that is torqued to approximately 33% of its 100,000 psi yield strength (2:1 lever ratio)'
HOLD_DOWN_REC_TENSION_N	11565.4	N	read	stud tension at that torque, 2600 lb: Carr Lane, Standard Strap Clamp Forces (https://www.carrlane.com/engineering-resources/technical-design-information/clamp-strap): 3/8-16 stud, recommended torque 16 ft lb, clamping force 1300 lb, tensile force in stud 2600 lb; 'based on a clean, dry clamping stud that is torqued to approximately 33% of its 100,000 psi yield strength (2:1 lever ratio)'
HOLD_DOWN_NUT_FACTOR	0.196923	-	derived	torque factor K = T/(F d) of Carr Lane's own table row (0.197: a clean, dry stud)
HOLD_DOWN_TORQUE_NM	10.8465	N m	authored	start state: each clamp nut was tightened to 8 ft lb, half the maker's recommended torque (about 55 N on a 200 mm wrench). Chosen below the recommended value because of the work: at 16 ft lb the 1 in wide toe, reaching 22 mm over the board, would bear on Douglas fir with 10 MPa on average, at 8 ft lb with 5.0 MPa, under the 800 lbf/in2 (5.5 MPa) compression strength perpendicular to grain of coast Douglas-fir at 12 % moisture (Wood Handbook Table 5-3b, as reproduced at https://engineersedge.com/civil_engineering/wood_engineering_properties_2_13744.htm). The torque itself is this environment's choice
HOLD_DOWN_PRELOAD_N	5782.69	N	derived	stud tension of a tightened clamp, F = T/(K d) = 5783 N (1300 lb); the strap shares it between the work and the heel block by its lever arms (HOLD_DOWN_HEEL_M: about 2.8 kN on the work per clamp; E21 measures the share)
HOLD_DOWN_E_PA	2e+11	Pa	literature	Young's modulus of carbon steel (stud and strap)
HOLD_DOWN_TOE_OVERLAP_M	0.022	m	authored	how far the strap's toe reaches over the work's end (start-state placement). The strap lies nose-down by a fraction of a degree, so it bears on the work with its toe's tip

name	value	unit	label	basis
HOLD_DOWN_STUD_OFFSET_M	0.03	m	authored	stud axis beyond the clamped part's end: as close to the work as the plywood backer's 15-20 mm overhang lets it stand (5 mm clear of the backer's edge in the timber stack)
HOLD_DOWN_HEEL_M	0.025, 0.038, 0.008	m	authored	heel block: length along the strap, width, and how much of it the strap's heel end covers. Its height is the stack's (a steel block of the work's height). The strap bears 49 mm in front of the stud (toe) and 45.6 mm behind it (heel): the lever puts 45.6/94.6 = 0.48 of the stud tension on the work (Carr Lane's table assumes 0.5)
HOLD_DOWN_BEARING_M	0.003, 0.0005	m	engine	the strap's two bearing lines (toe on the work, heel on the block) are rounded: two contact spheres of 3 mm radius each, 0.5 mm proud of the strap's flat underside (the flat boxes stand back by that much). A sphere on a face has that face's normal whatever the strap's tilt. With flat box faces as the bearing surfaces the strap, tilted 1.2 degrees by this engine's soft stack, touches with its edges, and MuJoCo's box-box contact normal jumps between the two bodies' face normals
LOOSE_SOLMIX	1000	-	engine	weight of a small loose part's own contact law where it meets a hardened face (MuJoCo solmix, at equal contact priority; hardened faces weigh 1): the softer of two bodies decides how a contact gives, as two springs in series do. The parts: the heads of the M8 and carriage bolts and the centre punch's body (scene.loose_shapes). With the hardened face's law for everything on it, a part of 18-137 g lies on a contact 0.02-1 micrometre deep, which one tick's error lifts it off: on the clamped boards with the stack itself quiet (other bodies below 0.012 mm/s) a carriage bolt hopped at 12 mm/s in bursts in 5 of 6 placements, and with its own law in none of 5. Friction is the larger of the two coefficients and the rolling resistance is the part's, as MuJoCo mixes equal priorities
HOLD_DOWN_SOLIMP	0.999, 0.9999, 0.0005	-	engine	contact impedance of the hold-down clamps (heel block, washer, strap toe, tail, rails, bearing spheres) and of the wooden parts of a clamped stack. MuJoCo's friction holds a load only with a slip acceleration proportional to 1 - impedance (SOLVER_IMPRATIO), and a strap clamp keeps hundreds of newtons of friction locked in under its washer, toe and heel: with STIFF_SOLIMP (0.99, 0.999, 0.0005) the main solve has the top board of the timber stack accelerating at 0.55 m/s2 and 13.9 rad/s2 before the no-slip pass with nothing on it, and a carriage bolt laid on it hops at 11 mm/s. With this value the board's residual is 0.002 m/s2 and 0.06 rad/s2 and the bolt lies at 0.02 mm/s (the set-down experiment). The stack shortens 0.04 mm under the two clamps, where a real one shortens about 0.14 mm and STIFF_SOLIMP gives 0.36 mm (HOLD_DOWN_STACK_STIFF). A numerical stiffness with no material meaning
HOLD_DOWN_BEARING_ROLLS	False	-	engine	whether the load-bearing faces of a strap clamp (toe, heel block, tail, the four bearing spheres) and the wooden parts of a clamped stack carry rolling and twisting resistance (MuJoCo condim 6, ROLLING_B_M). Hardened faces carry it so that a round part stops rolling on a metal face. On these faces nothing rolls, and the rows do harm: the main solve leaves them slack and the no-slip pass makes them rigid, so after that pass the clamp's balance leans on moments the main solve has not seen (the straps 465 rad/s2 and the top board 21 rad/s2 short of balance at impratio 100, 7 and 0.7 without the rows). The bearing faces also outrank the work (contact priority 2), so a toe on a steel sheet has none either. The washer on the strap's rails keeps its rows: without them its seat creeps 1.2 micro-rad a second and the clamp twitches every 0.32 s at 0.04 mm/s. Metal work faces keep theirs for the round parts that lie on them (centre punch, anchors)

name	value	unit	label	basis
HOLD_DOWN_STACK_STIFF	True	-	engine	the wooden parts of a clamped stack (plywood backer, boards) take the stiff contact of clamping faces (STIFF_SOLIMP, 2-tick time constant, contact priority 1: scene.stiff_shapes) instead of the default contact (clamps.stiffen_stack). Under the two clamps' 5.6 kN the stack's faces stand in each other by, summed over the stack: timber stack 2.26 mm with the default contact (0.77 + 0.81 + 0.68), 0.36 mm with this one (0.12 + 0.14 + 0.10); panel stack 0.96 mm (0.30 + 0.66) and 0.32 mm (0.24 between the steel sheet, which keeps SOLID_SOLIMP, and the backer, + 0.08). A real stack shortens under the toes by about 0.14 mm (timber) and 0.08 mm (panel): hand estimate from WOOD_ET, 2.78 kN per toe spreading at 45 degrees from a 25 x 6 mm bearing strip (0.10-0.18 and 0.06-0.10 mm for a 12-3 mm strip). Chosen as the nearer of the two; still 2.6 and 4 times the real figure. At rest for 30 s both settings pass E0's rule (this one: panel 0.0001 mm, timber 0.010 mm from hand-over, at most 0.013 mm/s; default: 0.0001 and 0.0025 mm), and panel-holes and timber-bolt pass end to end with either. With the default contact the timber stack also leaned 0.05 mm in 2 s under half its sideways capacity, and the stud tensions drifted 0.6 % in 30 s; with this one 0.001 mm and 0.01 %. Side effect: these faces' friction rules or ties their contacts (MuJoCo priority): the sheet on its backer 0.45, a clamp toe on a board 0.45. Measured with E21 under either setting, E0's rest check and E12
HOLD_DOWN_NUT_ARMATURE_KG	10	kg	engine	inertia on the nut's slide along the stud (a relative coordinate against the bench: it adds nothing to any moving part). The stud and strap are 3-5e7 N/m stiff; with critical damping on this inertia the 1 ms step integrates that spring stably (h^2 k < 4 m + 2 h c)
HOLD_DOWN_TIGHTEN	gain_per_s: 200; max_speed_m_s: 0.005; tol: 0.002; hold_ticks: 100; limit_s: 2.5	-	authored	start-state preparation (before hand-over only): the nut runs down the stud at up to 5 mm/s until the stud tension stays within 0.2 % of HOLD_DOWN_PRELOAD_N for 0.1 s (at most 2.5 s); from then on the nut's place on the stud is fixed and only the stud's and strap's elasticity holds the load
PUNCH_MASS_KG	0.07	kg	read	mass of the Rennsteig 430 130 automatic centre punch: retailers list 65 g and 78 g (search summary: rapidonline.com 57-7065, toolteam.com, farnell 4133602) and another listing 67 g; 70 g taken. A real punch is a tube round its spring and hammer (built of solid steel bars it would weigh 131 g); the barrel and cap carry the density that gives this total (tools.CentrePunch)
PUNCH_BODY_M	0.125, 0.014, 0.004	m	read	automatic centre punch: overall length, shaft diameter, tip diameter of the Rennsteig 430 130 ('Automatikkoerner 125 mm Schaftquerschnitt 14 mm 60-130 N', listings: https://www.testbericht.de/produkte/rennsteig-werkzeuge-automatischer-koerner-430-130 ; search-level)
PUNCH_RELEASE_RANGE_N	60, 130	N	read	adjustable release force of the Rennsteig 430 130 (same listings; also https://toolteam.com/en-GB/rennsteig-automatic-centre-punch-o-4-mm-60-130-n-430-130-t-25385-4049002002380). The cap's setting is a start state: each task layout draws it from this range with its seed (layout.py)
PUNCH_RELEASE_N	95	N	authored	release force of a punch built without a seeded setting (component probes): the middle of PUNCH_RELEASE_RANGE_N; a scene's punch takes its force from the tool's range and the seed
PUNCH_ENERGY_J	0.1425	J	derived	strike energy at the default setting: the main spring's energy at the release, 1/2 PUNCH_RELEASE_N x PUNCH_RELEASE_TRAVEL_M (3 mm); a punch with a seeded setting stores 1/2 F x of its own (tools.CentrePunch spends exactly this in each strike; the dimple takes the share the part's support does not absorb). The maker's energy and hammer mass are not published (open: 0.08-0.25 J swept)
PUNCH_TIP_HALF_ANGLE	0.785398	rad	read	60-90 deg tip; 90 deg used
PUNCH_TIP_R_M	0.00105	m	engine	radius of the point's rounded contact tip, the rod's own (a capsule). A real 90-degree point has a tip radius of ~0.1-0.3 mm (estimate), but a 0.3 mm tip sphere and a convex hull of the sharp cone are both pushed sideways once the soft contact lets them sink under the 100 N push (the bar thrown at 14.7 m/s, strikes without a dimple; sinking vs radius inferred). On a flat 2.1 mm disc the punch balances and buzzes for ever; a centre of mass 70 mm above a 1.05 mm rounded tip is unstable

name	value	unit	label	basis
PUNCH_POINT_DAMPING	1	x critical	estimate	damping of the point's spring for the punch body bouncing on it (at 0.26 of critical the body, pushed on a mark, chatters on the point at 0.2-0.4 m/s and the punch fires at a 50 N push); the internal hammer and point slide in oiled bores, no measured value
PUNCH_POINT_FRICTION_N	2	N	estimate	dry friction of the point and the internal hammer sliding in their bores (2 % of the release force, below what shifts the release): it damps the cocked point's spring, which otherwise kept a resting punch buzzing at 2-7 cm/s on its point
MH_A	1.085	-	read	torque factor A of the c/d = 0.18 row of MH Table 7 (Chisel Edge Factors), the row MH prescribes for drills of standard design; Machinery's Handbook 31st ed. (2020) pp. 1151-1153, read in the archive.org copy https://archive.org/details/machinerys-handbook-31st-v-1.0
MH_B	1.355	-	read	thrust factor B, MH Table 7 row c/d = 0.18 (with A, J and CHISEL_FRACTION)
MH_J	0.03	-	read	chisel-edge thrust factor J, MH Table 7 row c/d = 0.18 (the c/d = 0.20 row has A 1.105, B 1.380, J 0.040); 9 mm threshold Kd d^2 J = 231 N
KD_STEEL	95	MPa	derived	fitted to Rubenstein's AISI 1020 data (Strenkowski 2004 Table 1)
KD_ALUMINIUM	48.3	MPa	read	MH Table 6 (Work Material Factor Kd, sharp drill): 'Most Aluminum Alloys 7,000' (psi) = 48.3 MPa
CHISEL_FRACTION	0.18	x d	read	MH Table 7: 'For drills of standard design, use c/d = 0.18' (w/d 0.155), the row A, B and J are taken from
POINT_ANGLE	2.05949	rad	read	DIN 338 jobber point
RUB_MU	0.4	-	literature	chisel rubbing friction below the threshold
WEAR_PER_MM3	5e-05	1/mm3	derived	+25 % thrust after ~30 holes 6x6 mm (Subramanian via Audy)
STEEL_YIELD_MPA	250	MPa	literature	S235/1018 class
STEEL_HARDNESS_PA	1.3e+09	Pa	literature	Brinell ~130 HB
ALU_YIELD_MPA	276	MPa	literature	6061-T6
ALU_HARDNESS_PA	9.5e+08	Pa	literature	6061-T6 ~95 HB
COOK_COEF	0.4	-	read	Cook (1973) mean tool-chip interface temperature rise dT = 0.4 (u/rho C) (V t0/K)^(1/3), u specific energy, V cutting speed, t0 uncut chip thickness, rho C and K of the work; as reproduced in Groover, Fundamentals of Modern Manufacturing (lecture copy read: uomustansiriyah.edu.iq lecture 19)
STEEL_RHO_C	0.00382	J/mm3 K	literature	AISI 1018: 7870 kg/m3 x 486 J/kg K (MatWeb)
STEEL_DIFFUSIVITY	1.36e-05	m2/s	derived	AISI 1018 k 51.9 W/m K / rho c (MatWeb)
ALU_RHO_C	0.00242	J/mm3 K	literature	6061-T6: 2700 kg/m3 x 896 J/kg K (MatWeb)
ALU_DIFFUSIVITY	6.9e-05	m2/s	derived	6061-T6 k 167 W/m K / rho c (MatWeb)
WOOD_T0	13	N per 10 mm	read	MDF 10 mm drill thrust intercept (Szwajka 2019 uncoated)
WOOD_TF	66.7	N/(mm/rev) per 10 mm	read	MDF thrust slope
WOOD_M0	0.093	N m per (10 mm)^2	read	MDF torque intercept
WOOD_MF	0.747	N m/(mm/rev) per (10 mm)^2	read	MDF torque slope
KAPPA_WOOD	1	-	estimate	factor on the intercepts of the wood law for Douglas fir. The law's form and intercepts are Szwajka's MDF fit (742 kg/m3, 10 mm carbide twist drill, f 0.10-0.20 mm/rev, speed terms dropped), TRANSFERRED to solid softwood: no measurement of the thrust a brad point needs before it cuts at all was found (open)
WOOD_SOLID_DATUM	d_mm: 25.4; f_mm_rev: 1.02; thrust_n: 667.233; torque_nm: 3.84148	-	literature	the one solid-wood drilling measurement found: McMillin & Woodson 1972, Forest Prod. J. 22(11):55, southern pine SG 0.53, 25.4 mm spur machine bit, 1.02 mm/rev, across the grain, at 10.7 % moisture: 150 lb and 34 in-lb, i.e. 667 N and 3.84 N m (Table 1, read in the paper's scan, https://research.fs.usda.gov/download/treesearch/8240.pdf). The paper's averages over 0 to 80 % moisture are 618 N and 3.73 N m (138 lb, 33 in-lb), not the air-dry row. The same table: thrust peaks at 4.4 % (186 lb) and falls to 100-120 lb from 19 % up
WOOD_SOLID_TF	3.67008	-	derived	factor on the wood law's thrust slope for solid softwood, such that the law reproduces WOOD_SOLID_DATUM's thrust with the MDF intercept kept: 3.67 (3.39 with the all-moisture average); with it a 14.3 mm bit under 40 N feeds 0.061 mm/rev, where it would feed 0.22 with the MDF slope
WOOD_SOLID_MF	0.65941	-	derived	factor on the torque slope, from the same datum's torque: 0.66 (one point fixes one number per law; the intercepts stay transferred)
WOOD_GC	273	J/m2	read	Scots pine cutting toughness (Chuchala 2021) as stand-in
WOOD_ET	7.4e+08	Pa	derived	Douglas-fir E_T = 0.050 x 1.1 x 13.4 GPa (Wood Handbook)
WOOD_NU	0.4	-	read	Poisson ratio for the tear-out criterion

name	value	unit	label	basis
FD_SLOPE	0.006	1/N	derived	Cimperman particleboard: Fd 1.00 @60 N, 1.08 @82 N, 1.34 @117 N
FD_MAX	1.8	-	read	spade-bit exit Fd 1.55-1.82 (MDF)
WOOD_DENSITY	538	kg/m3	derived	Douglas-fir Coast SG 0.48 at 12 % MC
BRAD_SPUR_D_FRAC	0.2	x d	estimate	centre spur diameter of a brad-point bit as a fraction of the bit diameter (about 3 mm on the 9/16 in bit); no maker's drawing found
BRAD_SPUR_PIERCE_M	0.0002	m	estimate	depth at which a brad point's sharp centre spur has pierced the wood and seats the bit (a twist drill seats when its chisel edge is in, 0.87 mm for 14.3 mm)
BRAD_SPUR_LEN_M	0.003	m	estimate	length the centre spur leads the lips by, embedded ahead of them in the wood
WOOD_EMBED_K	0.082	N/mm2 per kg/m3	literature	embedment strength of timber for a dowel, f_h = 0.082 (1 - 0.01 d) rho (EN 1995-1-1 eq. 8.32, d in mm): the lateral load an embedded spur carries before it ploughs through the wood
WOOD_SCORCH_C	200, 300	deg C	read	Wood Handbook: pyrolysis 200-300, char front 300
TIMBER_AXIS_TOL_DEG	1.19349	deg	derived	a through-bolt hole square to the faces: a 1/2 in bolt in its 9/16 in hole through the 3 in (76.2 mm) two-board stack can lean 1.19 deg in the hole, so a hole within 1.19 deg of the face normal lets the bolt stand square to both faces (its head and washer bear flat). AWC NDS 2018 12.1.3.2: holes 1/32 to 1/16 in larger than the bolt, 'accurately aligned', bolts not forcibly driven (https://awc.org/wp-content/uploads/2026/08/AWC_NDS2018-withCommentary_20240204_AWCWebsite_Chapter-12-Dowel-Type-Fasteners.pdf); no published angular tolerance was found
CONCRETE_EV_BY_D	(8, 0.328), (12, 0.204), (16, 0.187)	mm, J/mm3	derived	energy of hammer blows per volume of concrete removed, by bit diameter: from a maker's rated drilling performance in medium-hard concrete, 28, 45 and 49 cm3/min at 8, 12 and 16 mm with 1.8 J blows at 5,100 per minute (Hilti TE 6-S operating instructions, technical data, p. 7 of https://staff.washington.edu/nelsonr/manuals/Hilti%20TE%206-S.pdf, read in the PDF): blow power 153 W over 467, 750 and 817 mm3/s. Interpolated linearly between the three diameters and held constant outside them (materials.concrete_ev): 0.198 J/mm3 at the 13.3 mm anchor bit. The energy per volume falls as the bit grows. Caveats: a brochure figure of another tool of nearly the modelled hammer's blow power (161 W), concrete grade and push not stated. One value for every diameter would drill the 13.3 mm hole at about half the maker's rate
CONCRETE_EV	0.34	J/mm3	derived	KIT's measured rates with a 10 mm bit: interpolated 0.32 (C20/25) - 0.39 (C50/60). Kept as the second, independent source and no longer used by the world: at 10 mm it is 28 % above CONCRETE_EV_BY_D (0.266 J/mm3), which the report states
AGGREGATE_EV_FACTOR	1.3	x	estimate	crushed-stone grains (open, swept 1-1.6)
STEEL_EV	30	J/mm3	estimate	carbide on rebar: effectively no penetration
CARBIDE_CHIP_J	50	J	open	blow energy on steel before the carbide head chips
CONCRETE_ROT_MU	0.45	-	estimate	rotation friction at the bit face
JAM_RATE_NM_PER_DEG	2	N m/deg	estimate	torque rise of an edge caught on a bar
CONCRETE_DENSITY	2350	kg/m3	literature	plain normal-weight concrete
AGGREGATE_FRACTION	0.38	-	estimate	coarse aggregate volume fraction
AGGREGATE_DMAX_M	0.019	m	literature	3/4 in maximum aggregate
REBAR_D_M	0.0127	m	read	#4 bar
REBAR_SPACING_M	0.3	m	authored	near-face mat each way (typical 200-300 mm)
REBAR_COVER_M	0.025	m	authored	near (room) face: 1 in, above ACI 318-19 Table 20.5.1.3.1's 3/4 in minimum for walls not exposed to weather or ground, No. 11 bar and smaller
FAR_MAT_COVER_M	0.0381	m	read	ACI 318-19 Table 20.5.1.3.1, concrete exposed to weather or in contact with ground: No. 5 bar and smaller 1-1/2 in (No. 6 through No. 18: 2 in), as reproduced at https://help.idecad.com/ideCAD/shear-wall-reinforcement-detailing (search-level). The wall's bars are No. 4
REBAR_CLEAR_MARGIN_M	0.002	m	authored	gap kept between the bit's edge and a near-mat bar when a wall mark is laid out after the scan, on top of the bit and bar radii, the 3 mm placement and the 3 deg axis tolerances (layout.rebar_clearance_m: 20.6 mm from a bar's axis on the bare wall, 22.6 mm through a ledger)
DUST_BULKING	1.6	x	estimate	loose drilling dust volume / solid volume

name	value	unit	label	basis
DUST_RESIDUAL_FRACTION	0.29	-	open	share of produced dust the turning flutes hold back (in transit; it settles when the bit stops or withdraws). Set so a 108 mm hole drilled without intermediate withdrawals keeps ~25 mm of dust and ~10 mm with them. No measured residual found. Only bound: Hilti's 2017 KB-TZ letter says dust left in an uncleaned hole 'can reduce the available anchor embedment depth, thus overdrilling of the hole may be required', i.e. it can exceed h0 - h_nom = 13 mm on the 1/2 in anchor; the E9 column after one withdrawal is 86 x this fraction mm (20.2 mm), so >= 0.15 is consistent with that statement. Effect measured (E9): uncleaned 20.2 mm, 18.4 mm after compaction under the anchor, anchor stops at 93.7 of 95 mm; cleaned 2.4 mm, 94.7 mm
VACUUM_PASS_REMOVES	0.85	-	open	loose dust removed per nozzle pass to the bottom, only while the vacuum runs (holes.vacuum_running); no per-pass removal data found. Effect (E9 clean): 4 passes and 2 strokes take a 20.2 mm column to 2.4 mm
BRUSH_STROKE_LOOSENS	0.6	-	open	packed dust loosened per brush stroke; no per-stroke data found (the KB-TZ2 procedure has no brush: ESR-4266 4.2.3)
DUST_COMPACT_RATIO	0.91	-	literature	loose / compacted bulk density of a granular material: 'usually between 0.87 and 0.96' (BS EN 1097-3 loose vs rodded, aggregate lecture notes, Mustansiriyah Univ., https://uomustansiriyah.edu.iq/media/lectures/5/5_2018_12_14!08_37_07_PM.pdf); mid value, transferred to drilling dust: the dust column under a driven anchor shortens to this fraction (holes._insertion_update)
DUST_COMPACT_PA	10000	Pa	estimate	pressure on the dust column above which it is pressed to its compacted density (rodding in BS EN 1097-3 is a light tamping); on a 13.3 mm hole 1.4 N, so a part resting on the dust under its own weight barely compacts it and any push or blow does
SPALL_DIAMETER_FACTOR	1.4	x d	estimate	entry crater (1.8 d with > 40 N lateral)
HSS_E	2.19e+11	Pa	read	Bohler S600
HSS_DENSITY	8100	kg/m3	read	Erasteel E M2
HSS_CP	433	J/kg K	read	Bohler S600
HSS_BEND_STRENGTH	4.7e+09	Pa	read	Erasteel E M2 4-point bend
SDS_BODY_BEND_STRENGTH	1.48e+09	Pa	derived	outer-fibre stress at which an SDS bit's fluted steel body yields (holes.sds_bending_limit): kwb HB2X SDS-plus bits have a 40Cr body hardened to HRC 45-52 (https://www.kwb.eu/en/detail/index/sArticle/3368982, search-level); HRC 45 converts to ~1.48 GPa tensile strength (ASTM A370 / ISO 18265 conversion tables, lower end). The HSS figure (4.7 GPa, 4-point bend of M2) does not apply to a tool-steel SDS body
HSS_SHEAR_ULT	2e+09	Pa	estimate	0.35-0.5 of the bend strength
FLUTE_I_FACTOR	0.4	-	derived	fluted second moment / solid (0.3-0.5)
STEEL_E_PA	2e+11	Pa	literature	Young's modulus of tool steel (a bit's bending stiffness 3 E I / L^3)
BEND_TAU_MAX_S	0.05	s	engine	upper bound of the time constant with which a bit's bending moment follows the wall forces on it (holes._bending_now): where neither the tool nor the workpiece can move (a press spindle over a clamped bar) there is no mass behind an impact and the time constant sqrt(m_eff / k_bit) has no meaning; a sustained load then registers in full within 0.15 s
HEAT_TO_TIP	0.15, 0.4	-	estimate	fraction of cutting power into the bit: metal, dry wood; for metal also the share by which a hot tip body raises the corners' interface temperature (Blok partition, holes._heat)
TIP_H_W_M2K	20	W/m2 K	estimate	still air; x4 with cutting oil
HSS_SOFTEN_C	450	deg C	estimate	wear acceleration onset inside Movchan's reversible band
HSS_BURN_C	550	deg C	read	irreversible loss above ~820 K; Bohler tempers at 550-570
CHIP_DOUBLE_DEPTH_D	3.5	x d	estimate	torque doubles at 3.5 d packed depth (peck rule 3 d)
BIT_HOLE_CLEARANCE_M	3e-05	m	literature	radial running clearance of a drill in its own hole: drilled holes are 0.02-0.24 mm oversize (OSG)
FEED_MAX_D	0.2	x d mm/rev	estimate	largest chip per revolution a twist drill can cut (thin-sheet grabs reach ~1 mm/rev on 6 mm, OSG)
FEED_GRAB_MAX_D	5.4414	mm/rev per mm of d	derived	after the exit cap collapses the lips catch the remaining ring like a thread: the chip per revolution is bounded by the flute's helix lead pi d/tan(30 deg), not by the lips' chip capacity (the lunge and grab of a drill breaking through; with the 0.2 d cap a 6 mm drill through 1.5 mm sheet peaked at 1.0 N m and never grabbed: E6)
CHIP_ONSET_D	2	x d	estimate	packing torque starts beyond 2 d of unretracted depth (chips 'cluster beyond about 3 d', Mellinger)
LIP_ERROR_M	3e-05	m	estimate	lip-height error of a hand-ground jobber
ASYM_GAIN	5	-	open	lateral force from lip asymmetry (2-10)
TIP_MU	0.4	-	literature	spinning tip lateral friction coefficient

name	value	unit	label	basis
KBTZ2	d: 0.0127; h_nom: 0.095; h0: 0.108; torque: 68	m, m, m, N m	read	Hilti KWIK Bolt TZ2 1/2 in carbon steel: stud diameter 1/2 in; nominal embedment 3-3/4 in; hole depth 4-1/4 in; installation torque 50 ft lb (ICC-ES ESR-4266 Table 1 and the Hilti technical supplement's setting table)
ANCHOR_CLIP_N	500	N	open	KB-TZ2 expansion clip drag on insertion in the nominal 13.3 mm hole (the split clip springing against the wall, CLIP_WALL_MU x its ring force); it sets the clip's ring stiffness (parts.WedgeAnchor), so the drag falls in a larger hole and is zero beyond the clip's free diameter. No published value in newtons was found. At 150 N a steady push overcomes it, seating an anchor with 160 N in 0.4 s, where the approval document requires hammering ('The anchor must be hammered into the predrilled hole until hnom is achieved', ICC-ES ESR-4266, 4.2.3), a setting-tool patent says the clip 'is slightly larger than the pre-drilled hole and requires that a significant force be applied' (US 8,424,180) and an industrial arm of 13 kg payload pushed a 12 mm wedge anchor 7 mm in before it stuck (arXiv 2403.19948, IV-B; no force given). 500 N is chosen between two bounds, neither a measurement: above what one FR3 pushes steadily (136-158 N) and what a hand presses on a stud's end, and such that an installer's 2 lb hammer (KIT_HAMMER) drives the anchor in a number of blows a person would accept (E20, E12). A split steel ring of the clip's size pressed 0.4 mm smaller gives 200 to 2,600 N for a band 0.8 to 1.5 mm thick (piston-ring formula, mu 0.6): the order, not a value. It is the model's only resistance to driving the anchor and does not depend on speed. In a hole worn or drilled oversize the drag is smaller in proportion and an anchor can again be pushed in (E9's hole, 13.61 mm: 113 N): such a hole also lets the anchor spin or pull up (E20)
KBTZ2_PITCH_M	0.00195385	m	read	1/2-13 UNC thread: 13 threads per inch, 1.954 mm per turn (ASME B1.1; the KB-TZ2 1/2 in stud carries a 1/2-13 thread and an ASTM A563 Grade A hex nut, ICC-ES ESR-4266 (rev. June 2026), https://icc-es.org/wp-content/uploads/report-directory/ESR-4266.pdf section 3.1)
KBTZ2_D2_M	0.0114309	m	derived	basic pitch diameter of 1/2-13 UNC, d - 0.649519 P = 0.4500 in = 11.43 mm (ASME B1.1 thread form)
KBTZ2_THREAD_LEN_M	G: 0.060325; L: 0.12065	m	read	thread length from the top: 1/2 x 4-1/2 (G) 2-3/8 in, 1/2 x 7 (L) 4-3/4 in (Hilti KB-TZ2 Technical Supplement, Dec 2020, Table 47); the nut cannot run past it
NUT_BEARING_R_M	0.0081375	m	derived	mean radius of the nut's bearing annulus on the washer: from the washer's 13.5 mm hole (0.53 in, estimate) to the nut's 3/4 in across flats (ASME B18.2.2), (D_o + D_i)/4
NUT_FACTOR_K	0.2	-	estimate	torque factor K (T = K F d): the generic handbook value for zinc-plated threads, Shigley's Mechanical Engineering Design Table 8-15 (non-plated black 0.30, zinc-plated 0.20, lubricated 0.18; not re-opened), used as an estimate for the KB-TZ2, whose own factor is not published: its carbon-steel stud is zinc-nickel plated (ESR-4266 3.1) and 'nuts and bolts are finished with a proprietary coating. Only Hilti KB-TZ2 nuts can be used with KB-TZ2 bolts' (Hilti KB-TZ2 Technical Supplement, Dec 2020, p. 3, https://images.thdstatic.com/catalog/pdfImages/a3/a3a0c225-ae97-40b5-92b3-745632435551.pdf). It decides the preload: the thread and bearing friction below are solved from it, so the stud tension at 68 N m is 68/(K d) = 26.8 kN by arithmetic (63 % of the neck's 42.2 kN yield), unconfirmed by any measurement on this anchor
THREAD_MU	0.151294	-	derived	thread (mu_G) and nut-bearing (mu_K) friction of the zinc-plated nut on the stud and washer, taken equal: the value for which the ISO 16047 / VDI 2230 torque split T = F (P/2 pi + 0.577 mu_G d2 + mu_K r_b) gives NUT_FACTOR_K on the 1/2-13 thread; the solved preload at 68 N m is then T/(K d) = 26.8 kN
BEARING_MU	0.151294	-	derived	nut-on-washer bearing friction, equal to THREAD_MU (see there)
CONE_ANGLE_DEG	3.5	deg	literature	taper of the expansion cone (mandrel) against the axis: 'The optimum angle is 3.5 deg', range 2.5-4.5, for a 1/2 in expansion stud with a 1.0 in wedge tip and a 0.5 in sleeve (US Patent 4,052,925, US Expansion Bolt Co., 1977, https://patents.google.com/patent/US4052925); the KB-TZ2's own taper is not published (estimate for it)

name	value	unit	label	basis
CONE_MU	0.16	-	estimate	static friction of the expansion element on the cone (mandrel): the value ASSUMED as an input by the finite-element study of a torque-controlled sleeve anchor (a different anchor) of Noh, Kim and Joo, Trans. Korean Nuclear Society, Spring 2014 (https://www.kns.org/files/pre_paper/31/601 .pdf); not a measurement, and not of a KB-TZ2. The study shows that follow-up expansion needs this to stay below the clip-on-concrete friction. Here it enters only the friction that keeps the stud from turning with the nut (parts.WedgeAnchor)
CLIP_WALL_MU	0.6	-	estimate	static friction of the expansion element on the concrete hole wall: an assumed input of the same finite-element study (Noh, Kim and Joo 2014, as CONE_MU), not a measurement. Here it only converts the clip's drag ANCHOR_CLIP_N into its ring force (which sets the stud's spin friction before the cone grips)
CONE_R_M	0.005425	m	derived	mean radius of the cone under the clip: from the 9.0 mm neck (ESR-4266 effective tensile stress area 63.6 mm2 for the 1/2 in carbon-steel anchor, d = sqrt(4 A/pi)) to the 12.7 mm body
WEDGE_BETA_N_M	2.77725e+07	N/m	read	axial stiffness of the installed 1/2 in KB-TZ2 in the service load range, uncracked concrete, beta_uncr = 158,585 lb/in = 27.8 kN/mm (ICC-ES ESR-4266 (rev. June 2026), https://icc-es.org/wp-content/uploads/report-directory/ESR-4266.pdf, design table, 1/2 in column; cracked 113,515 lb/in). Used as the stiffness of the cone being drawn into the clip (follow-up expansion), i.e. the stud rise per newton of stud tension while setting: no published stud rise during setting was found, so the setting displacement F/beta (about 1 mm at 26.8 kN) is derived, not checked against an independent measurement
WEDGE_STEEL_N	49998	N	read	nominal steel strength in tension of the 1/2 in carbon-steel KB-TZ2, N_sa = 11,240 lb = 50.0 kN (ICC-ES ESR-4266 (rev. June 2026), https://icc-es.org/wp-content/uploads/report-directory/ESR-4266.pdf, Table 4: f_uta 114,000 psi on A_se,N 0.099 in2 = 63.6 mm2; yield f_y 96,300 psi, 42.2 kN on that area). The capacity of the tie that models the expansion (parts.WedgeAnchor): at this stud tension the neck breaks and the anchor has failed for good. With NUT_FACTOR_K it corresponds to 127 N m at the nut, 1.87 x the installation torque. The same table gives no pullout strength for the 1/2 in anchor (N/A: it does not govern) and k_uncr = 24 at h_ef 3-1/4 in for concrete breakout (not modelled: see the class docstring)
KIT_HAMMER	head_kg: 0.907; head_len_m: 0.1; handle_len_m: 0.34; handle_r_m: 0.013; head_x_m: 0.179	kg, m	read	the kit's hand hammer for driving wedge anchors: a 32 oz (2 lb, 907 g) ball-peen, 14-1/2 in (368 mm) overall (Estwing E3-32BP as listed by distributors, search summary: https://www.drapertools.com/product/23578/estwing-e3-32bp-ball-pein-hammer-32oz-907g/). Built as a steel cylinder of that mass 100 mm long (face 38 mm) on a hickory handle; head_x is the head's centre from the handle's middle. The head collides as that plain cylinder (it is drawn with its ball and its eye section): dropped, it can roll on its side for more than 10 s and can stand on the end where its ball is drawn (E14's sweep). Against the clip's 500 N drag (ANCHOR_CLIP_N) a 16 oz hammer (0.45 kg head) would need three times the blows
HAMMER_SWING_M_S	5	m/s	authored	speed the phantom hand gives the kit hammer before a blow on an anchor (E12, E20, E9): a firm one-hand swing, 13 J with the 1.04 kg hammer. An operator's choice, not a property of the world; no measurement of installers' hammer speeds was sought
CLIP_FREE_D_M	0.0137	m	estimate	outside diameter of the unexpanded clip, free: above the largest hole a standard 1/2 in carbide bit drills (ANSI B212.15 tip 0.520-0.530 in, 13.21-13.46 mm), so the clip drags in every standard hole. A hole larger than this leaves the clip loose: the anchor cannot be set (spins or pulls up). Not published; its effect is measured in E20 (oversize holes)
CLIP_LEN_M	0.02	m	estimate	length of the expansion clip along the stud (2-22 mm from the tip)
CLIP_TRAVEL_M	0.0127	m	estimate	travel of the cone into the clip before the clip runs off the taper: the 1.0 in wedge tip less the 0.5 in sleeve of US 4,052,925 (at 3.5 deg, 0.78 mm of radial expansion)
CLIP_PLAY_M	0.0005	m	estimate	axial play of the clip between the collar it is driven by and the start of the taper (ESR-4266 3.1: 'restrained by the mandrel taper and by a collar'); the stud rises this far before the cone meets the clip. Not published

name	value	unit	label	basis
ANCHOR_SET_ON_N	20	N	authored	thread load, and washer load on the fixture, above which the nut is taken to bear (parts.WedgeAnchor: only then can the stud's tension draw the cone into the clip; the evaluator counts the stud's turning as spin only while the nut bears)
NUT_SLIDE_ARMATURE_KG	10	kg	engine	inertia on the nut's slide along the stud (a relative coordinate; parts.WedgeAnchor.bind): it keeps the nut on its thread position through a hammer blow (0.002-0.007 mm of slip; without it 0.03-0.35 mm for a few ticks at every blow, with no fault and no stud tension: E20 seating_blows and real_hammer)
NUT_TURN_ARMATURE	0.0001	kg m2	engine	inertia on the nut's turn relative to the stud (parts.WedgeAnchor.bind), 100 x the nut's own and about that of the anvil and socket that drive it (8e-5 kg m2). Without it the nut's own friction stops the bare nut within a tick whenever the socket's flats lose contact, and in a run-down at 2300 rpm the socket then stands up to 11.3 deg into the nut's hex for one tick (E20 dusty; 4.6 deg with it; module at 2300 rpm). Cost: it reacts on the stud, which turns 2-6 deg in its hole during a setting in a nominal hole, 10-12 deg in E9 and 15 deg in a 13.6 mm hole (0.25 deg without; the evaluator's limit is 45 deg). The wrench's blows do not need it (E20 sound without it at 1500 rpm: 0 corners passed, hex at most 5.9 deg, the set nut 0.06 deg and 12 N back in 2 s)
CLIP_END_BAND_M	2e-05	m	engine	the clip counts as against its collar, or the cone as against the clip, within this distance of that end of the clip's play (parts.WedgeAnchor.pre_step): with 0.1 um the 5 % velocity reversal of the friction row that stops the stud took the stud off the end every other tick and the clip was dragged out 0.3 mm at half its drag; 20 um is 4 % of the play
CLIP_WRONG_WAY_N	1	N	engine	the clip's drag is taken off for a tick when the last tick's drag held the stud with more than this force in the direction that end of the play cannot hold (parts.WedgeAnchor.pre_step); 0.7 % of the drag, above the solver's residual on a resting stud
NUT_AF_M	0.01905	m	literature	1/2-13 hex nut, width across flats 3/4 in (ASME B18.2.2 hex nuts; from memory, not re-read)
NUT_H_M	0.0111	m	literature	1/2-13 hex nut, thickness 7/16 in (ASME B18.2.2; from memory)
WASHER_T_M	0.0032	m	estimate	thickness of the KB-TZ2's washer. The carbon-steel anchor's 'washers meet the requirements of ASTM F844' (Hilti KB-TZ2 Technical Supplement, Dec 2020, p. 3; for the stainless anchors the same page names 'ANSI B18.22.1, Type A, plain'). Neither standard was opened and the supplement does not give the size: a 1/2 in Type A wide plain washer is 0.086-0.132 in thick (from memory)
WASHER_R_M	0.0172	m	estimate	outer radius of the KB-TZ2's washer (an ASTM F844 washer, see WASHER_T_M; size not given by Hilti: a 1/2 in Type A wide plain washer is 1.375 in across, 17.5 mm radius; from memory)
NUT_CHAMFER_H_M	0.00085	m	derived	depth of the hex nut's top chamfer at its corners (parts.anchor: over this depth the nut's top is a hex with its corners on the across-flats circle). ASME B18.2.2 (as summarized by engineersedge.com and torqbolt.com): chamfer circle between the width across flats F and 0.95 F, chamfer length at the corners 5-15 % of the thread diameter (0.64-1.9 mm for 1/2 in); with a 30 degree chamfer from the circle F to the corners G: (21.99 - 19.05)/2 x tan 30 = 0.85 mm
NUT_PREVAILING_NM	0.02	N m	estimate	torque that turns the free (unloaded) nut on the stud and the washer under it, so that a loose anchor's nut turns only under an applied torque; below the clip's spin hold in a hole that fits it (CONE_MU x ring force x CONE_R_M, 0.22 N m in the nominal hole), above it when the clip is loose
NUT_HEX_SOLIMP	0.9999, 0.9999, 0.0005	-	engine	contact impedance of the nut's hex and of the socket's hex walls (parts.WedgeAnchor.bind, tools.ImpactWrench.bind): MuJoCo's stiffest. With the hardened faces' STIFF_SOLIMP the socket stands 2 degrees into the nut's flats at 57 N m (a contact stiffness of about 1,600 N m/rad about the axis, softer than the wrench's own jaw spring of 5,400): each blow loses T^2/2k in it and the wrench stalls at 59 N m after 188 blows (E20 sound)
NUT_TURNING_RAD_S	0.2	rad/s	authored	nut speed on the stud above which the nut counts as being turned on (parts.WedgeAnchor records the friction torque it turns against as the installation torque only then): 2 % of the speed a blow of the wrench turns it at

name	value	unit	label	basis
ANCHOR_SPIN_MAX_DEG	45	deg	authored	the evaluator's limit on the stud turning in its hole while its nut is loaded: an anchor that turns with the nut has spun (the installer's sign of a bad hole); no published threshold
T_INST_TOL	0.1	-	authored	the evaluator accepts an installation torque from 0.9 T_inst (ISO 6789 torque-wrench accuracy is 4-6 %, plus margin); above 1.7 T_inst is over-torque: the setting torque of ACI 355.2's splitting test, 'greater than the smaller of 1.7 T_inst and T_inst + 100 ft-lb' (test 9.3, as quoted by R. Morgan, STRUCTURE magazine, Apr 2017, https://www.structuremag.org/article/how-installation-torque-can-affect-expansion-anchors/; 1.7 T_inst is the smaller for this anchor). For the 1/2 in KB-TZ2 with NUT_FACTOR_K that is 115.6 N m and 45.5 kN, between the neck's yield (42.2 kN) and its strength (WEDGE_STEEL_N, 50.0 kN, 127 N m), where the model's anchor fails
IMPACT_TOOL_KG	1.04	kg	read	Hilti SIW 6AT-A22 technical data: 1.04 kg (tool), 1/2 in square drive, 3 gears 1000/1500/2300 rpm, max torque 90/135/213 N m, up to 3,500 impacts/min, 228 x 154 x 94 mm (hire listing https://www.speedyservices.com/hire/power-tools/02_0288-h-Hilti-SIW-6AT-A22--Impact-Wrench-22v-104kg, Hilti product-page search snippet)
IMPACT_BATTERY_KG	0.83	kg	read	Hilti B 22/5.2 Li-ion pack (22 V, 5.2 Ah), weight 0.830 kg: a distributor's data table for the Hilti item (https://www.oglaend-system.com/product-variants/battery-pack-b22-5-2-li-ion-article39899-52088.html?pid=42260, Hilti's own pages refuse automated reading)
AT_MODULE_KG	0.26	kg	read	the SI-AT-A22 Adaptive Torque module between tool and battery: 0.26 kg (EPTA Procedure 01/2003, without battery; Hilti's product pages for the module as returned by a search: https://www.hilti.co.nz/r4838537 and regional copies; the page itself could not be opened). With these two the wrench weighs 1.04 + 0.83 + 0.26 + 0.20 (socket) = 2.33 kg, more than one FR3 carries beyond its hand (2.27 kg)
SOCKET_KG	0.2	kg	estimate	3/4 in deep impact socket on a 1/2 in drive (63 mm long, 28 mm OD) with the anvil: the turning mass the blows accelerate
SOCKET_CLEARANCE_M	0.00015	m	estimate	clearance of the socket's hex on each flat of the 3/4 in nut (a 19.35 mm opening on 19.05 mm)
SOCKET_LEAD_IN_M	0.001	m	estimate	the 45 degree chamfer at the mouth of the socket's hex (tools.ImpactWrench): the socket finds a nut up to this far off its axis and takes the nut's corners in before the hexes are in line; not published for Hilti's sockets (impact sockets carry a lead-in of about a millimetre)
IMPACT_MAX_RPM	2300	rpm	read	Hilti SIW 6AT-A22 technical data: 1.04 kg (tool), 1/2 in square drive, 3 gears 1000/1500/2300 rpm, max torque 90/135/213 N m, up to 3,500 impacts/min, 228 x 154 x 94 mm (hire listing https://www.speedyservices.com/hire/power-tools/02_0288-h-Hilti-SIW-6AT-A22--Impact-Wrench-22v-104kg, Hilti product-page search snippet)
IMPACT_MAX_IPM	3500	1/min	read	Hilti SIW 6AT-A22 technical data: 1.04 kg (tool), 1/2 in square drive, 3 gears 1000/1500/2300 rpm, max torque 90/135/213 N m, up to 3,500 impacts/min, 228 x 154 x 94 mm (hire listing https://www.speedyservices.com/hire/power-tools/02_0288-h-Hilti-SIW-6AT-A22--Impact-Wrench-22v-104kg, Hilti product-page search snippet)
IMPACT_MAX_TORQUE_NM	213	N m	read	Hilti SIW 6AT-A22 technical data: 1.04 kg (tool), 1/2 in square drive, 3 gears 1000/1500/2300 rpm, max torque 90/135/213 N m, up to 3,500 impacts/min, 228 x 154 x 94 mm (hire listing https://www.speedyservices.com/hire/power-tools/02_0288-h-Hilti-SIW-6AT-A22--Impact-Wrench-22v-104kg, Hilti product-page search snippet)
IMPACT_CAM_NM	5	N m	estimate	anvil torque at which the hammer's spring lets the cam ride off and blows begin (below it the anvil turns with the motor, as a nut runs down its thread); not published
IMPACT_ROTOR_INERTIA	5e-05	kg m2	estimate	motor rotor and gear train reflected to the anvil while the hammer is engaged (running)
IMPACT_SPINUP_S	0.1	s	estimate	time the motor takes from the trigger's pull to its set speed (the electronic soft start of a brushless tool; tools.ImpactWrench ramps the anvil's speed linearly); not published for the SIW 6AT-A22

name	value	unit	label	basis
IMPACT_GEAR_RPM	1000, 1500, 2300	rpm	read	no-load speeds of the wrench's three gears; their rated torques 90/135/213 N m are 0.090, 0.090 and 0.093 N m per rpm, proportional to speed (Hilti SIW 6AT-A22 technical data: 1.04 kg (tool), 1/2 in square drive, 3 gears 1000/1500/2300 rpm, max torque 90/135/213 N m, up to 3,500 impacts/min, 228 x 154 x 94 mm (hire listing https://www.speedyservices.com/hire/power-tools/02_0288-h-Hilti-SIW-6AT-A22--Impact-Wrench-22v-104kg, Hilti product-page search snippet)). The plain wrench (no Adaptive Torque module) runs at its gear's speed
IMPACT_HAMMER_INERTIA	0.0001452	kg m2	estimate	moment of inertia of the wrench's hammer about the anvil's axis: a 0.30 kg steel ring with a 22 mm radius of gyration (it has to fit the 60 mm head of a 1.04 kg tool); not published. It sets the energy of a blow, I w^2/2 = 4.2 J at 2300 rpm (245 W at 3,500 blows/min), and with IMPACT_JAW_STIFFNESS how many blows a setting takes (the count is proportional to 1/I)
IMPACT_JAW_STIFFNESS	5386.17	N m/rad	derived	one-way torsion spring between the hammer and the anvil (jaws, anvil, socket): the value for which a blow at the largest speed peaks at the maker's largest torque against a nut that does not turn, w sqrt(k I) = IMPACT_MAX_TORQUE_NM at IMPACT_MAX_RPM (5,384 N m/rad with the hammer above; only the product k I is calibrated). Same order as the torsional stiffness of a 1/2 in square drive 25 mm long, G J/L = 80 GPa x 0.1406 x (12.7 mm)^4 / 25 mm = 11,700 N m/rad, in series with the jaws and the socket
IMPACT_DILATION	16	-	engine	time dilation of a blow (tools.ImpactWrench, declared device): the hammer's inertia x 16^2 at 1/16 of its speed on the same spring, and the anvil's and socket's inertia x 16^2 for the duration of a blow. Blow energy and peak torque are unchanged; a blow lasts 5-50 ms instead of 0.3-3 ms and the hammer moves 0.9 deg per 1 ms tick at 2300 rpm (13.8 deg undilated; the anvil turns at most 21 rad/s, 1.2 deg a tick, while impacting, E20). The explicit spring needs 16^2 x the anvil's 3.06e-5 kg m2 >= k dt^2 = 5.4e-3 (factor >= 13.3): with 10 the peak against a held nut came out at 86 N m for 139. Checked against a 2 microsecond undilated solution of the same equations (hammer, spring, a nut whose friction torque rises 21.9 N m/rad): 65 blows and 1.74 s to 68 N m at 1500 rpm for 63 and 1.66 s; held nut 138.9 N m for 138.9; gear 3 unmoderated 154 N m after 3 s for 166
AT_FIRST_RPM	2300	rpm	authored	motor speed the Adaptive Torque module sets for the scanned 1/2 in KB-TZ2 (the patent's 'first rotational speed'; US 10,589,407 B2, https://patents.google.com/patent/US10589407B2: 'rotational speeds assigned to different tightening torques are stored', Hilti's table is not published): the wrench's full speed, where a blow carries 4.2 J and peaks at 213 N m against a held nut, so the module's stop on the sensed torque does the limiting. Chosen on the one timing the patent gives, 'approximately 2 seconds' for an impact setting: at 1500 rpm (1.8 J a blow) the setting took 3.2 s and 111 blows on a concrete face and ran into the 5 s time-out through a wood ledger (183 blows, 67.3 N m; E20). The speed whose blows peak at 68 N m itself (734 rpm) is no program at all: the torque then approaches 68 N m only asymptotically (reduced model: 0.9 of it after about 390 blows, 21 s)
AT_TIMEOUT_S	5	s	authored	impacting time after which the module stops without green (red); the patent puts a whole impact setting at about 2 s. The module's stop itself has no parameter of its own: it stops (green) when the torque a blow passed to the socket reaches the anchor's installation torque (the threshold is authored: the patent's 'predetermined threshold value' is not published)
KBTZ2_WALL_LENGTH_M	0.1143	m	read	KB-TZ2 1/2 x 4-1/2 (letter G): fixture 0 to 1/4 in at h_nom 3-3/4 in (Hilti KWIK Bolt TZ2 Concrete and Masonry Technical Supplement, Dec 2020, Table 47 (https://images.thdstatic.com/catalog/pdfImages/a3/a3a0c225-ae97-40b5-92b3-745632435551.pdf)); clean-and-set seats its washer on the bare wall
KBTZ2_LEDGER_LENGTH_M	0.1778	m	read	KB-TZ2 1/2 x 7 (letter L), the shortest that fastens a 1-1/2 in ledger at h_nom 3-3/4 in: fixture 0 to 2-3/4 in (1/2 x 5-1/2 takes only 1-1/4 in) (Hilti KWIK Bolt TZ2 Concrete and Masonry Technical Supplement, Dec 2020, Table 47 (https://images.thdstatic.com/catalog/pdfImages/a3/a3a0c225-ae97-40b5-92b3-745632435551.pdf)). Its washer seats h_nom + the ledger thickness from the tip (a 146 mm anchor would seat 133 mm from the tip and stop at 91 mm)

name	value	unit	label	basis
ABORTED_HOLE_DISTANCE_FACTOR	2	x aborted depth	read	a new hole at least twice the depth of the aborted hole away from it, or closer if the aborted hole is filled with high-strength mortar (filling is the alternative, not a requirement): ETAG 001 Part 1 section 7.3 (https://umwelt-online.de/recht/bau/bauprod/etag001_1b.htm); ETA-06/0168 Annex B1 'In case of aborted hole: new drilling at a minimum distance away of twice the depth of aborted hole or smaller distance if the aborted hole is filled with high strength mortar and if under shear or oblique tension load it is not in the direction of the load application' (https://www.dibt.de/pdf_storage/2025/ETA-06!0168(8.06.01-128!25)e.pdf). Measured from the aborted hole; ESR-4266 (US) has no aborted-hole text
RELOCATE_MAX_M	0.15	m	authored	farthest a relocated anchor may move along its strut channel from the aborted hole without the layout being redrawn (half the 300 mm anchor spacing); excludes the line's other anchor hole
LEDGER_THICK_M	0.038	m	literature	2x lumber dressed 1-1/2 in (38 mm)
LEDGER_DEPTH_M	0.184	m	literature	2x8 dressed 7-1/4 in (184 mm)
LEDGER_EDGE_TOP_M	0.0508	m	read	IRC Table R507.9.1.3(2) 'Placement of lag screws and bolts in deck ledgers' (ledger: top edge 2 in, bottom edge 3/4 in, ends 2 in, row spacing 1-5/8 in; staggered top to bottom), reproduced at https://gocodebook.com/us/california/california-residential-code/floors/slabs-decks-and-steel-framing/exterior-decks-attachment
LEDGER_EDGE_BOTTOM_M	0.01905	m	read	IRC Table R507.9.1.3(2) 'Placement of lag screws and bolts in deck ledgers' (ledger: top edge 2 in, bottom edge 3/4 in, ends 2 in, row spacing 1-5/8 in; staggered top to bottom), reproduced at https://gocodebook.com/us/california/california-residential-code/floors/slabs-decks-and-steel-framing/exterior-decks-attachment
LEDGER_ENDS_M	0.0508	m	read	IRC Table R507.9.1.3(2) 'Placement of lag screws and bolts in deck ledgers' (ledger: top edge 2 in, bottom edge 3/4 in, ends 2 in, row spacing 1-5/8 in; staggered top to bottom), reproduced at https://gocodebook.com/us/california/california-residential-code/floors/slabs-decks-and-steel-framing/exterior-decks-attachment
LEDGER_ROW_SPACING_M	0.041275	m	read	IRC Table R507.9.1.3(2) 'Placement of lag screws and bolts in deck ledgers' (ledger: top edge 2 in, bottom edge 3/4 in, ends 2 in, row spacing 1-5/8 in; staggered top to bottom), reproduced at https://gocodebook.com/us/california/california-residential-code/floors/slabs-decks-and-steel-framing/exterior-decks-attachment
SDS_BIT_WORKING_M	0.1524	m	read	Bosch HC2083 1/2 in: working length 6 in, total 8 in (Bosch Bulldog SDS-plus bits (https://www.boschtools.com/us/en/brocas-para-martillo-perforador-bulldog-sds-plus%C2%AE-43549-ocs-ac/)); the 4 in working-length bit (HC2081) cannot drill the 108 mm anchor hole
SDS_BIT_STICKOUT_M	0.16	m	derived	the modelled bit ahead of the chuck sleeve: its tip reaches stickout - 10 mm (SDS slide in the idle-stroke window) = 150 mm, the HC2083's 152 mm working length within 2 mm
SDS_LONG_BIT_WORKING_M	0.254	m	read	Bosch HC2084 1/2 in: working length 10 in, total 12 in (Bosch Bulldog SDS-plus bits (https://www.boschtools.com/us/en/brocas-para-martillo-perforador-bulldog-sds-plus%C2%AE-43549-ocs-ac/)): through a 38 mm ledger to 108 mm needs 146 mm; the 6 in bit leaves 6 mm, and in this model (tip reach 150 mm) the full-diameter depth through the ledger would stop at 108.9 mm
SDS_LONG_BIT_STICKOUT_M	0.2616	m	derived	HC2084 stickout: the HC2083's plus the 4 in of extra working length

name	value	unit	label	basis
ROLLING_B_M	0.0001	m	literature	rolling-resistance lever arm (MuJoCo condim-6 rolling friction, torque = b N) of the round rims of loose steel parts (punch barrel and cap, anchor washer, carriage-bolt dome): b = 0.1 mm, 'hardened steel ball bearings on steel' (Crr 0.0010-0.0015) in the rolling-resistance table of https://en.wikipedia.org/wiki/Rolling_resistance (secondary source). Without it (condim 3) round parts roll 0.7-1.8 m on the steel plate and are still moving after 20 s. The metal faces (world.py, contact priority 1) and the hole module's face pads (priority 2) carry the same lever arm, since the higher-priority geom's contact parameters rule. Machinery's Handbook 31st ed. p. 174 gives 0.02 in (0.51 mm) for iron on iron and calls published values approximate: 0.1-0.51 mm brackets it. Round parts can also rock on the plate when their shank's face support switches on and off (holes.face_near_segment); that is not this value
MU	default: 0.5; pad: 0.8; steel_steel: 0.2; steel_wood: 0.4; steel_concrete: 0.45; wood_wood: 0.35; rubber: 0.9; bit_wall: 0.1; floor: 0.6; steel: 0.3; aluminium: 0.3; douglas_fir: 0.45; plywood: 0.45; concrete: 0.6	-	estimate	friction coefficients (MuJoCo: one coefficient, static = sliding). Machinery's Handbook 31st ed. p. 173, Bowden and Tabor's tables: steel on steel 0.8 static and 0.4 sliding clean, 0.16 static and 0.03 sliding oiled; iron on iron 1.0 clean, 0.15-0.20 lubricated; aluminium on aluminium 1.35 clean, 0.30 lubricated; wood on metal 0.2-0.6 clean; wood on wood 0.25-0.5 clean. Shop steel is lightly oiled: steel 0.3 and steel_steel 0.2 lie BETWEEN the oiled values (up to 0.16) and the clean sliding value (0.4), inside neither range, chosen, not measured (open). steel_concrete 0.45 is below a published 0.57-0.70 for steel on concrete (a search summary; not re-read) and is kept as a chosen value for a dusty face (open). The Franka pad 0.8 is the repository's qualified value (barcode_scanning)
CONCRETE_ROT_EV_J_MM3	5	J/mm3	estimate	rotation-only scraping of concrete: why the hammer mode must be on
REAM_ZONE_D	1	x d	estimate	axial length, in bit diameters, of the hole wall a side-loaded spinning bit wears (holes._ream): the wall the bit's friction work at CONCRETE_ROT_EV_J_MM3 removes is spread over it as a crescent. With it a 0.2 deg misfit growing at 0.9 mm/s of drilling is worn away at a ~40 N side load; no measured value found (open in substance)
KAPPA_SDS_WOOD	2	-	estimate	SDS carbide head cutting timber rotary-only (scraping)
SDS_IN_METAL_FACTOR	1.5	x W	estimate	SDS head in steel (negative rake)
HAMMER_INDEX_DEG	30	deg	estimate	rotation between blows needed for a fresh strike (full trigger indexes ~86 deg per blow)
HAMMER_NO_INDEX_EFF	0.2	-	estimate	removal efficiency of a blow on the spot the last blow crushed (hammer-only mode)
DUST_CUSHION_M	0.003	m	estimate	dust column under the bit that halves a blow ~ ln2 x this
WITHDRAW_CARRIES	0.5	-	open	share of loose dust a withdrawal of the bit carries out
NOZZLE_BOTTOM_TOL_M	0.005	m	authored	a vacuum pass counts if the nozzle reaches within 5 mm of the floor; a threshold of this environment, no published value
BRUSH_DEPTH_FRACTION	0.9	-	authored	a brush stroke reaches >= 90 % depth
SPALL_DIAMETER_FACTOR_LATERAL	1.8	x d	estimate	entry crater when the bit is pushed sideways during the first blows
SPALL_LATERAL_N	40	N	estimate	lateral load that enlarges the entry spall
SPALL_CRATER_SLOPE_DEG	35	deg	literature	slope of the entry spall crater's wall from the face: the concrete breakout cone of the CCD method (projected radius 1.5 h_ef, about 35 deg; ACI 318 R17.6.2.1, EN 1992-4), used as the crater shape. A 1.4 d crater on the 13.3 mm bit is then 1.86 mm deep; the hole's wall proxies start below it (holes._place_hole)
BURR_GROSS_M	0.0016	m	read	exit burr that must be removed before bolting: RCSC Specification for Structural Joints (2004) section 3.4, 'Burrs that extend 1/16 in. or less above the surface are permitted to remain on the faying surfaces of snug-tightened joints ... Burrs that extend over 1/16 in. ... shall be removed' (https://www.boltcouncil.org/files/2004RCSCSpecification.pdf). Under BURR_CAP_FRACTION a 9 mm drill in S235 leaves 0.19 mm eased, 0.42 mm at 1.1 kN, 0.49 mm at 1.5 kN: this limit is not reached by any push the press allows
BURR_MIN_M	5e-05	m	estimate	uniform minimum exit burr of a backed or eased exit

name	value	unit	label	basis
BURR_CAP_FRACTION	0.25	x cap thickness	derived	exit burr height / remaining thickness at cap collapse sqrt(2T/(pi sigma_y)): geometric mean of nine St37 + conventional HSS twist-drill burr heights (Alrabii 2009, J. Eng. Univ. Baghdad 15(4) 4219, Figs 5-6: 0.12-1.15 mm for 5-12.5 mm drills; 9 mm at 0.03 mm/rev 0.24 mm) against the cap thickness this law gives at their thrust; ratios 0.10-0.77 (geometric s.d. x1.8). The cap is the drill cap pushed out; the burr is the rim left behind (Kim & Dornfeld's uniform burr with cap)
WOOD_SCORCH_K_PER_J	1.5	K/J	estimate	lumped wall-segment temperature rise per joule of rubbing (0.5 mm segment, k 0.12 W/m K)
WOOD_RUB_RECOVER_J_S	5	J/s	estimate	rubbing-heat loss of a wall segment
OIL_H_FACTOR	4	x	estimate	cutting oil raises tip heat transfer
HSS_K	21.8	W/m K	read	Bohler S600 conductivity
BURN_WEAR_STEP	0.3	W	estimate	burnt corners: irreversible wear step above 550 C
SDS_WEAR_PER_MM3	8.9e-08	W/mm3	derived	KIT: ROP -15 % after 21 m of 10 mm holes, i.e. W +0.7 %/m = 0.007 per 78,540 mm3 removed
REFLEX_EFFORT_FRACTION	0.95	-	authored	drilling contact setting: a joint stops above 95 % of its effort limit
RECOIL_DUTY	0.3	x blow period	derived	width of the half-sine recoil on the housing per blow, as a share of the blow period: the half-sine with the same impulse and peak as the air-spring (upper-chamber gauge) pressure pulse of an electro-pneumatic hammer, digitised from Wang 2014 Fig. 4.9 (Concordia MASc, Bosch 11313 EVS model, 1830 bpm, 75 N push; https://spectrum.library.concordia.ca/978749/1/Wang_MSc_F2014.pdf): equivalent half-sine 9.8-10.1 ms of the 32.8 ms period (0.30), FWHM 5.2 ms (0.16); scaled with the crank angle (an assumption) to the hammer's blow rate
RECOIL_PULSE_S	0.00391304	s	derived	recoil half-sine width at the full blow rate (RECOIL_DUTY x 13.0 ms = 3.9 ms; at a partial trigger the pulse widens with the blow period)
PRESS_STAND_M	0.76	m	authored	steel stand under the benchtop press; puts the table at 0.964 m (target 0.97 m)
PRESS_BUTTON_N	8	N	estimate	push-button actuation force
PRESS_BUTTON_TRAVEL_M	0.003	m	estimate	push-button travel
PRESS_BUTTON_SWITCH_FRACTION	0.5	-	derived	the switch point is where the button spring (PRESS_BUTTON_N over the travel, half preloaded) pushes back with PRESS_BUTTON_N; at 0.7 of the travel a steady push needed 9.6 N, not the 8 N actuation force
VACUUM_TANK_D_M	0.46	m	estimate	tank diameter of a 14 gal (53 l) drum vac; the manual gives no dimensions (RIDGID RV2400HF 14 gal wet/dry vac with certified HEPA filter, owner's manual SP6775 (https://images.thdstatic.com/catalog/pdfImages/24/24e4f496-2f24-43d8-88fd-97a2df60ab0a.pdf): p. 10 'turn the unit on by pushing the switch from position "O" to position "I"'; p. 16 parts 830564 'Actuator, Switch', 73180 'Switch, Toggle')
VACUUM_TANK_H_M	0.45	m	estimate	tank height on its caster base (14 gal class)
VACUUM_HEAD_H_M	0.12	m	estimate	power head height above the tank (14 gal class)
VACUUM_ROCKER_SIZE_M	0.032, 0.025, 0.008	m	read	on/off rocker 1.25 x 1 in (RIDGID 83797 rocker switch for the WD1660 vac, retailer listing https://empirerigging.com/ridgid-83797, search-level); its 8 mm actuator thickness estimated. That is another RIDGID model's switch: the modelled RV2400HF's own parts list names a toggle switch with an actuator (vacuum.py), whose size was not found, so the modelled switch is a rocker of the WD1660's size (a cosmetic difference)
VACUUM_ROCKER_THROW_RAD	0.174533	rad	estimate	rocker tilt either side of its middle (typical rocker switch)
VACUUM_ROCKER_PRESS_N	8	N	estimate	push on the middle of a rocker half that turns it over; the push-button value (PRESS_BUTTON_N), no published figure for this switch. Measured on the built switch: a vertical push on the raised half's top face (6.5 mm from the pivot) turns it over between 9.0 and 10.5 N; 4 N does not
VACUUM_ROCKER_DETENT_NM	0.064	N m	derived	detent torque at the rocker's middle: VACUUM_ROCKER_PRESS_N at a quarter of its length from the pivot (the middle of a half)
VACUUM_ROCKER_ARMATURE	0.0001	kg m2	estimate	reflected inertia of the toggle mechanism under the rocker (keeps omega dt < 1 at 1 ms)
VACUUM_ROCKER_DAMPING	0.002	N m s/rad	estimate	rocker joint damping
PRESS_ROTOR_INERTIA	0.005	kg m2	estimate	3/4 hp rotor (~0.002 kg m2) reflected through a 1.7:1 belt
PRESS_STOP_PITCH_M	0.0015	m	estimate	depth-stop rod M12 x 1.5
PRESS_COAST_S	1	s	estimate	spindle coast-down time constant after switching off
VISE_JAW_MU	0.25	-	estimate	ground steel jaw plates on steel
VISE_CRANK_R	0.08	m	estimate	crank knob radius
VISE_DRY_FRICTION_NM	0.05	N m	estimate	unloaded screw drag
VISE_SCREW_RM	0.0065	m	estimate	mean thread radius of the vise screw
PUNCH_RELEASE_TRAVEL_M	0.003	m	estimate	point travel to the release force

name	value	unit	label	basis
PUNCH_POINT_ARMATURE	0.03	kg	estimate	internal hammer and spring mass reflected to the point (keeps omega dt < 1)
PUNCH_RETURN_N	5	N	estimate	light spring that pushes the point back out after the release, at full point travel (half of it preloaded); once the internal hammer has struck, the main spring's stored energy (1/2 k x^2 = PUNCH_ENERGY_J at the release) has gone into the strike and no longer pushes the point
PUNCH_RECOCK_M	5e-05	m	authored	the trip pin re-seats (cocked again) once the point is back out within 0.05 mm, where the main spring's force (1.7 N) is below the return spring's (2.5 N): re-cocking adds no energy
PHANTOM_LATERAL_K	24000	N/m	read	KIT hand-arm model lateral stiffness 24 N/mm (compliant holder of the walking bench E5)
VISE_PRECLAMP_NM	5	N m	authored	crank torque of the start-state clamp (a firm hand on an 80 mm crank: 62 N)
WRENCH_MAX_N	150	N	authored	largest cartesian_wrench force accepted (the drilling contact setting's hand-force range)
WRENCH_MAX_NM	25	N m	authored	largest cartesian_wrench moment accepted
SDS_BIT_D_MM	13.3	mm	read	1/2 in SDS-plus carbide tip: ANSI B212.15 allows 0.520 to 0.530 in, the middle is 13.3 mm
PRESS_BITS	press-drill: 9, 0.081, 1000; press-blind: 6.8, 0.069, 2400; match-bolt: 9, 0.081, 1000	mm, m, rev/min	authored	the bit in the press's chuck, its stick-out and the spindle speed the belt is set to, per press task: a 9 mm clearance drill for an M8 bolt (ISO 273, medium) and a 6.8 mm tapping drill for M8 (ISO 2306). The speeds are round numbers chosen here (28 m/min at the 9 mm drill's rim, 51 m/min at the 6.8 mm drill's), not steps read from the G7943's belt chart
WALL_MARK_TOL_M	0.003	m	authored	the evaluator passes a wall hole whose entry is within 3 mm of its mark (a 1/2 in anchor in a 9/16 in fixture hole has 1.6 mm of clearance all round)
WALL_AXIS_TOL_DEG	3	deg	authored	the evaluator passes a wall hole whose axis is within 3 degrees of the wall's normal
MAX_LAYOUT_SHIFT_M	0.15	m	authored	rebar-relocate: the largest shift of a mark the task allows, half the 300 mm bar spacing

References

[1] *Guide to Machinery's Handbook*, 27th ed., pp. 200–201, 237 (Shaw–Oxford thrust and torque; A, B, J; worked example). http://www.iqytechnicalcollege.com/Machinery%20Handbook.pdf

[2] Strenkowski, Hsieh and Shih 2004 (Rubenstein's AISI 1020 thrust and torque data). https://career.engin.umich.edu/wp-content/uploads/sites/51/2013/08/04_MTM_drilling_modeling.pdf

[3] Audy 2009, Annals FEH (Micheletti and Levi; Subramanian). https://annals.fih.upt.ro/pdf-full/2009/ANNALS-2009-1-04.pdf

[4] Ema, effects of twist drill point geometry on torque and thrust. https://www.ed.gifu-u.ac.jp/file/3627.pdf

[5] OSG drilling technical guide (walking, thin plate, oversize). https://www.uscti.com/uscti-pages/HOLEMAKING%20-%20Technical%20Data.pdf

[6] Grizzly G7943/G7944 drill press manual. https://cdn0.grizzly.com/manuals/g7943_m.pdf

[7] Böhler S600 datasheet; Movchan and Chornoivanenko 2020 (HSS heat capacity, tempering). https://ufm.imp.kiev.ua/article/v21/i01/

[8] Rennsteig 430 130 automatic centre punch (length, body, adjustable release force; a retailer's listing, read at search level). https://www.testbericht.de/produkte/rennsteig-werkzeuge-automatischer-koerner-430-130

[9] DeWalt DCD800/DCD805 instruction manual (speeds, torque, mass). https://assets.dewalt.com/GLOBALBOM/QU/DCD800B/1/Instruction_Manual/EN/NA229093_DCD800_DCD805_T1_NA.pdf

[10] DeWalt DCH273 instruction manual (blow energy and rate, speed, mass). https://service.blackanddecker.com.au/i/DEWALT/GLOBALBOM/XE/DCH273/2/Instruction_Manual/ZH/DCH273_T2_ASIA.pdf

[11] Szwajka et al. 2019 (MDF drilling forces). https://pmc.ncbi.nlm.nih.gov/articles/PMC6384622/

[12] Sedlacek and Slany 2010 (Hocheng–Dharan delamination; chisel by pre-drilling). https://mmscience.eu/journal/issues/june-2010/articles/analysis-of-delamination-in-drilling-of-composite-materials/download

[13] Chuchała et al. 2021 (Scots pine cutting toughness). https://df.tuzvo.sk/sites/default/files/07-01-21.pdf

[14] USDA FPL, *Wood Handbook*, FPL-GTR-282 (2021).
https://research.fs.usda.gov/download/treesearch/62200.pdf

[15] Botti et al. 2020 (feed force against drilling rate and vibration).
https://stacks.cdc.gov/view/cdc/225793/cdc_225793_DS1.pdf

[16] Rempel, Barr and Antonucci 2017 (hammer-drill test bench; bit wear).
https://stacks.cdc.gov/view/cdc/49915/cdc_49915_DS1.pdf

[17] Uhl et al. 2022 and Zimprich et al. 2026, KIT (drilling rate, push, wear, hand-arm model).
https://publikationen.bibliothek.kit.edu/1000151192/149581508;
https://publikationen.bibliothek.kit.edu/1000191284/176624221

[18] ICC-ES ESR-4266, Hilti Kwik Bolt TZ2 (dimensions, h_{nom}, hole cleaning).
https://icc-es.org/wp-content/uploads/report-directory/ESR-4266.pdf

[19] OSHA 29 CFR 1926.1153, Table 1 (vacuum dust collection for drilling in concrete).
https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.1153

[20] EN 1992-1-1, Table 3.1 (C25/30), via eurocodeapplied.
https://eurocodeapplied.com/design/en1992/concrete-design-properties

[21] Franka Research 3 datasheet and product manual; libfranka collision behaviour defaults.
https://raw.githubusercontent.com/frankaemika/libfranka/master/include/franka/robot.h

[22] Hilti US5914882 (anti-rotation clutch). https://patents.google.com/patent/US5914882

[23] Hilti, *KWIK Bolt TZ2 Concrete and Masonry Technical Supplement*, Dec. 2020 (anchor lengths, fixture thickness, washer and nut, setting table).
https://images.thdstatic.com/catalog/pdfImages/a3/a3a0c225-ae97-40b5-92b3-745632435551.pdf

[24] Hilti SIW 6AT-A22 impact wrench, technical data (mass, gears, blow rate, maximum torque; a hire listing and the maker's product page at search level).
https://www.speedyservices.com/hire/power-tools/02_0288-h-Hilti-SIW-6AT-A22--Impact-Wrench-22v-104kg

[25] Hilti, US 10,589,407 B2 (the adaptive-torque module: a first phase that ends when a sensed torque reaches a threshold, then a counted follow-on). https://patents.google.com/patent/US10589407B2

[26] ISO 16047 and VDI 2230 (the split of a tightening torque into pitch, thread friction and bearing friction); cited for the relation only.

[27] Shigley, *Mechanical Engineering Design*, Tables 8-5 and 8-6 (thread and collar friction), as reproduced at
https://amesweb.info/Screws/screw-nut-collar-friction-table.aspx

[28] Carr Lane, *Standard Strap Clamp Forces* (stud torque, tension and clamping force).
https://www.carrlane.com/engineering-resources/technical-design-information/clamp-strap

[29] TE-CO catalogue: plain strap clamp 30402, spherical flange nuts and washers.
https://catalog.te-co.com/viewitems/machined-clamps/plain-clamps

[30] RCSC, *Specification for Structural Joints Using ASTM A325 or A490 Bolts*, 2004 (burrs up to 1/16 in may remain).
https://www.boltcouncil.org/files/2004RCSCSpecification.pdf

[31] Alrabii 2009, *J. Eng. Univ. Baghdad* 15(4), Figs. 5–6 (exit burr heights of HSS twist drills in St37).

[32] Segonds, Masounave, Songmene and Bès 2013, *J. Mater. Process. Technol.* (burr type at breakthrough in ductile materials; accepted manuscript). https://espace2.etsmtl.ca/id/eprint/5829/1/A-simple-analytical-model-for-burr-type-prediction-in-drilling-of-ductile-materials.pdf

[33] Schneider, *Cutting Tool Applications*, chapter on drills and drilling operations (breakthrough in thin material; read at search level).

[34] Black & Decker, US 7,410,006 (a drill catching at breakthrough turns in the operator's grasp).
https://patents.google.com/patent/US7410006

[35] McMillin and Woodson 1972, *Forest Prod. J.* 22(11):55 (thrust and torque of a 25.4 mm spur machine bit in southern pine; Table 1 read in the scan). https://research.fs.usda.gov/download/treesearch/8240.pdf

[36] Hilti TE 6-S operating instructions, technical data (blow energy and rate, drilling performance in medium-hard concrete at 8, 12 and 16 mm; read in the document). https://staff.washington.edu/nelsonr/manuals/Hilti%20TE%206-S.pdf

[37] Bordinassi et al., COBEM 2003, COB03-1926 (thrust and torque of 6 mm HSS drills in ABNT 1020; three points read off Fig. 7). https://abcm.org.br/anais/cobem/2003/html/pdf/COB03-1926.pdf

[38] US 8,424,180 (a setting tool for wedge anchors; the clip "requires that a significant force be applied").

[39] arXiv:2403.19948, Sec. IV-B (an industrial arm of 13 kg payload pushing a 12 mm wedge anchor: 7 mm in before it stuck; no force given).

[40] Hilti B 22/5.2 battery pack, 0.830 kg (a distributor's data table) and SI-AT-A22 module, 0.26 kg (the maker's product page at search level). https://www.oglaend-system.com/product-variants/battery-pack-b22-5-2-li-ion-article39899-52088.html?pid=42260

[41] Estwing E3-32BP ball-peen hammer, 32 oz (907 g) head, 368 mm overall (a distributor's listing, read at search level). https://www.drapertools.com/product/23578/estwing-e3-32bp-ball-pein-hammer-32oz-907g/

[42] Wang 2014, MASc thesis, Concordia University, Fig. 4.9 (an electro-pneumatic hammer's blow on a bench). https://spectrum.library.concordia.ca/978749/1/Wang_MSc_F2014.pdf

[43] EN 1995-1-1, eq. 8.32 (embedment strength of timber under a dowel).

[44] Standards cited for dimensions and tolerances only: ANSI B212.15 (carbide-tipped masonry bits), ISO 4017 (hexagon head screws), ASME B18.5 (carriage bolts), ISO 273 (clearance holes), ISO 2306 (tap drills), EN 10056-1 (equal angles), EN 10058 (flat bars), ASTM F844 (washers), ACI 318-19 Table 20.5.1.3.1 (cover), EN 206 (strength classes).

[45] JET JDP-17 drill press (quill travel per handle revolution, transferred to the modelled press: see `PRESS_PINION_R` in the ledger).

[46] WEN DPA424 4 in drill-press vise. https://wenproducts.com/products/wen-dpa424-4-in-drill-press-vise

[47] The Wood Database, Douglas-fir (the photograph the shavings' colour was measured on).

[48] Poly Haven (CC0): `Drill_01`, `drill_press_01`, `concrete`, the dressing models and textures listed with their SHA-256 in `drilling/visual_assets.json`. https://polyhaven.com

Sources that the text does not cite by number where it uses them: [4, 3, 37] are data rows of E2 (Table 12); [5] is the guidance on walking and thin plate behind E5 and E6; [15, 16] are the percussion data behind E8; [19] the vacuum requirement of the dust step; [23] the anchor lengths; [14, 13, 47] wood properties and the shavings' colour; [45, 46] the press's pinion and the vise; [40] the wrench's pack and module; [44] dimensions and tolerances. Their values are in the ledger.

Every web address the parameter ledger cites is in the ledger's *basis* column (Appendix A); sources read only at search level (a result snippet, not the opened document) are marked so there and above.